FINITE-ORDER EXPONENTIAL MIXING DOES NOT IMPLY THE CENTRAL LIMIT THEOREM

BERNAT BASSOLS CORNUDELLA[1] AND MATHEUS M. CASTRO[2]

Abstract. We show that finite-order exponential mixing does not imply the central limit theorem for smooth diffeomorphisms. For every integer $R \geq 2$, we construct a $\mathcal{C}^\infty$ diffeomorphism $T : M \to M$ of a compact smooth manifold M and a T-invariant probability measure μ such that (T, μ) is exponentially mixing of order R for $\mathcal{C}^1(M)$ observables, but not of order $2R + 2$. For some centred observable $f_0 \in \mathcal{C}^\infty(M)$, no subsequence of the normalised sums $\frac{1}{\sqrt{n}} \sum_{i=0}^{n-1} f_0 \circ T^i$ converges in distribution to a Gaussian law. In particular, exponential decay of two-point correlations implies neither exponential mixing of all orders nor the central limit theorem. We realise (T, μ) in two ways. In the first, T is a fixed linear Anosov diffeomorphism of $\mathbb{T}^2$ and μ is singular with respect to Lebesgue measure. In the second, M can be any compact smooth surface without boundary and μ is a physical measure whose statistical basin contains a nonempty open set. The main theorems have been formalised in Lean 4.

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1. Introduction

The statistics of a given transformation $T : M \to M$ are encoded by the behaviour of the partial sums

$$S_n f := \sum_{i=0}^{n-1} f \circ T^i$$

for suitable observables $f : M \to \mathbb{R}$. When T preserves a probability measure μ, the sequence of functions $(f \circ T^i : (M, \mu) \to \mathbb{R})_{i \in \mathbb{N}}$ may be seen as a stationary process and, in particular, the random variables $f \circ T^i$ are identically distributed.

If T is μ-ergodic and $f \in L^1(M, \mu)$, Birkhoff's ergodic theorem [3] yields

$$\frac{S_n f(x)}{n} \xrightarrow{n\to\infty} \mu(f) := \int f \, \mathrm{d}\mu \quad \text{for } \mu\text{-almost every } x. \tag{1}$$

This is the dynamical analogue of the strong law of large numbers for independent and identically distributed (i.i.d.) random variables. While (1) determines the long-term average behaviour of the systems, this convergence is not effective. Hence, Birkhoff's ergodic theorem provides no information about the fluctuations of $S_n f - n\mu(f)$, their appropriate scaling, or their limiting distribution.

For i.i.d. random variables with finite variance, the classical Lindeberg–Lévy central limit theorem (CLT) states that their centred sums, normalised by $\sqrt{n}$, converge in distribution to a centred normal distribution [22]. The corresponding question in dynamics concerns whether $(S_n f - n\mu(f))/\sqrt{n}$ also converges to a normal distribution. Replacing f by $f - \mu(f)$, we may assume that $\mu(f) = 0$. We say that f satisfies a CLT if

$$\frac{S_n f}{\sqrt{n}} \xRightarrow{n\to\infty} \mathcal{N}(0, \sigma_f^2), \tag{2}$$

for some $\sigma_f^2 \geq 0$, where $\Rightarrow$ denotes convergence in distribution with respect to μ.

The random variables $(f \circ T^i)_{i \in \mathbb{N}}$ are, in general, far from being independent. Therefore, it is natural to ask which conditions on the dependence of observations $(f \circ T^i)_{i \in \mathbb{N}}$ ensure the CLT for regular observables. A basic measure of this dependence is the *correlation* $\mu(g\, h \circ T^i) - \mu(g)\mu(h)$ between observations separated by i iterates. On a compact Riemannian manifold M, we say that correlations decay at rate $\rho(i) \to 0$ for $\mathcal{C}^r$ observables if

$$\left|\mu(g\, h \circ T^i) - \mu(g)\mu(h)\right| \leq C\rho(i)\|g\|_{\mathcal{C}^r(M)}\|h\|_{\mathcal{C}^r(M)}$$

for all $g, h \in \mathcal{C}^r(M)$ and $i \geq 1$, with C independent of g, h and i. *Exponential decay of correlations* corresponds to $\rho(i) = e^{-ci}$ for some $c > 0$.

A remarkable theorem of Björklund and Gorodnik [4] establishes exponential mixing of all orders as a sufficient condition for the CLT. This condition controls joint correlations of arbitrarily many observables. We say that (T, μ) is *exponentially mixing of all orders* for $\mathcal{C}^1$ observables[1] if, for every integer $s \geq 2$, there exist constants $C_s, c_s > 0$ such that

$$\left|\mu\left(\prod_{i=1}^{s} f_i \circ T^{n_i}\right) - \prod_{i=1}^{s} \mu(f_i)\right| \leq C_s e^{-c_s \min_{1\leq i<s}(n_{i+1}-n_i)} \prod_{i=1}^{s} \|f_i\|_{\mathcal{C}^1(M)} \tag{3}$$

for all $f_1, \ldots, f_s \in \mathcal{C}^1(M)$ and integers $0 \leq n_1 < \cdots < n_s$. In this setting, if T is a $\mathcal{C}^1$ diffeomorphism, the result of Björklund and Gorodnik [4, Theorem 1.1] implies that every centred real observable $f \in \mathcal{C}^1(M)$ satisfies the CLT (2).

In this paper, we show that the hypothesis of exponential mixing of all orders in the theorem of Björklund and Gorodnik is sharp with respect to the number of mixing orders. This result holds even for systems with physical invariant measures. Our main results have been formalised in Lean 4 (see Section 9).

For every nonempty compact smooth surface M without boundary and every integer $R \geq 2$, we construct a $\mathcal{C}^\infty$ diffeomorphism $T : M \to M$ preserving a physical probability measure μ such that (T, μ) is exponentially mixing of order R for $\mathcal{C}^1$ observables, while the normalised sums

[1] By [18, Theorem B.1], given $T : M \to M \in \mathcal{C}^r$, with $r \geq 1$, we have that T is μ-exponentially mixing of order R for $\mathcal{C}^\beta$ observables for some $\beta \in (0, r)$ if and only if T is μ-exponentially mixing of order R for $\mathcal{C}^\alpha$ observables for all $\alpha \in (0, r]$. The choice of $\beta = 1$ in this paper is just for simplicity.

$S_n f/\sqrt{n}$ have no Gaussian weak subsequential limit for some centred $f \in \mathcal{C}^\infty(M)$. Moreover, the statistical basin of μ contains a nonempty open set.

The same mixing and limit-law properties can also be realised by a fixed real-analytic linear Anosov diffeomorphism of $\mathbb{T}^2$. Both constructions fail exponential mixing at order $2R+2$. As a corollary, we obtain, to the best of our knowledge, the first examples in smooth dynamics of exponential mixing without exponential mixing of all orders, including examples with physical measures for $\mathcal{C}^\infty$ diffeomorphisms.

To place these results in context, we briefly recall some classical central limit theorems. Beyond the i.i.d. setting, the theorems of Lyapunov [53] and Lindeberg [48] give sufficient conditions for (non-identical) independent random variables to satisfy a CLT after normalisation by the standard deviation of the sum. For dependent observations, central limit theorems hold under suitable moment and dependence assumptions. Early results for Markov chains were obtained by Nagaev [55]. For stationary mixing processes, classical results include those of Rosenblatt [62], Ibragimov [32] and Doukhan, Massart and Rio [19].

In dynamics, early central limit theorems were obtained by Fortet [23] and Kac [35] for a special class of expanding endomorphisms of the circle. These results already highlighted the importance of regularity of the observable, only covering suitable classes of Hölder continuous functions. Further early results include Sinai's CLT for geodesic flows on manifolds of constant negative curvature [67] and Leonov's theorem for suitable classes of observables of ergodic endomorphisms of compact commutative groups [47].

Exponential decay of correlations and central limit theorems for sufficiently regular observables have been established in several important classes of dynamical systems. These include mixing Axiom A diffeomorphisms with Gibbs measures [7, 27] and classes of piecewise expanding interval maps [49, 25]. For nonuniformly hyperbolic systems, Young [72] developed a framework in which suitable assumptions on return times yield both exponential decay of correlations and the CLT. For geodesic flows on compact manifolds of negative curvature, central limit theorems were established by Ratner [60], while exponential decay of correlations was obtained by Dolgopyat [16] and Liverani [51]. For surveys of central limit theorems in dynamics and their proof methods, we refer to [15, 25] and the references therein.

1.1. **Decay of correlations and the central limit theorem.** For Markov chains, quantitative convergence to equilibrium provides classical criteria for the CLT. For example, uniform ergodicity ensures the CLT for every square-integrable observable, without any continuity assumption [55, 34]. More generally, for stationary processes, sufficiently fast decay of the α-mixing coefficients, together with suitable moment assumptions, implies the CLT [62, 32, 19]. In particular, exponential α-mixing suffices when a moment of order $2+\delta$ is finite for some $\delta > 0$.

The corresponding question in smooth dynamics is more delicate, since correlation estimates are usually available only for observables with prescribed regularity, such as $\mathcal{C}^1(M)$, and their bounds depend on the corresponding regularity norms. Such estimates do not directly give the uniform control over measurable events required by α-mixing criteria. Nevertheless, suitable correlation estimates yield the CLT in several dynamical settings. For ergodic noninvertible maps, a sufficient condition for the CLT is summable decay of correlations for regular observables against bounded measurable observables [24, 50]. For example, suppose that there exist $C > 0$ and $\rho(n) \geq 0$, with $\sum_{n\geq 1} \rho(n) < \infty$, such that

$$|\mu(f\, g \circ T^n) - \mu(f)\mu(g)| \leq C\rho(n)\|f\|_{\mathcal{C}^1(M)}\|g\|_{L^\infty(\mu)} \tag{4}$$

for every $f \in \mathcal{C}^1(M)$, $g \in L^\infty(\mu)$, and $n \geq 1$. Then every centred $f \in \mathcal{C}^1(M)$ satisfies the CLT through martingale approximation (see [50, Theorem 1.1]). The estimate (4) requires no regularity of the second observable and therefore gives stronger control than an estimate involving $\|f\|_{\mathcal{C}^1(M)}\|g\|_{\mathcal{C}^1(M)}$. The estimate (4) cannot hold for invertible maps unless μ is a point mass.

Other criteria yield the CLT by controlling correlations between functions of blocks of observations. Under such decorrelation assumptions, Le Borgne and Pène [45] and Pène [57] also establish Berry–Esseen-type rates of convergence in the CLT. The theorem of Björklund and Gorodnik [4] instead gives a sufficient condition in terms of exponential mixing of all orders for regular observables without effective rates of convergence.

Our results show that exponential decay of correlations for $\mathcal{C}^1$ observables, even when extended to exponential mixing of any prescribed finite order, does not suffice to guarantee the CLT. Indeed, for every $R \geq 2$, we construct systems that are exponentially mixing of order R for $\mathcal{C}^1$

observables, yet admit centred smooth observables for which the CLT fails. Thus, no fixed finite number of exponential mixing estimates can replace the all-orders hypothesis in the theorem of Björklund and Gorodnik.

1.2. **Rokhlin's problem and its quantitative counterparts.** Rokhlin's classical multiple mixing problem [61] asks whether mixing implies mixing of all orders. To state the question precisely, for ordered times $0 \le n_1 < \cdots < n_s$, write

$$\Delta(n_1, \dots, n_s) := \min_{1 \le i < s} (n_{i+1} - n_i).$$

Here, the order s counts the number of observables or sets involved.

Question 1.1 (Rokhlin's multiple mixing problem)**.** Let T be an invertible probability-preserving transformation of a standard probability space. If $\mu(A \cap T^{-n}B) \xrightarrow{n\to\infty} \mu(A)\mu(B)$ for all measurable A, B, must

$$\mu\left(\bigcap_{i=1}^{s} T^{-n_i} A_i\right) \longrightarrow \prod_{i=1}^{s} \mu(A_i) \quad \text{as } \Delta(n_1, \dots, n_s) \to \infty$$

hold for every $s \ge 3$ and every collection of measurable sets $A_1, \dots, A_s$?

Rokhlin's problem and its variants have motivated extensive work on multiple mixing (see [36, 31, 63, 66, 21, 39, 1, 38] and the references therein). For an overview and further references, see [64]. Rokhlin's problem was recently answered affirmatively in [56, Theorem 1.1 and Corollary 8.1][2], where it has been shown that every mixing probability-preserving transformation is mixing of all finite orders.

In smooth dynamics, exponential mixing has strong qualitative consequences. Dolgopyat, Kanigowski and Rodriguez–Hertz [18] prove that any $\mathcal{C}^{1+\alpha}$ diffeomorphism of a compact manifold preserving a smooth probability measure is Bernoulli whenever it is exponentially mixing. Maldonado [54, Theorem A] further extends this conclusion to invariant SRB measures. Since the Bernoulli property implies mixing of all orders, these results provide a partial answer to Question 1.1.

A finer quantitative version of Rokhlin's problem asks whether exponential mixing for $\mathcal{C}^1$ observables implies exponential mixing of all orders for $\mathcal{C}^1$ observables [18]. Notice that the Bernoulli conclusion alone does not provide these higher-order exponential mixing estimates.

A related question concerns the CLT. Dolgopyat, Dong, Kanigowski and Nándori [17, Appendix C] ask whether exponential mixing implies the classical CLT for smooth observables, in the setting of smooth systems preserving a smooth probability measure. By the theorem of Björklund and Gorodnik [4], a counterexample to this implication would provide a negative answer to the quantitative (exponential) version of Rokhlin's problem.

The following question summarises the discussion of the paragraphs above. Figure 1 highlights the implications discussed thus far and the contributions of this paper.

Question 1.2 (Exponential mixing and the CLT [18, 17])**.** Let T be a smooth diffeomorphism of a compact manifold preserving a smooth probability measure μ. Suppose that (T, μ) has exponential decay of correlations for smooth observables.

(1) Does (T, μ) have exponential mixing of all orders for smooth observables?
(2) Does every centred smooth observable satisfy the CLT with respect to μ?

Here, exponential mixing of order s means that (3) holds for this value of s, with constants $C_s, c_s > 0$ independent of the observables and the observation times.

While Question 1.2 concerns smooth invariant measures, our results show that both implications fail in the broader setting of invariant probability measures, including realisations with physical measures. The case of smooth invariant measures is not resolved by our constructions.

Similarly, we may extend Question 1.2 to the setting of equilibrium states for Hölder continuous potentials:

Question 1.3 (Exponential mixing and CLT for equilibrium states)**.** Let $T : M \to M$ be a smooth diffeomorphism of a compact smooth manifold, and let μ be an equilibrium state for a Hölder continuous potential $\varphi : M \to \mathbb{R}$. Suppose that (T, μ) has exponential decay of correlations for $\mathcal{C}^1(M)$ observables.

[2]The main theorem was also formally verified in Lean.

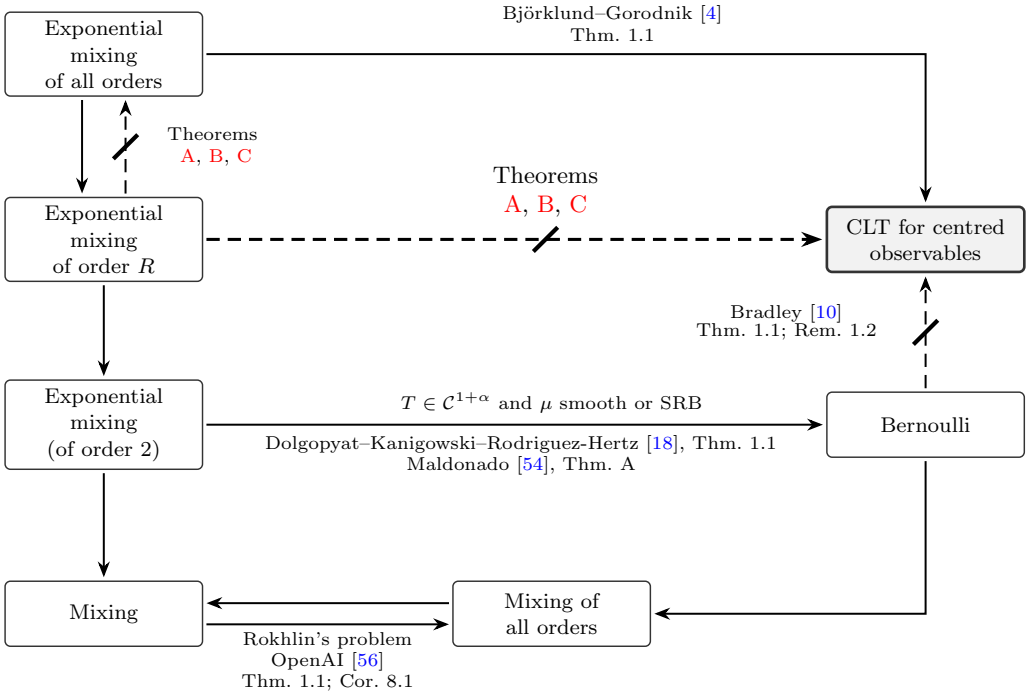


Figure 1. Implications (solid arrows) and non-implications (crossed dashed arrows) between mixing properties and the CLT.

(1) Is (T, μ) exponentially mixing of all orders for $\mathcal{C}^1(M)$ observables?
(2) Does every centred smooth observable satisfy the CLT with respect to μ?

For $\mathcal{C}^\infty$ diffeomorphisms of closed surfaces with positive topological entropy, both conclusions hold for mixing measures of maximal entropy ($\varphi \equiv 0$) [12], and more generally for mixing equilibrium states of high-temperature Hölder potentials, i.e. satisfying $\sup_M \varphi - \inf_M \varphi < h_{\mathrm{top}}(T)$ [12, 52].

1.3. **Stationary processes without a CLT.** The failure of the CLT for dependent stochastic processes has been studied for many decades. A recurring question is how much independence can be retained while the CLT still fails. Janson constructed stationary pairwise-independent counterexamples to the CLT [33]. Bradley obtained a β-mixing counterexample [8], which he later showed to be triplewise independent [9]. For each $R \in \mathbb{N}$, Pruss constructed R-tuplewise independent counterexamples in which the CLT fails [59], and Bradley and Pruss subsequently obtained this property for strictly stationary sequences [11].

Failure of the CLT can also coexist with stronger ergodic properties. For each $R \in \mathbb{N}$, Nguyen constructed a strictly stationary counterexample that is R-tuplewise independent and mixing in the ergodic-theoretic sense [71]. Bradley constructed a stationary $\{-1, 1\}$-valued process that is five-tuplewise independent and Bernoulli, but whose normalised sums have no Gaussian weak subsequential limit [10].

These examples show that arbitrary finite orders of independence can coexist with failure of the CLT, even for Bernoulli processes. Our main probabilistic contribution is to add quantitative (exponential) finite-order mixing to this picture. For every $R \geq 2$, Theorem C constructs a stationary symmetric $\{-1, 1\}$-valued process which is $(2R+1)$-tuplewise independent and exponentially mixing of order R, while the CLT fails. In fact, its $\sqrt{n}$-normalised partial sums admit no Gaussian weak limit along any subsequence. Moreover, exponential mixing fails at order $2R+2$. Thus, arbitrarily high finite orders of independence and of exponential mixing are compatible with failure of the CLT.

Theorems A and B provide smooth realisations of the stationary processes constructed in Theorem C. In particular, this implies that exponential mixing of any prescribed order R for $\mathcal{C}^1(M)$ observables does not imply the CLT. Hence the all-orders hypothesis in the theorem of Björklund and Gorodnik [4] is sharp with respect to the number of mixing orders.

For every integer $R \geq 2$, we construct a smooth diffeomorphism $T : M \to M$ preserving a probability measure μ such that there are constants $C_R, c_R > 0$ satisfying

$$\left| \mu\left(\prod_{i=1}^{R} f_i \circ T^{n_i} \right) - \prod_{i=1}^{R} \mu(f_i) \right| \leq C_R e^{-c_R \Delta(n_1,\dots,n_R)} \prod_{i=1}^{R} \|f_i\|_{\mathcal{C}^1(M)}$$

for all $f_1, \dots, f_R \in \mathcal{C}^1(M)$ and all integers $0 \leq n_1 < \cdots < n_R$, while there exists $f_0 \in \mathcal{C}^\infty(M)$ with $\mu(f_0) = 0$ such that

$$\frac{1}{\sqrt{n}} \sum_{i=0}^{n-1} f_0 \circ T^i \quad \text{has no Gaussian weak subsequential limit,}$$

including degenerate Gaussian laws (Dirac masses). The construction also gives constants $c, C > 0$ such that, for every sufficiently large integer N, there are times $0 \leq n_1 < \cdots < n_{2R+2} < N$, satisfying

$$\Delta(n_1, \dots, n_{2R+2}) \geq \frac{N}{C} \quad \text{and} \quad \left| \mu\left(\prod_{i=1}^{2R+2} f_0 \circ T^{n_i} \right) \right| \geq cN^{-R-1}.$$

This polynomial lower bound rules out exponential mixing of order $2R+2$ for $\mathcal{C}^1(M)$ observables. Thus (T, μ) is exponentially mixing of order R, but not exponentially mixing of all orders.

We realise the dynamical system (T, μ) described above in two ways. In Theorem A, T is a fixed linear Anosov diffeomorphism of $\mathbb{T}^2$ and μ is supported on a fixed two-symbol horseshoe Λ. The same pair (T, Λ) applies to every order of mixing R chosen, whereas the invariant measure μ depends on R. In Theorem B, for every order R, T is a $\mathcal{C}^\infty$ diffeomorphism of any prescribed nonempty compact smooth surface without boundary, and μ is a physical measure.

Both realisations come from the shift (σ, ν) on $\{-1, 1\}^{\mathbb{Z}}$, with the invariant measure ν constructed in Theorem C. Under ν, the shift is exponentially mixing of order R, but not of order $2R + 2$, and the $\sqrt{n}$-normalised Birkhoff sums of the coordinate observable $f_0(\omega) := \omega_0$ have no Gaussian weak subsequential limit. These properties pass to (T, μ) through a regular conjugacy π between the shift and the restriction of T to the horseshoe, with $\mu = \pi_* \nu$.

In the linear Anosov realisation of Theorem A, we take T to be a fixed power of the classical cat map and use standard hyperbolic coding [68, 6, 40]. The map, the horseshoe and the smooth observable are independent of the order R. For each invariant measure μ, the failure of the CLT persists under small $\mathcal{C}^1$ perturbations of the observable, so the observable can also be chosen to be a real trigonometric polynomial.

The physical realisation in Theorem B is based on methods developed by Colli and Vargas [13] and Kiriki, Li, Nakano, Soma and Vargas [43] (see also [41, 42]). In Appendix A, we adapt these methods to a symmetric affine horseshoe and prove Proposition 3.11, which realises every ergodic invariant measure of the full two-symbol shift as a physical measure of a $\mathcal{C}^\infty$ diffeomorphism of any prescribed compact smooth surface M without boundary.

1.4. **Organisation of the paper.** The paper is divided into two parts. Part I concerns the dynamical realisations, i.e. Theorems A and B and one appendix. Section 2 states the main results. Section 3 proves both theorems assuming Theorem C. The analytic (Theorem A) and physical (Theorem B) realisations then follow from hyperbolic coding and the smooth physical realisation of symbolic measures, respectively (Subsections 3.3 and 3.4). The latter construction is proved in Appendix A.

Part II concerns the proof of Theorem C. In Section 4, Proposition 4.2 constructs random stationary nested interval partitions of $\mathbb{Z}$. Using these partitions, we construct a particular probability measure ν on $\{-1, 1\}^{\mathbb{Z}}$ in Sections 5 and 6, under which the coordinate process $Z = (Z_i)_{i \in \mathbb{Z}}$, defined by $Z_i(\omega) := \omega_i$, is stationary (Proposition 6.3). In Section 7, Propositions 7.1 and 7.2 establish independence of sufficiently separated coordinates of Z and exponential mixing of order R for (σ, ν). Finally, Section 8 proves failure of the CLT for Z by showing that its normalised partial sums $n^{-1/2} \sum_{i=0}^{n-1} Z_i$ have no Gaussian weak subsequential limit (Corollary 8.4). Section 8 also proves failure of exponential mixing at order $2R+2$ (Corollary 8.6). Both conclusions follow from the estimate in Proposition 8.1. Figure 2 summarises the main proof dependencies.

Appendix A proves Proposition 3.11, which realises every ergodic invariant probability measure of the full two-symbol shift as a physical measure of a $\mathcal{C}^\infty$ diffeomorphism of any prescribed

nonempty compact smooth surface without boundary. This proposition is used in the proof of Theorem B.

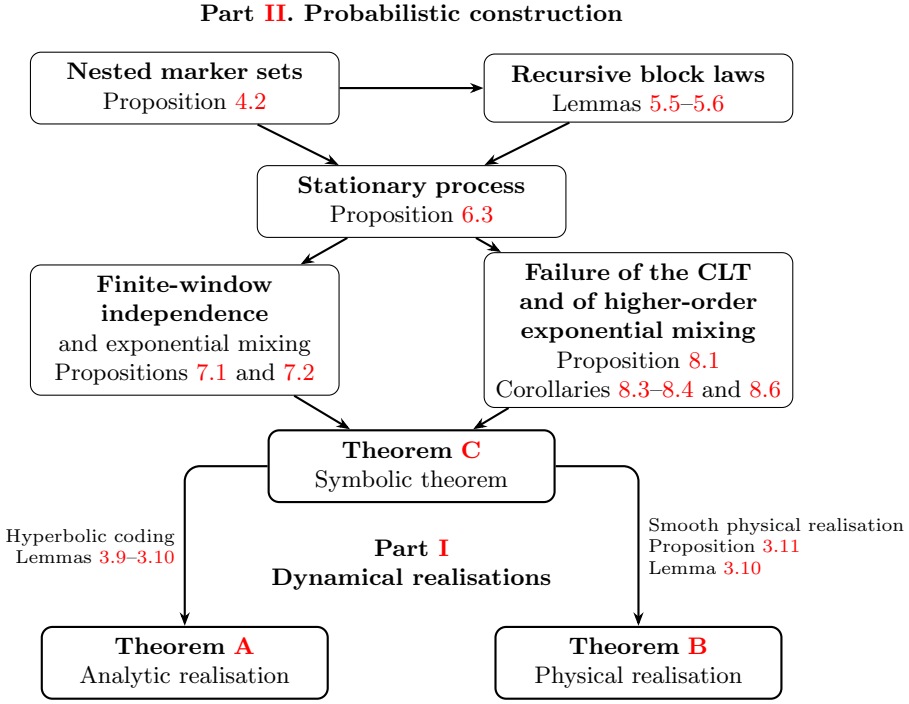


Figure 2. Main proof dependencies. Part II proves Theorem C, which Part I assumes to obtain Theorems A and B. The dynamical realisations are presented before the probabilistic construction.

Part I. Dynamical realisations

In this part we prove Theorems A and B using Theorem C, whose proof is given in Part II. We first establish stability of CLT failure under perturbations of the observable, then construct the fixed linear Anosov realisation and the physical realisation, the latter using Proposition 3.11. The proofs of Theorems A, B and C have been formalised in Lean 4 (see Section 9).

2. Main results

We use the notation $S_n f$ and $\Delta(n_1, \dots, n_s)$ as in Section 1. On a compact Riemannian manifold M, set $\|f\|_{\mathcal{C}^1(M)} := \|f\|_\infty + \|Df\|_\infty$. We recall two notions of convergence which are used in the statements and proofs below.

Definition 2.1. For probability measures λ_n and λ on a metric space E, we write $\lambda_n \xRightarrow{n\to\infty} \lambda$ and say that λ_n *converges weakly* to λ if

$$\int_E u \, \mathrm{d}\lambda_n \longrightarrow \int_E u \, \mathrm{d}\lambda \quad \text{for every } u \in \mathcal{C}_b(E),$$

where $\mathcal{C}_b(E)$ is the space of bounded continuous real-valued functions on E.

For real-valued random variables Y_n on (M, μ), we write $Y_n \xRightarrow{n\to\infty} \lambda$ and say that Y_n *converges in distribution* to a probability measure λ on $\mathbb{R}$ if $\mathrm{Law}_\mu(Y_n) \xRightarrow{n\to\infty} \lambda$. Here $\mathrm{Law}_\mu(Y_n)(A) := \mu\{x \in M : Y_n(x) \in A\}$ for every Borel set $A \subset \mathbb{R}$. Thus convergence in distribution of random variables means weak convergence of their laws [2, Section 3, p. 25].

Definition 2.2. Let $X = (X_i)_{i\in\mathbb{Z}}$ be a real-valued process on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. For an integer $p \geq 1$, the process is *p-tuplewise independent* if every collection of at most p coordinates with distinct indices is jointly independent.

All assertions that an observable is $\{-1,1\}$-valued are understood almost surely with respect to the stated measure. Gaussian laws include point masses. We use $\not\Rightarrow$ to denote failure of convergence in distribution.

2.1. **Linear Anosov realisation of finite-order exponential mixing without the CLT.** The first result realises finite-order exponential mixing without the CLT on a single real-analytic Anosov diffeomorphism. Both the map and the smooth observable are fixed. Varying the invariant measure produces examples with arbitrarily high prescribed orders of exponential mixing.

Theorem A (Linear Anosov realisation)**.** *Let $A:\mathbb{T}^2\to\mathbb{T}^2$ be the cat map*

$$A(x,y)=(2x+y,x+y)\pmod 1.$$

There exist an integer $k_0\geq 2$, a two-symbol horseshoe Λ for $T=A^{k_0}$, and a real-valued observable $f_0\in\mathcal{C}^\infty(\mathbb{T}^2)$ such that, for every integer $R\geq 2$, there is a T-invariant probability measure μ with the following properties.

(i) $\operatorname{supp}\mu=\Lambda$.

(ii) *There are constants $C_R,c_R>0$ such that*

$$\left|\mu\left(\prod_{i=1}^{R} f_i\circ T^{n_i}\right)-\prod_{i=1}^{R}\mu(f_i)\right|\leq C_R e^{-c_R\Delta(n_1,\dots,n_R)}\prod_{i=1}^{R}\|f_i\|_{\mathcal{C}^1(\mathbb{T}^2)}$$

for all $f_1,\dots,f_R\in\mathcal{C}^1(\mathbb{T}^2)$ and all integers $0\leq n_1<\cdots<n_R$.

(iii) *Under μ, the stationary process $(f_0\circ T^i)_{i\in\mathbb{Z}}$ is $\{-1,1\}$-valued, satisfies $\mu(f_0=-1)=\mu(f_0=1)=1/2$, and is $(2R+1)$-tuplewise independent. In particular $\mu(f_0)=0$.*

(iv) *The laws of $S_nf_0/\sqrt{n}$ under μ have no Gaussian weak subsequential limit; i.e. for every strictly increasing sequence of positive integers $n_k\to\infty$ and every $\sigma^2\geq 0$,*

$$\frac{1}{\sqrt{n_k}}\sum_{i=0}^{n_k-1} f_0\circ T^i \not\Rightarrow \mathcal{N}(0,\sigma^2)\quad \text{as } k\to\infty,$$

where convergence in distribution is taken with respect to μ.

(v) *There is an open neighbourhood $\mathcal{O}$ of f_0 in $\mathcal{C}^1(\mathbb{T}^2)$ such that, for every $g\in\mathcal{O}$, the laws of $S_n(g-\mu(g))/\sqrt{n}$ under μ have no Gaussian weak subsequential limit. In particular, such a g can be chosen to be a real trigonometric polynomial.*

(vi) *There are constants $c,C>0$ and $N_0\geq 1$ such that, for every integer $N\geq N_0$, there are integers $0\leq n_1<\cdots<n_{2R+2}<N$ satisfying*

$$\Delta(n_1,\dots,n_{2R+2})\geq\frac{N}{C}\quad\text{and}\quad\left|\mu\left(\prod_{i=1}^{2R+2} f_0\circ T^{n_i}\right)\right|\geq cN^{-R-1}. \tag{5}$$

Hence, (T,μ) is not exponentially mixing of order $2R+2$ in $\mathcal{C}^1(\mathbb{T}^2)$.

The proof of Theorem A is given in Subsection 3.3, which transports the process of Theorem C to a horseshoe using the Birkhoff–Smale theorem and Hölder symbolic coding [40, Sections 6.5 and 18.7]. The coding transfers the correlation estimates, while a smooth function separating the two symbol cylinders realises the coordinate process exactly. Proposition 3.5 then gives stability under perturbations of the observable and allows approximation by a real trigonometric polynomial.

2.2. **Physical realisation of finite-order exponential mixing without the CLT.** Fix a smooth volume m on M. The *statistical basin* of an invariant probability measure μ is

$$B_T(\mu):=\left\{x\in M:\frac{1}{n}\sum_{i=0}^{n-1} f\circ T^i(x)\xrightarrow{n\to\infty}\int f\,\mathrm{d}\mu,\ \text{for every } f\in\mathcal{C}^0(M)\right\}.$$

The measure is *physical* if $m(B_T(\mu))>0$. This definition is independent of the choice of smooth volume, since any two smooth positive densities are mutually absolutely continuous.

The next theorem shows that finite-order exponential mixing without the CLT can also occur for a physical measure. The construction applies to every nonempty compact smooth surface without boundary and yields a $\mathcal{C}^\infty$ diffeomorphism. Moreover, the statistical basin contains a nonempty open set on which the time averages of every continuous observable converge uniformly

to its integral against μ. Thus μ describes the long-term averages of an open set of initial conditions, while the normalised sums of a smooth observable have no Gaussian weak subsequential limit with respect to μ.

Theorem B (Physical realisation)**.** *Let M be a nonempty compact smooth surface without boundary. For every integer $R \geq 2$, there exist a $\mathcal{C}^\infty$ diffeomorphism $T : M \to M$, a two-symbol horseshoe Λ, a T-invariant probability measure μ, and a real-valued observable $f_0 \in \mathcal{C}^\infty(M)$ with the following properties.*

(i) $\operatorname{supp} \mu = \Lambda$.

(ii) *There are constants $C_R, c_R > 0$ such that*

$$\left| \mu\left(\prod_{i=1}^{R} f_i \circ T^{n_i} \right) - \prod_{i=1}^{R} \mu(f_i) \right| \leq C_R e^{-c_R \Delta(n_1,\dots,n_R)} \prod_{i=1}^{R} \|f_i\|_{\mathcal{C}^1(M)}$$

for all $f_1, \dots, f_R \in \mathcal{C}^1(M)$ and all integers $0 \leq n_1 < \cdots < n_R$.

(iii) *Under μ, the stationary process $(f_0 \circ T^i)_{i\in\mathbb{Z}}$ is $\{-1,1\}$-valued, satisfies $\mu(f_0 = -1) = \mu(f_0 = 1) = 1/2$, and is $(2R+1)$-tuplewise independent.*

Moreover, the laws of $S_n f_0/\sqrt{n}$ under μ have no Gaussian weak subsequential limit; i.e. for every strictly increasing sequence of positive integers $n_k \to \infty$ and every $\sigma^2 \geq 0$,

$$\frac{1}{\sqrt{n_k}} \sum_{i=0}^{n_k-1} f_0 \circ T^i \not\Rightarrow \mathcal{N}(0,\sigma^2) \quad \text{as } k \to \infty,$$

where convergence in distribution is taken with respect to μ.

(iv) *There is an open neighbourhood $\mathcal{O}$ of f_0 in $\mathcal{C}^1(M)$ such that, for every $g \in \mathcal{O}$, the laws of $S_n(g - \mu(g))/\sqrt{n}$ under μ have no Gaussian weak subsequential limit.*

(v) *There is a nonempty open set $W \subset M$ such that, for every $g \in \mathcal{C}^0(M)$,*

$$\lim_{n\to\infty} \sup_{y\in W} \left| \frac{1}{n} S_n g(y) - \mu(g) \right| = 0. \tag{6}$$

Thus $W \subset B_T(\mu)$, and hence μ is physical.

(vi) *The observable f_0 satisfies (5), with constants depending on this realisation. Hence, (T,μ) is not exponentially mixing of order $2R+2$ in $\mathcal{C}^1(M)$.*

The proof in Subsection 3.4 combines Theorem C with Proposition 3.11. The proposition gives the uniform convergence of time averages in (6) and shows that the invariant measure is physical. Lemma 3.10 transfers the statistical properties through the horseshoe conjugacy.

The preceding theorems show that exponential mixing of any prescribed order R can coexist with failure of exponential mixing at a higher order. In particular, exponential decay of correlations does not, in general, imply exponential mixing of all orders.

Corollary 2.3. *There exists a real-analytic Anosov diffeomorphism $T : \mathbb{T}^2 \to \mathbb{T}^2$ such that, for every integer $R \geq 2$, there is a T-invariant probability measure μ_R for which (T,μ_R) is exponentially mixing of order R, but not of order $2R+2$, for $\mathcal{C}^1(\mathbb{T}^2)$ observables.*

For every nonempty compact smooth surface M without boundary and every integer $R \geq 2$, there also exists a $\mathcal{C}^\infty$ diffeomorphism of M preserving a physical measure with the same mixing properties for $\mathcal{C}^1(M)$ observables.

Proof. Apply items (ii) and (vi) of Theorem A, and items (ii) and (vi) of Theorem B. The lower bound in (5) is incompatible with an exponential upper bound at times whose minimum gap is at least N/C. □

3. Dynamical realisations under Theorem C

The proofs in this section use a stationary process with the properties in Theorem C. We state that theorem first, then establish stability under changes of the observable, the fixed analytic realisation and the physical realisation. The existence of the process is proved afterwards.

3.1. **Symbolic realisation of finite-order exponential mixing without the CLT.** We state the probabilistic result first and postpone its proof to Sections 4–8. Throughout the present section, we assume this theorem and use only its stated properties.

Notation 3.1. We use the following notation for the symbolic system and its coordinate process.

- Let $\Sigma := \{-1,1\}^{\mathbb{Z}}$ and let $\sigma : \Sigma \to \Sigma$ be the left shift, $(\sigma\omega)_i := \omega_{i+1}$. For $i \in \mathbb{Z}$, let $Z_i(\omega) := \omega_i$, and write $\varphi := Z_0$.
- For a finite integer interval $I = [u,v] \cap \mathbb{Z}$, write $Z[I] := (Z_u, \dots, Z_v)$. For finite $I, J \subset \mathbb{Z}$ we use $\operatorname{dist}(I,J) := \min\{|i-j| : i \in I,\ j \in J\}$.
- For $0 < \theta < 1$, equip Σ with the metric $d_\theta(\omega,\omega') := \theta^{\min\{|i|:\omega_i \neq \omega'_i\}}$, and let $\mathcal{F}_\theta(\Sigma)$ denote the space of functions that are Lipschitz for d_θ, with $\|F\|_\theta := \|F\|_\infty + \operatorname{Lip}_\theta(F)$. Since $d_\theta^\eta = d_{\theta^\eta}$, varying θ already covers every Hölder exponent, and we use no separate exponent.
- For $0 \le n_1 < \cdots < n_s$ we abbreviate $\mathbf{n} := (n_1, \dots, n_s)$ and set
$$\Delta(\mathbf{n}) := \min_{1 \le i < s} (n_{i+1} - n_i).$$
- Write $S_n := \sum_{i=0}^{n-1} Z_i$.

Theorem C constructs a stationary $\{-1,1\}$-valued process with exponential mixing of any prescribed order R, but whose normalised sums have no Gaussian weak subsequential limit. It is the main result from which Theorems A and B are derived.

Theorem C. *For every integer $R \ge 2$, there exists a full-support σ-invariant Borel probability measure ν on Σ with the following properties.*

(i) *Under ν, the coordinate process $(Z_i)_{i\in\mathbb{Z}}$ is stationary, is $\{-1,1\}$-valued, satisfies $\nu(Z_0 = -1) = \nu(Z_0 = 1) = 1/2$, and is $(2R+1)$-tuplewise independent. Thus any collection of at most $2R+1$ coordinates with distinct indices is jointly independent.*

(ii) *There exists $K_R < \infty$ such that, for all disjoint nonempty finite intervals $I_1, \dots, I_R$ ordered from left to right,*
$$\min_{i<R} \operatorname{dist}(I_i, I_{i+1}) > K_R \max_i |I_i| \quad \text{then} \quad Z[I_1], \dots, Z[I_R] \text{ are jointly independent.} \tag{7}$$

(iii) *For every $0 < \theta < 1$, there exist constants $C_{R,\theta}, c_\theta > 0$ such that*
$$\left| \nu\left(\prod_{i=1}^R F_i \circ \sigma^{n_i} \right) - \prod_{i=1}^R \nu(F_i) \right| \le C_{R,\theta} e^{-c_\theta \Delta(\mathbf{n})} \prod_{i=1}^R \|F_i\|_\theta$$
for all $F_1, \dots, F_R \in \mathcal{F}_\theta(\Sigma)$ and all integers $0 \le n_1 < \cdots < n_R$.

(iv) *There exist $\delta > 0$ and $n_0 \ge 1$ such that*
$$\mathbb{E}_\nu \left(\frac{S_n}{\sqrt{n}} \right)^{2R+2} \le (2R+1)!! - \delta \quad \text{for every } n \ge n_0.$$
In particular, the laws of $S_n/\sqrt{n}$ under ν have no Gaussian weak subsequential limit. Explicitly, for every strictly increasing sequence of positive integers $n_k \to \infty$ and every $\sigma^2 \ge 0$,
$$\frac{S_{n_k}}{\sqrt{n_k}} \not\Rightarrow \mathcal{N}(0,\sigma^2) \quad \text{as } k \to \infty,$$
where convergence in distribution is taken with respect to ν.

(v) *There exist constants $c, C > 0$ and $N_0 \ge 1$ such that, for every integer $N \ge N_0$, there are integers $0 \le n_1 < \cdots < n_{2R+2} < N$ satisfying*
$$\Delta(\mathbf{n}) \ge \frac{N}{C} \quad \text{and} \quad -\mathbb{E}_\nu \left[\prod_{i=1}^{2R+2} Z_{n_i} \right] \ge c N^{-R-1}.$$
Hence, (Σ, σ, ν) is not exponentially mixing of order $2R+2$ in $\mathcal{F}_\theta(\Sigma)$ for any $0 < \theta < 1$.

The proof of Theorem C is carried out in Part II and completed at the end of Section 8. We now assume Theorem C and prove Theorems A and B. We first record a consequence of the finite-interval independence assertion that will be used for the physical realisation.

Corollary 3.2. *Every stationary probability measure on Σ satisfying (7) is mixing, and hence ergodic.*

Proof. Let A, B be cylinder events. They depend on coordinates in two fixed finite intervals. For every sufficiently large n, those intervals, after translating the second by n, satisfy the separation condition in (7). Therefore $\nu(A \cap \sigma^{-n}B) = \nu(A)\nu(B)$ for all sufficiently large n. Since the cylinder sets form an algebra generating the Borel σ-algebra of Σ, the standard approximation argument for mixing completes the proof (see [70, Section 1.7]). □

3.2. **Stability of CLT failure under perturbations of the observable.** We prove that the absence of Gaussian weak subsequential limits persists under small $\mathcal{C}^1$ perturbations of the observable. The argument requires only summable two-point correlations and will be used in both smooth realisations, including analytic approximation on the torus. We use the following distance to compare the laws of the normalised sums and measure their separation from Gaussian laws.

Notation 3.3 (Bounded-Lipschitz distance)**.** For probability measures $\mathbb{P}, \mathbb{Q}$ on $\mathbb{R}$, set

$$d_{\mathrm{BL}}(\mathbb{P}, \mathbb{Q}) := \sup_{\|u\|_\infty \le 1,\ \mathrm{Lip}(u) \le 1} |\mathbb{P}(u) - \mathbb{Q}(u)|.$$

Let $\mathcal{G}$ denote the set of all Gaussian probability measures on $\mathbb{R}$, including point masses, and write

$$\mathrm{dist}_{\mathrm{BL}}(\mathbb{P}, \mathcal{G}) := \inf_{\mathbb{Q} \in \mathcal{G}} d_{\mathrm{BL}}(\mathbb{P}, \mathbb{Q}).$$

The following lemma gives a uniform second-moment bound for the normalised sums. We use this bound to obtain tightness and to control how their laws change when the observable is perturbed.

Lemma 3.4. *Suppose T preserves a probability measure μ on a compact manifold and*

$$\left| \int f g \circ T^n \, \mathrm{d}\mu - \int f \, \mathrm{d}\mu \int g \, \mathrm{d}\mu \right| \le C\gamma(n) \|f\|_{\mathcal{C}^1} \|g\|_{\mathcal{C}^1} \quad \text{and} \quad \sum_{n \ge 1} \gamma(n) < \infty, \tag{8}$$

where $\gamma(n) \ge 0$. Then every centred $h \in \mathcal{C}^1(M)$ satisfies

$$\sup_{n \ge 1} \frac{\|S_n h\|_{L^2(\mu)}}{\sqrt{n}} \le C_0 \|h\|_{\mathcal{C}^1} \quad \text{with} \quad C_0^2 := 1 + 2C \sum_{i \ge 1} \gamma(i). \tag{9}$$

Proof. For every $n \ge 1$, invariance of μ and (8) give

$$\begin{aligned} \frac{1}{n} \|S_n h\|_{L^2(\mu)}^2 &= \mu(h^2) + 2 \sum_{i=1}^{n-1} \left(1 - \frac{i}{n}\right) \mu(h \, h \circ T^i) \le \|h\|_\infty^2 + 2C \sum_{i \ge 1} \gamma(i) \|h\|_{\mathcal{C}^1(M)}^2 \\ &\le C_0^2 \|h\|_{\mathcal{C}^1(M)}^2. \end{aligned}$$

Taking square roots and the supremum over $n \ge 1$ proves (9). □

Proposition 3.5. *Assume (8). Let $f \in \mathcal{C}^1(M)$ be centred and suppose that $S_n f/\sqrt{n}$ has no Gaussian weak subsequential limit. Then there are $\delta > 0$, $c_0 > 0$ and $n_0 \ge 1$ such that*

$$\mathrm{dist}_{\mathrm{BL}} \left(\mathrm{Law}_\mu \left(\frac{S_n(g - \mu(g))}{\sqrt{n}} \right), \mathcal{G} \right) \ge c_0 \tag{10}$$

for every $n \ge n_0$ and every $g \in \mathcal{C}^1(M)$ with $\|g - f\|_{\mathcal{C}^1} < \delta$.

Proof. Write $\mathbb{P}_n := \mathrm{Law}_\mu(S_n f/\sqrt{n})$. Lemma 3.4 gives

$$\sup_{n \ge 1} \int x^2 \, \mathbb{P}_n(dx) \le C_0^2 \|f\|_{\mathcal{C}^1(M)}^2,$$

so $(\mathbb{P}_n)$ is tight. We first show that

$$\liminf_{n \to \infty} \mathrm{dist}_{\mathrm{BL}}(\mathbb{P}_n, \mathcal{G}) > 0. \tag{11}$$

Otherwise, there would be $n_j \to \infty$ and $\mathbb{G}_j \in \mathcal{G}$ with $d_{\mathrm{BL}}(\mathbb{P}_{n_j}, \mathbb{G}_j) \to 0$. By Prohorov's theorem [2, Theorem 5.1], pass to a subsequence for which $\mathbb{P}_{n_j}$ converges in the weak topology to a probability measure $\mathbb{P}$. Since d_{BL} metrises this topology [20, Theorem 11.3.3],

$$d_{\mathrm{BL}}(\mathbb{G}_j, \mathbb{P}) \le d_{\mathrm{BL}}(\mathbb{G}_j, \mathbb{P}_{n_j}) + d_{\mathrm{BL}}(\mathbb{P}_{n_j}, \mathbb{P}) \longrightarrow 0.$$

The Gaussian laws, including point masses, are closed in the space of probability measures on $\mathbb{R}$ with this weak topology (see [58, Lemma B.1, item 1] or [5, Exercise 2.7.49]). Thus $\mathbb{P}$ would be a Gaussian weak subsequential limit of $(\mathbb{P}_n)$, contrary to the hypothesis. This proves (11). Choose $c_0 > 0$ and n_0 so that the distance is at least $2c_0$ for every $n \ge n_0$.

Now let $g \in \mathcal{C}^1(M)$ and put $h := g - \mu(g) - f$. Because $\mu(f) = 0$,

$$\mu(h) = 0, \quad |\mu(g)| = |\mu(g - f)| \le \|g - f\|_\infty \quad \text{and} \quad \|h\|_{\mathcal{C}^1(M)} \le 2\|g - f\|_{\mathcal{C}^1(M)}.$$

Write $\mathbb{Q}_n := \mathrm{Law}_\mu(S_n(g - \mu(g))/\sqrt{n})$. For a test function u with $\|u\|_\infty \le 1$ and $\mathrm{Lip}(u) \le 1$, evaluate both sums on the same probability space. Their difference is $S_n h/\sqrt{n}$, so

$$|\mathbb{P}_n(u) - \mathbb{Q}_n(u)| \le \frac{\|S_n h\|_{L^1(\mu)}}{\sqrt{n}} \le \frac{\|S_n h\|_{L^2(\mu)}}{\sqrt{n}} \le C_0 \|h\|_{\mathcal{C}^1(M)} \le 2C_0 \|g - f\|_{\mathcal{C}^1(M)}.$$

Taking the supremum over u gives

$$d_{\mathrm{BL}}(\mathbb{P}_n, \mathbb{Q}_n) \le 2C_0 \|g - f\|_{\mathcal{C}^1(M)}.$$

Choose $\delta < c_0/(2C_0)$. For $n \ge n_0$ and $\|g - f\|_{\mathcal{C}^1(M)} < \delta$,

$$\mathrm{dist}_{\mathrm{BL}}(\mathbb{Q}_n, \mathcal{G}) \ge \mathrm{dist}_{\mathrm{BL}}(\mathbb{P}_n, \mathcal{G}) - d_{\mathrm{BL}}(\mathbb{P}_n, \mathbb{Q}_n) > 2c_0 - 2C_0\delta > c_0.$$

This proves (10). □

Corollary 3.6. *In the setting of Proposition 3.5 on $\mathbb{T}^d$, a real trigonometric polynomial can be chosen whose centred normalised sums have no Gaussian weak subsequential limit.*

Proof. Real trigonometric polynomials are dense in $\mathcal{C}^1(\mathbb{T}^d)$ in the $\mathcal{C}^1$ norm (see [26, Corollary 3.1.11 and its proof]). Choose such a polynomial in the neighbourhood from Proposition 3.5. It is real analytic, and subtracting its mean preserves that class. □

The neighbourhood in Proposition 3.5 is for a fixed map and a fixed measure. It need not be uniform in the requested mixing order.

3.3. **Realisation on a fixed analytic diffeomorphism.**

3.3.1. *Hyperbolic coding.* For Theorem A, the geometric construction is a two-symbol horseshoe in a fixed iterate of the cat map. We use the following classical results of the theory of hyperbolic dynamical systems.

Theorem 3.7 (Homoclinic horseshoe theorem [40])**.** *Let a $\mathcal{C}^1$ diffeomorphism have a hyperbolic periodic point with a transverse homoclinic intersection. A positive iterate has a compact invariant hyperbolic Cantor set on which the iterate is conjugate to the full two-sided shift on two symbols.*

Theorem 3.8 (Regularity of hyperbolic coding [40, Chapter 18])**.** *For a finite Markov coding π of a uniformly hyperbolic invariant set, there are $C > 0$ and $0 < \varrho < 1$ such that the image of each cylinder specified on $[-\ell, \ell]$ has diameter at most $C\varrho^\ell$. In the full-shift horseshoe case, the coding may be taken to be a homeomorphism.*

The following lemma is standard. Smale's horseshoe construction [68, Section I.5, Proposition 5.3 and Theorem 5.5], applied to a transverse homoclinic point of the cat map, gives a two-symbol horseshoe after passing to a suitable iterate $T = A^{k_0}$. The estimate (12) follows from Theorem 3.8, and the zero-area conclusion follows from ergodicity of T with respect to Haar area. These choices are independent of R.

Lemma 3.9. *There are $k_0 \ge 2$, a compact two-symbol horseshoe Λ for $T = A^{k_0}$, and a coding homeomorphism $\pi : \Sigma \to \Lambda$ such that $T \circ \pi = \pi \circ \sigma$. Moreover, for some $0 < \theta < 1$ and $C_1 < \infty$,*

$$\|f \circ \pi\|_\theta \le C_1 \|f\|_{\mathcal{C}^1(\mathbb{T}^2)} \quad \text{for every } f \in \mathcal{C}^1(\mathbb{T}^2). \tag{12}$$

Finally, Λ has zero Haar area.

3.3.2. *Transporting the measure and the coordinate observable.* The following lemma transfers the symbolic correlation bounds to the horseshoe and realises the coordinate process by a smooth observable. We use the same argument in both realisations to preserve the laws of the sums and the higher-order correlations.

Lemma 3.10. *Let T be a $\mathcal{C}^1$ diffeomorphism of a compact smooth Riemannian manifold M, and let $\pi : \Sigma \to \Lambda$ be a homeomorphism onto a compact invariant set, with $T \circ \pi = \pi \circ \sigma$. Suppose that, for some $0 < \theta < 1$ and $C_1 < \infty$, $\|f \circ \pi\|_\theta \le C_1 \|f\|_{\mathcal{C}^1(M)}$ for $f \in \mathcal{C}^1(M)$. For $R \ge 2$, let ν be as in Theorem C and set $\mu := \pi_* \nu$. Then the following hold.*

(i) *The measure μ is T-invariant and has support Λ. There are constants $C_R, c_R > 0$ such that*

$$\left| \mu\left(\prod_{i=1}^{R} f_i \circ T^{n_i} \right) - \prod_{i=1}^{R} \mu(f_i) \right| \le C_R e^{-c_R \Delta(\mathbf{n})} \prod_{i=1}^{R} \|f_i\|_{\mathcal{C}^1(M)} \tag{13}$$

for all $f_1, \dots, f_R \in \mathcal{C}^1(M)$ and $0 \le n_1 < \cdots < n_R$, where $C_R = C_{R,\theta} C_1^R$ and $c_R = c_\theta$.

(ii) *There is $f_0 \in \mathcal{C}^\infty(M)$, depending only on π, such that*

$$(f_0 \circ T^i) \circ \pi = Z_i \quad (i \in \mathbb{Z}) \quad \text{and} \quad S_n f_0 \circ \pi = S_n. \tag{14}$$

Thus the process $(f_0 \circ T^i)_{i \in \mathbb{Z}}$ under μ has the same law as the coordinate process under ν.

Proof. For every bounded Borel function f, the conjugacy and σ-invariance of ν give

$$\int f \circ T \, \mathrm{d}\mu = \int (f \circ \pi) \circ \sigma \, \mathrm{d}\nu = \int f \circ \pi \, \mathrm{d}\nu = \int f \, \mathrm{d}\mu.$$

Hence μ is T-invariant. Also $\mu(\Lambda) = 1$. Every open neighbourhood of a point of Λ has a nonempty open preimage under the homeomorphism π. Since ν has full support, that neighbourhood has positive μ-measure, proving $\operatorname{supp} \mu = \Lambda$.

Fix $f_1, \dots, f_R \in \mathcal{C}^1(M)$ and put $F_i = f_i \circ \pi$. Applying Theorem C(iii) yields

$$\left| \mu\left(\prod_{i=1}^{R} f_i \circ T^{n_i} \right) - \prod_{i=1}^{R} \mu(f_i) \right| = \left| \nu\left(\prod_{i=1}^{R} F_i \circ \sigma^{n_i} \right) - \prod_{i=1}^{R} \nu(F_i) \right| \le C_{R,\theta} e^{-c_\theta \Delta(\mathbf{n})} \prod_{i=1}^{R} \|F_i\|_\theta$$

$$\le C_{R,\theta} C_1^R e^{-c_\theta \Delta(\mathbf{n})} \prod_{i=1}^{R} \|f_i\|_{\mathcal{C}^1(M)}.$$

This gives (13) with the stated constants.

For the observable, the sets $\Lambda_\pm := \pi\{\omega : \omega_0 = \pm 1\}$ are disjoint and compact. Choose smooth cutoffs $\chi_\pm$, equal to one near $\Lambda_\pm$ and supported in disjoint neighbourhoods, and set $f_0 := \chi_+ - \chi_-$. Then $f_0 \circ \pi = Z_0$. For every $i \in \mathbb{Z}$,

$$(f_0 \circ T^i) \circ \pi = f_0 \circ \pi \circ \sigma^i = Z_0 \circ \sigma^i = Z_i.$$

Summing over $i = 0, \dots, n-1$ proves (14). Because $\mu = \pi_* \nu$, these identities identify every finite-dimensional distribution of the two processes. The cutoffs depend only on π, so the same f_0 works for all R and ν. □

Proof of Theorem A. Fix T, Λ, π from Lemma 3.9 and the observable f_0 from Lemma 3.10. For a given $R \ge 2$, take ν from Theorem C and set $\mu := \pi_* \nu$. Lemma 3.10(i) gives the support and correlation conclusions (i)–(ii).

By (14), the process of f_0 under μ has the same law as the symbolic coordinates under ν. Thus Theorem C(i), (iv) gives (iii)–(iv), including $\mu(f_0) = 0$. The same identity gives, for any observation times,

$$\mu\left(\prod_{i=1}^{2R+2} f_0 \circ T^{n_i} \right) = \nu\left(\prod_{i=1}^{2R+2} Z_{n_i} \right).$$

Item (v) of Theorem C therefore provides the separated-time lower bound in (vi).

Finally, (ii) implies exponential mixing and hence (8). Proposition 3.5 gives the neighbourhood in (v), and Corollary 3.6 provides a real trigonometric polynomial in it. The map, coding and original observable were all fixed before R was chosen. □

3.4. **Realisation with a physical measure.** We use the following smooth physical realisation result. To avoid interrupting the flow of the argument, its proof is given in Appendix A.

Proposition 3.11 (Smooth physical realisation of symbolic measures)**.** *Let M be a nonempty compact smooth surface without boundary. There exist a $\mathcal{C}^\infty$ diffeomorphism $F_0 : M \to M$, a two-symbol horseshoe Λ for F_0, and a homeomorphism $\pi : \Sigma \to \Lambda$ satisfying $F_0 \circ \pi = \pi \circ \sigma$, such that the following holds.*

For every ergodic σ-invariant Borel probability measure ν on Σ, there are a $\mathcal{C}^\infty$ diffeomorphism $T : M \to M$, equal to F_0 in a neighbourhood of Λ, and a nonempty open set W such that $\mu = \pi_*\nu$ is physical and W is contained in its basin. In fact, for every $v \in \mathcal{C}^0(M)$,

$$\lim_{N\to\infty} \sup_{y\in W} \left| \frac{1}{N} \sum_{j=0}^{N-1} v(T^j y) - \int v \, d\mu \right| = 0. \tag{15}$$

Moreover, $T(\Lambda) = \Lambda$, $T \circ \pi = \pi \circ \sigma$, and there exist $0 < \theta < 1$ and $C_1 < \infty$ such that

$$\|v \circ \pi\|_\theta \le C_1 \|v\|_{\mathcal{C}^1(M)} \quad \text{for every } v \in \mathcal{C}^1(M).$$

Proof of Theorem B. Fix M and $R \ge 2$. Let F_0, Λ and π be given by Proposition 3.11. Take ν from Theorem C. It is ergodic by Corollary 3.2.

Proposition 3.11 gives a $\mathcal{C}^\infty$ diffeomorphism $T : M \to M$ and a nonempty open set W such that $\mu = \pi_*\nu$ satisfies (v). The same proposition gives the conjugacy and the pullback bound required by Lemma 3.10. That lemma proves (i)–(iii) and (vi) by transferring the symbolic estimates and the coordinate-process law to M. Finally, Proposition 3.5 gives (iv). □

Part II. Probabilistic construction

In this part, we construct the stationary process $(Z_i)_{i\in\mathbb{Z}}$ used in Part I and prove Theorem C. We begin with a sketch of the construction. The details and proofs follow in Sections 4–8.

Sketch of the construction. Fix an integer $R \ge 2$ and set $L := 2R + 2$. We construct the stationary process $(Z_i)_{i\in\mathbb{Z}}$ by defining its probability measure ν on $\Sigma = \{-1, 1\}^{\mathbb{Z}}$. The construction has three steps, followed by references to the proofs of its properties.

Step 1. Random interval partitions. We first construct a nested sequence of random interval partitions of $\mathbb{Z}$. Starting from $\mathcal{M}_0 = \mathbb{Z}$, we choose random sets of markers $\mathcal{M}_k \subset \mathcal{M}_{k-1} \subset \mathbb{Z}$. The intervals between consecutive markers in $\mathcal{M}_k$ form a partition $\mathcal{P}_k$ of $\mathbb{Z}$. Its elements, called level-k blocks, are the intervals $[a, b) \cap \mathbb{Z}$ with $a, b \in \mathcal{M}_k$, $a < b$, and $(a, b) \cap \mathcal{M}_k = \varnothing$. Since the marker sets are nested, every level-k block is a union of consecutive level-$(k-1)$ blocks, which we call its children.

The sets $\{\mathcal{M}_k\}_{k\ge0}$ are constructed so that, with probability one, every level-k block $B \in \mathcal{P}_k$ satisfies

$$(4L)^k < |B| \le 3(4L)^k, \quad \text{for every } k \ge 1,$$

(Lemma 4.4(iii)), and has at least L children (Lemma 4.4(iv)). Thus, the block lengths grow at a controlled rate, and each parent has enough children for the interaction introduced in the next step.

The nested marker sets $\mathcal{H} = (\mathcal{M}_k)_{k\ge0}$ are stationary, in the sense that translating all of them by the same integer leaves their joint law unchanged (Lemma 4.5), i.e.

$$(\mathcal{M}_k + m)_{k\ge0} \overset{d}{=} (\mathcal{M}_k)_{k\ge0} \quad \text{for every } m \in \mathbb{Z},$$

where $\mathcal{M}_k + m := \{x + m : x \in \mathcal{M}_k\}$.

Step 2. Recursive construction. Fix a realisation h of the nested marker sets $\mathcal{H}$, and write $\mathcal{P}_k(h)$ for the corresponding level-k partition. We recursively define a probability measure ρ_B on $\{-1, 1\}^B$ for each block $B \in \mathcal{P}_k(h)$.

At level 0, every block is a singleton $B = \{j\}$, and ρ_B assigns probability $1/2$ to each of the values $X_j = -1$ and $X_j = 1$.

Let $k \ge 1$ and $B \in \mathcal{P}_k(h)$. Write $B_1, \dots, B_r \in \mathcal{P}_{k-1}(h)$ for its children (i.e. $B = B_1 \cup B_2 \cup \dots \cup B_r$), ordered from left to right. Assume that the laws $\rho_{B_1}, \dots, \rho_{B_r}$ have already been defined. Observe that, from Step 1, that $r \ge L$.

For $1 \le i \le L$, Lemma 5.2 gives an odd function $\psi_i = \psi_{\rho_{B_i}}$ taking values in $[-1,1]$ such that

$$\mathbb{E}_{\rho_{B_i}}\big[\psi_i(X[B_i])\big] = 0 \quad \text{and} \quad \mathbb{E}_{\rho_{B_i}}\big[X_j\psi_i(X[B_i])\big] = \frac{1}{\sqrt{3|B_i|}} \quad \text{for every } j \in B_i.$$

Here $X_j(x) = x_j$ and $X[B_i] = (X_j)_{j\in B_i}$. With coordinates ordered from left to right, we use the same function ψ_ρ whenever the same law ρ occurs.

We define the probability ρ_B on $\{-1,1\}^B$ as

$$\rho_B(dx) = \left(1 - \frac{1}{2}\prod_{i=1}^{L}\psi_i(x[B_i])\right)\bigotimes_{i=1}^{r}\rho_{B_i}(dx[B_i]).$$

The product measure includes all r blocks, whereas the density depends only on the first L. The density lies between $1/2$ and $3/2$ and integrates to one, so ρ_B is a probability measure with full support (Lemma 5.5(i)). Integrating over any one of the first L blocks eliminates the product term because the corresponding ψ_i has mean zero. Thus, under ρ_B, any collection of at most $L-1$ vectors $X[B_i]$ is independent, and each has law ρ_{B_i} (Lemma 5.5(iii)).

The law ρ_B remains invariant under $x \mapsto -x$, and its coordinates remain $(L-1)$-tuplewise independent and uniform on $\{-1,1\}$ (Lemma 5.5(i)–(ii)). These properties allow Lemma 5.2 to be applied at every level. Choosing one coordinate in each of the first L blocks gives a negative L-point correlation (Lemma 5.6).

Step 3. The stationary process. Once again we fix a realisation h of the nested marker sets $\mathcal{H}$. For each $k \ge 0$, define the probability $\nu_h^{(k)}$ on $\{-1,1\}^{\mathbb{Z}}$ as

$$\nu_h^{(k)} := \bigotimes_{B\in\mathcal{P}_k(h)} \rho_B.$$

For every nonempty finite set $E \subset \mathbb{Z}$, choose k such that $(4L)^k \ge \max E - \min E + 1$. Then

$$\nu_h^{(\ell)}\big|_E = \nu_h^{(k)}\big|_E \quad \text{for every } \ell \ge k$$

(Lemma 6.2). These stabilised finite-dimensional distributions are consistent and, by the Kolmogorov extension theorem, determine a probability measure ν_h on Σ (Proposition 6.3(i)).

Averaging over the nested marker sets gives

$$\nu(A) := \int \nu_h(A)\,\mathbb{P}_{\mathcal{H}}(dh) \quad \text{for } A \in \mathcal{B}(\Sigma),$$

where $\mathbb{P}_{\mathcal{H}}$ is the law of $\mathcal{H}$. On the probability space $(\Sigma, \mathcal{B}(\Sigma), \nu)$, define

$$Z_i(\omega) := \omega_i \quad \text{for every } \omega \in \Sigma \text{ and } i \in \mathbb{Z}.$$

Properties of the process. The coordinate process $(Z_i)_{i\in\mathbb{Z}}$ is stationary, and its law ν has full support on Σ (Proposition 6.3(ii)). Item (i) of Theorem C follows from Proposition 6.3(iii). Items (ii) and (iii) of Theorem C are proved in Propositions 7.1 and 7.2, respectively.

The key properties of the process are stated in items (iv)–(v). Item (iv) gives the failure of the CLT for any subsequence (Corollaries 8.3 and 8.4). Item (v) shows that exponential mixing does not extend to all orders. It holds at order R but fails at order $L = 2R+2$ (Corollary 8.6).

4. Stationary nested marker sets and interval partitions

Fix an integer $R \ge 2$ and set $L := 2R+2$, so that $L \ge 6$ is even. We construct stationary nested random marker sets and the associated interval partitions on which the coordinate process will be built. At level k, the block lengths and the dependence range of the marker configurations are comparable to $(4L)^k$. The factor 4 in this scale ensures that each parent contains at least L children, allowing the interaction in Section 5 to involve L distinct children. The bounds on block lengths and dependence ranges will yield finite-window independence and exponential correlation bounds. Proposition 4.2 records the four properties used in the later sections; those sections do not use the particular marker selection rule.

We use the finitely dependent colourings of Holroyd and Liggett [29, Corollary 20] to construct nested marker sets. Our selection of markers is similar to the construction of nets in [30, proof of Corollary 11]. Here we repeat the selection within each preceding marker set and establish the block-length and finite-dependence bounds needed for our construction.

4.1. **Nested marker sets and associated partitions.** We consider a sequence $\mathcal{H} = (\mathcal{M}_k)_{k\geq 0}$ of nested bi-infinite subsets of $\mathbb{Z}$, with

$$\mathbb{Z} = \mathcal{M}_0 \supset \mathcal{M}_1 \supset \mathcal{M}_2 \supset \cdots .$$

We use the following notation.

- The elements of $\mathcal{M}_k$ are the *level-k markers*. The intervals between consecutive markers form the partition
$$\mathcal{P}_k := \big\{[u,v)\cap\mathbb{Z} : u,v\in\mathcal{M}_k,\ u<v,\ \mathcal{M}_k\cap(u,v)=\varnothing\big\}$$
of $\mathbb{Z}$ into finite intervals, called *level-k blocks*.
- For $k \geq 1$, each block $B \in \mathcal{P}_k$ is a union of consecutive blocks $B_1, \ldots, B_r \in \mathcal{P}_{k-1}$, because $\mathcal{M}_k \subset \mathcal{M}_{k-1}$. We call these blocks the *children* of B and write $N_k(B) := r$ for their number. Thus $B = B_1 \cup \cdots \cup B_{N_k(B)}$, with $B_i \in \mathcal{P}_{k-1}$.
- The *configuration* of $\mathcal{H}$ at $i \in \mathbb{Z}$ through level k is
$$\mathcal{H}_i^{(k)} := \big(\mathbf{1}_{\{i\in\mathcal{M}_0\}}, \ldots, \mathbf{1}_{\{i\in\mathcal{M}_k\}}\big).$$
We write $\mathcal{H}^{(k)} := (\mathcal{H}_i^{(k)})_{i\in\mathbb{Z}}$ for the process recording marker membership through level k.
- For a finite interval $I = [u,v]\cap\mathbb{Z}$, write
$$\mathcal{H}^{(k)}[I] := (\mathcal{H}_u^{(k)}, \ldots, \mathcal{H}_v^{(k)}).$$
It determines all blocks and their subdivisions in I up to level k.
- For an interval I and $r \geq 0$, write
$$I^{+r} := \{j \in \mathbb{Z} : \operatorname{dist}(j, I) \leq r\}. \tag{16}$$

Finite dependence will allow us to make sufficiently separated local marker configurations independent.

Definition 4.1 (Finite dependence)**.** For $d \geq 0$, a process indexed by $\mathbb{Z}$ is *d-dependent* if its restrictions to any two sets of indices at distance greater than d generate independent σ-algebras.

Figures 3 and 4 illustrate the notation introduced above.

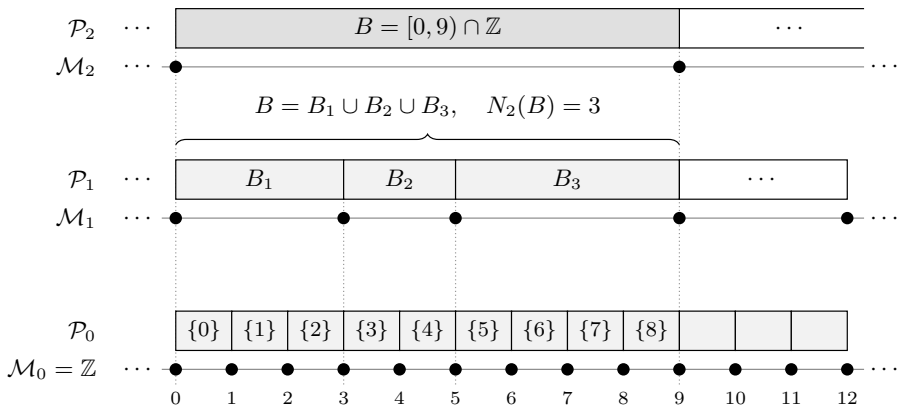


Figure 3. Visual representation of $\mathcal{M}_0, \mathcal{M}_1, \mathcal{M}_2, \mathcal{P}_0, \mathcal{P}_1, \mathcal{P}_2$. The level-2 block B has the three level-1 children $B_1 = [0,3)\cap\mathbb{Z}$, $B_2 = [3,5)\cap\mathbb{Z}$ and $B_3 = [5,9)\cap\mathbb{Z}$. Each level-1 block is a union of level-0 singletons. The rows continue in both directions.

The following proposition gives the required stationary nested marker sets.

Proposition 4.2. *There exist a finite constant $D = D(L)$ and a random sequence of nested marker sets $\mathcal{H} = (\mathcal{M}_k)_{k\geq 0}$, with law $\mathbb{P}_{\mathcal{H}}$, i.e. the distribution of $\mathcal{H}$ on $\big(\{0,1\}^{\mathbb{Z}}\big)^{\mathbb{Z}_{\geq 0}}$ with its product σ-algebra, such that the following hold.*

(H1) *Almost surely, for every $k \geq 1$ and every $B \in \mathcal{P}_k$, $(4L)^k < |B| \leq 3(4L)^k$.*
(H2) *Almost surely, for every $k \geq 1$ and every $B \in \mathcal{P}_k$, $N_k(B) > 4L/3 > L$.*

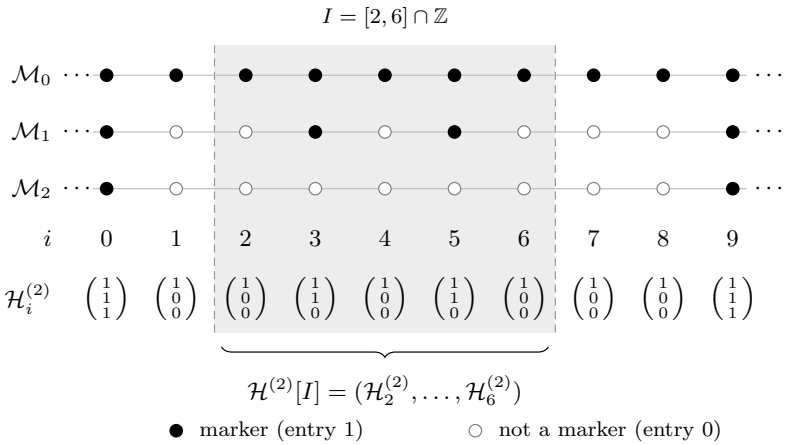


Figure 4. Configurations of the nested marker sets through level 2. For each integer i, the tuple $\mathcal{H}_i^{(2)}$ records membership in $\mathcal{M}_0$, $\mathcal{M}_1$ and $\mathcal{M}_2$, with its entries displayed vertically in this order. The shaded columns form the restriction $\mathcal{H}^{(2)}[I]$ to the interval $I = [2,6] \cap \mathbb{Z}$.

(H3) *For every $k \geq 0$, $p \geq 1$ and $i_1, \dots, i_p, h \in \mathbb{Z}$,*

$$\mathrm{Law}\big(\mathcal{H}_{i_1+h}^{(k)}, \dots, \mathcal{H}_{i_p+h}^{(k)}\big) = \mathrm{Law}\big(\mathcal{H}_{i_1}^{(k)}, \dots, \mathcal{H}_{i_p}^{(k)}\big).$$

(H4) *For every $k \geq 0$, the process $\mathcal{H}^{(k)}$ is $(4L)^k D$-dependent.*

We now construct the marker sets and prove Proposition 4.2. A reader who assumes the proposition may proceed directly to Section 5.

4.2. **A stationary finitely dependent colouring.** Recall that $L = 2R+2$ is fixed throughout. We use the following consequence of [29, Corollary 20].

Theorem 4.3 ([29, Corollary 20])**.** *There are an integer $q = q(L) \geq 2$ and a process $V = (V_i)_{i \in \mathbb{Z}}$ taking values in the set of colours $\{1, \dots, q\}$ with the following properties.*

(i) *The process is strictly stationary, meaning that for every $p \geq 1$ and every $j_1, \dots, j_p, h \in \mathbb{Z}$, $\mathrm{Law}(V_{j_1+h}, \dots, V_{j_p+h}) = \mathrm{Law}(V_{j_1}, \dots, V_{j_p})$.*

(ii) *The process is $(4L)$-dependent.*

(iii) *Almost surely,*

$$V_i \neq V_j \quad \text{whenever} \quad 0 < |i-j| \leq 4L. \tag{17}$$

Following [29], we refer to the values of V as colours. Property (17) says that nearby indices receive different colours, which is the feature used in the selection rule below. Fix this process once and for all. Only the fact that $q < \infty$ is used.

4.3. **The marker selection rule.** We construct the marker sets recursively. For each level $\ell \geq 0$, we denote the increasing enumeration of $\mathcal{M}_\ell$ by $(t_j^{(\ell)})_{j \in \mathbb{Z}}$, chosen so that $t_0^{(\ell)} \leq 0 < t_1^{(\ell)}$.

- *Initial level.* Set $\mathcal{M}_0 := \mathbb{Z}$, so that $t_j^{(0)} = j$ and $\mathcal{P}_0 = \big\{\{j\} : j \in \mathbb{Z}\big\}$.
- *Enumerate the old markers.* Fix $k \geq 1$ and suppose that $\mathcal{M}_{k-1}$ has been constructed. Write its increasing enumeration as

$$\mathcal{M}_{k-1} = \Big\{t_j^{(k-1)} : j \in \mathbb{Z}\Big\} \quad \text{with} \quad \cdots < t_{-1}^{(k-1)} < t_0^{(k-1)} \leq 0 < t_1^{(k-1)} < t_2^{(k-1)} < \cdots .$$

- *Assign colours.* Let $V^{(k)} = (V_i^{(k)})_{i \in \mathbb{Z}}$ be an independent copy of the process in Theorem 4.3, independent of all randomness used at the previous levels, and assign the colour $V_j^{(k)}$ to the marker $t_j^{(k-1)}$. The colour attached to $t_j^{(k-1)}$ is $V_j^{(k)}$, not $V_{t_j^{(k-1)}}^{(k)}$.
- *Select the new markers.* Set $A_0 := \varnothing$, with $\mathrm{dist}(t, \varnothing) := +\infty$, and, for $a = 1, \dots, q$, define

$$A_a := A_{a-1} \cup \Big\{t_j^{(k-1)} \in \mathcal{M}_{k-1} : V_j^{(k)} = a,\ \mathrm{dist}\Big(t_j^{(k-1)}, A_{a-1}\Big) > (4L)^k\Big\}. \tag{18}$$

Set $\mathcal{M}_k := A_q$. In particular, $\mathcal{M}_k \subset \mathcal{M}_{k-1}$.

The enumeration $t_0^{(k-1)} \le 0 < t_1^{(k-1)}$ is used only to assign the colouring.

The following lemma controls the geometry of the partitions at every level. The block lengths are comparable to $(4L)^k$, and each parent $B \in \mathcal{P}_k$ contains at least L children $B_i \in \mathcal{P}_{k-1}$, allowing the recursive construction of the block laws.

Lemma 4.4. *For every $k \ge 1$, the marker set $\mathcal{M}_k$ is almost surely a bi-infinite subset of $\mathcal{M}_{k-1}$. Moreover, the following hold.*

(i) *The set $\mathcal{M}_k$ is $(4L)^k$-separated, that is,*

$$|u - v| > (4L)^k \quad \text{for all distinct } u, v \in \mathcal{M}_k.$$

(ii) *The set $\mathcal{M}_k$ is maximal with this property inside $\mathcal{M}_{k-1}$, meaning that for every $t \in \mathcal{M}_{k-1}$ there exists $u \in \mathcal{M}_k$ with $|t - u| \le (4L)^k$. Equivalently, $\mathcal{M}_k$ is a $(4L)^k$-net in $\mathcal{M}_{k-1}$.*

(iii) *The family $\mathcal{P}_k$ is a partition of $\mathbb{Z}$ into finite intervals, and every $B \in \mathcal{P}_k$ satisfies*

$$(4L)^k < |B| \le 2(4L)^k + 3(4L)^{k-1} \le 3(4L)^k.$$

(iv) *Every $B \in \mathcal{P}_k$ is the union of consecutive blocks of $\mathcal{P}_{k-1}$, and*

$$N_k(B) > \frac{4L}{3}.$$

In particular, every level-k block has at least L children.

Proof. We argue by induction on $k \ge 1$, recalling that $\mathcal{M}_0 = \mathbb{Z}$ and $\mathcal{P}_0 = \big\{\{j\} : j \in \mathbb{Z}\big\}$.

Step 1. *The conclusions (i)–(iv) hold for $k = 1$.*

Proof of Step 1. Write $\mathcal{M}_0 = \{t_j^{(0)} : j \in \mathbb{Z}\}$ with $t_j^{(0)} = j$. We first prove (i). We claim that if $V_i^{(1)} = V_j^{(1)}$ with $i \ne j$, then $|t_i^{(0)} - t_j^{(0)}| > 4L$. Indeed, if $|t_i^{(0)} - t_j^{(0)}| \le 4L$, then $|i - j| = |t_i^{(0)} - t_j^{(0)}| \le 4L$, and hence $V_i^{(1)} \ne V_j^{(1)}$ by (17). Let $u \ne v$ belong to $\mathcal{M}_1$. If they are selected at the same stage of (18), they have the same colour, so the claim gives $|u - v| > 4L$. If they are selected at different stages, the marker selected later is added only if its distance from every previously selected marker is greater than $4L$. Thus $\mathcal{M}_1$ is $4L$-separated, proving (i). We next prove (ii). Take $t_j^{(0)} \in \mathcal{M}_0$. If $t_j^{(0)} \in \mathcal{M}_1$, then $\operatorname{dist}(t_j^{(0)}, \mathcal{M}_1) = 0$. Otherwise, let $a := V_j^{(1)}$. Since $t_j^{(0)} \notin A_a$, the definition of A_a gives $u \in A_{a-1}$ with $|t_j^{(0)} - u| \le 4L$. As $A_{a-1} \subset \mathcal{M}_1$, we obtain

$$\operatorname{dist}\left(t_j^{(0)}, \mathcal{M}_1\right) \le 4L. \tag{19}$$

This proves (ii), and it implies that $\mathcal{M}_1$ is unbounded in both directions. We now prove (iii). Since $\mathcal{M}_1$ is bi-infinite, its consecutive points determine a partition $\mathcal{P}_1$ of $\mathbb{Z}$ into finite intervals. Let $B = [u, v) \cap \mathbb{Z} \in \mathcal{P}_1$. By (i), $v - u > 4L$. Suppose, for a contradiction, that $v - u > 2(4L) + 3$. Then $(u + 4L, v - 4L)$ has length greater than 3 and therefore contains an integer $t \in \mathcal{M}_0$. Since u, v are consecutive points of $\mathcal{M}_1$, this gives $\operatorname{dist}(t, \mathcal{M}_1) > 4L$, contradicting (19). Thus

$$4L < |B| \le 2(4L) + 3 \le 3(4L),$$

which proves (iii). Finally, the level-0 blocks are singletons, so $N_1(B) = |B| > 4L > 4L/3$, proving (iv). ∎

Step 2. *Assume the conclusions at level $k - 1$, with $k \ge 2$. Then (i)–(ii) hold at level k.*

Proof of Step 2. Write the increasing enumeration of $\mathcal{M}_{k-1}$ as $(t_j^{(k-1)})_{j \in \mathbb{Z}}$. We first claim that if $V_i^{(k)} = V_j^{(k)}$ with $i \ne j$, then $|t_i^{(k-1)} - t_j^{(k-1)}| > (4L)^k$. Indeed, suppose that $|t_i^{(k-1)} - t_j^{(k-1)}| \le (4L)^k$. By (iii) at level $k - 1$, every level-$(k-1)$ block has length greater than $(4L)^{k-1}$, so

$$(4L)^k \ge \left|t_i^{(k-1)} - t_j^{(k-1)}\right| > |i - j|(4L)^{k-1},$$

whence $|i - j| < 4L$ and, by (17), $V_i^{(k)} \ne V_j^{(k)}$. Let $u \ne v$ belong to $\mathcal{M}_k$. If they are selected at the same stage, the claim gives $|u - v| > (4L)^k$. If they are selected at different stages, the marker selected later is retained only if its distance from every previously selected marker is greater than $(4L)^k$. This proves (i). For (ii), take $t_j^{(k-1)} \in \mathcal{M}_{k-1}$. If $t_j^{(k-1)} \in \mathcal{M}_k$ there is nothing to prove.

Otherwise, let $a := V_j^{(k)}$. Since $t_j^{(k-1)} \notin A_a$, (18) gives $u \in A_{a-1}$ with $|t_j^{(k-1)} - u| \le (4L)^k$. As $A_{a-1} \subset \mathcal{M}_k$, we obtain

$$\operatorname{dist}\left(t_j^{(k-1)}, \mathcal{M}_k\right) \le (4L)^k. \tag{20}$$

In particular $\mathcal{M}_k$ is maximal among the $(4L)^k$-separated subsets of $\mathcal{M}_{k-1}$, and is unbounded in both directions. ■

Step 3. *Under the same induction hypothesis, (iii)–(iv) hold at level k.*

Proof of Step 3. Since $\mathcal{M}_k$ is bi-infinite, its consecutive points partition $\mathbb{Z}$ into finite intervals, so $\mathcal{P}_k$ is a partition. Let $B = [u, v) \cap \mathbb{Z} \in \mathcal{P}_k$. From (i), $v - u > (4L)^k$. Suppose, for a contradiction, that

$$v - u > 2(4L)^k + 3(4L)^{k-1}.$$

Then $\left(u + (4L)^k, v - (4L)^k\right)$ has length greater than $3(4L)^{k-1}$. By (iii) at level $k-1$, every level-$(k-1)$ block has length at most $3(4L)^{k-1}$, so this interval contains some $t \in \mathcal{M}_{k-1}$. As u and v are consecutive points of $\mathcal{M}_k$, this gives $\operatorname{dist}(t, \mathcal{M}_k) > (4L)^k$, contradicting (20). Therefore $v - u \le 2(4L)^k + 3(4L)^{k-1} \le 3(4L)^k$, which proves (iii). For (iv), write $u = t_{j_0}^{(k-1)} < t_{j_0+1}^{(k-1)} < \cdots < t_{j_1}^{(k-1)} = v$ for the level-$(k-1)$ markers between u and v, which exist because $\mathcal{M}_k \subset \mathcal{M}_{k-1}$. Then B is the union of the $N_k(B) = j_1 - j_0$ consecutive level-$(k-1)$ blocks $[t_j^{(k-1)}, t_{j+1}^{(k-1)}) \cap \mathbb{Z}$. Each has length at most $3(4L)^{k-1}$ by (iii) at level $k-1$, so $|B| \le 3(4L)^{k-1} N_k(B)$, and the lower bound in (iii) gives

$$N_k(B) > \frac{(4L)^k}{3(4L)^{k-1}} = \frac{4L}{3}.$$

■

Steps 2–3 complete the induction. □

4.4. **Stationarity of the nested marker sets.** The following lemma shows that, for every $k \ge 0$, the process $(\mathcal{H}_i^{(k)})_{i \in \mathbb{Z}}$ is stationary.

Lemma 4.5. *For every $k \ge 0$, $p \ge 1$ and $i_1, \dots, i_p, h \in \mathbb{Z}$,*

$$\operatorname{Law}\left(\mathcal{H}_{i_1+h}^{(k)}, \dots, \mathcal{H}_{i_p+h}^{(k)}\right) = \operatorname{Law}\left(\mathcal{H}_{i_1}^{(k)}, \dots, \mathcal{H}_{i_p}^{(k)}\right). \tag{21}$$

Proof. We prove by induction on k that

$$\left(\mathcal{M}_0 + h, \dots, \mathcal{M}_k + h\right) \overset{d}{=} \left(\mathcal{M}_0, \dots, \mathcal{M}_k\right) \quad \text{for every } h \in \mathbb{Z}. \tag{22}$$

The case $k = 0$ is immediate since $\mathcal{M}_0 = \mathbb{Z}$. Assume the assertion through level $k-1$ and fix $h \in \mathbb{Z}$. If $(t_j^{(k-1)})_{j \in \mathbb{Z}}$ is the increasing enumeration of $\mathcal{M}_{k-1}$ anchored at zero, then $(t_j^{(k-1)} + h)_{j \in \mathbb{Z}}$ enumerates $\mathcal{M}_{k-1} + h$, but need not be anchored at zero. The anchored enumeration of the translated set has the form $(t_{j+r}^{(k-1)} + h)_{j \in \mathbb{Z}}$, where r is determined by $\mathcal{M}_{k-1}$ and h. Conditional on $\mathcal{M}_0, \dots, \mathcal{M}_{k-1}$, the integer r is fixed. Independence and stationarity of $V^{(k)}$ give

$$\operatorname{Law}\left((V_{i+r}^{(k)})_{i \in \mathbb{Z}} \mid \mathcal{M}_0, \dots, \mathcal{M}_{k-1}\right) = \operatorname{Law}\left((V_i)_{i \in \mathbb{Z}}\right).$$

In this way, the conditional law of the colouring on the translated markers is the same whether we use the anchored enumeration or $(t_j^{(k-1)} + h)_{j \in \mathbb{Z}}$. We may therefore compare the selections from $(t_j^{(k-1)}, V_j^{(k)})$ and $(t_j^{(k-1)} + h, V_j^{(k)})$ on the same probability space. Let $A_0, \dots, A_q$ be obtained from $(t_j^{(k-1)}, V_j^{(k)})$ by (18), and let $A'_0, \dots, A'_q$ be obtained from $(t_j^{(k-1)} + h, V_j^{(k)})$. We claim that

$$A'_a = A_a + h \quad \text{for each } 0 \le a \le q. \tag{23}$$

For $a = 0$ this is immediate. If $A'_{a-1} = A_{a-1} + h$, then, since $\operatorname{dist}(\cdot, \cdot)$ is translation invariant,

$$A'_a = \left(A_{a-1} \cup \left\{t_j^{(k-1)} : V_j^{(k)} = a,\ \operatorname{dist}\left(t_j^{(k-1)}, A_{a-1}\right) > (4L)^k\right\}\right) + h = A_a + h.$$

Thus (23) holds for every a, and in particular $\mathcal{M}_k + h$ is the marker set produced from the translated old markers with the same colours. Since $V^{(k)}$ is independent of the previous levels and $(\mathcal{M}_0, \dots, \mathcal{M}_{k-1})$ has the same law after translation, (22) follows. Finally, $\mathcal{H}_i^{(k)}$ is determined by the memberships $\mathbf{1}_{\{i \in \mathcal{M}_j\}}$, $j \le k$, so (22) gives (21). □

4.5. **Finite dependence of the marker sets at each scale.** Recall that I^{+r} denotes the set of integers at distance at most r from I, as defined in (16). The following lemma shows that $\mathcal{M}_k \cap I$ is determined by $\mathcal{M}_{k-1} \cap I^{+q(4L)^k}$ and the colours attached to those markers.

Lemma 4.6. *Fix $k \geq 1$ and an interval $I \subset \mathbb{Z}$. Consider two level-$(k-1)$ marker sets $\mathcal{M}_{k-1}$ and $\widetilde{\mathcal{M}}_{k-1}$, with increasing enumerations*

$$\mathcal{M}_{k-1} = \left\{ t_j^{(k-1)} : j \in \mathbb{Z} \right\} \quad \text{and} \quad \widetilde{\mathcal{M}}_{k-1} = \left\{ \widetilde{t}_j^{(k-1)} : j \in \mathbb{Z} \right\},$$

and attach to their markers the colours $(V_i^{(k)})_{i\in\mathbb{Z}}$ and $(\widetilde{V}_i^{(k)})_{i\in\mathbb{Z}}$ respectively. Let $\mathcal{M}_k$ and $\widetilde{\mathcal{M}}_k$ be the marker sets produced from these two configurations by (18). If

$$\left\{ \left(t_j^{(k-1)}, V_j^{(k)} \right) : t_j^{(k-1)} \in I^{+q(4L)^k} \right\} = \left\{ \left(\widetilde{t}_j^{(k-1)}, \widetilde{V}_j^{(k)} \right) : \widetilde{t}_j^{(k-1)} \in I^{+q(4L)^k} \right\}, \tag{24}$$

then $\mathcal{M}_k \cap I = \widetilde{\mathcal{M}}_k \cap I$.

Proof. We prove by induction on $a = 0, \dots, q$ that

$$A_a \cap I^{+(q-a)(4L)^k} = \widetilde{A}_a \cap I^{+(q-a)(4L)^k}. \tag{25}$$

For $a = 0$ this follows from $A_0 = \widetilde{A}_0 = \varnothing$. Fix $1 \leq a \leq q$ and assume (25) with $a-1$ in place of a. Take $t \in A_a \cap I^{+(q-a)(4L)^k}$. If $t \in A_{a-1}$, then $t \in I^{+(q-a+1)(4L)^k}$ and the induction hypothesis gives $t \in \widetilde{A}_{a-1} \subset \widetilde{A}_a$. Suppose instead that $t \notin A_{a-1}$. By the definition of A_a there is j with $t = t_j^{(k-1)}$, $V_j^{(k)} = a$ and $\operatorname{dist}(t, A_{a-1}) > (4L)^k$. Since $t \in I^{+q(4L)^k}$, assumption (24) provides m with $\widetilde{t}_m^{(k-1)} = t$ and $\widetilde{V}_m^{(k)} = a$. We claim that $\operatorname{dist}(t, \widetilde{A}_{a-1}) > (4L)^k$. Otherwise some $u \in \widetilde{A}_{a-1}$ satisfies $|u - t| \leq (4L)^k$, hence $u \in I^{+(q-a+1)(4L)^k}$, and the induction hypothesis puts $u \in A_{a-1}$, contradicting $\operatorname{dist}(t, A_{a-1}) > (4L)^k$. The definition of $\widetilde{A}_a$ therefore gives $t \in \widetilde{A}_a$. This proves one inclusion in (25). The reverse is obtained by interchanging the two configurations. Taking $a = q$ gives $\mathcal{M}_k \cap I = \widetilde{\mathcal{M}}_k \cap I$. □

The following lemma is a standard consequence of the properties of conditional independence [37, Theorems 8.9 and 8.12]. We state and prove it in the form needed below.

Lemma 4.7. *Let $\mathcal{H}$ be a random element in a measurable space, and let $H_1, \dots, H_s, X_1, \dots, X_s$ be random elements taking values in finite sets. Suppose that $H_1, \dots, H_s$ are jointly independent and measurable with respect to $\sigma(\mathcal{H})$, and that $X_1, \dots, X_s$ are conditionally independent given $\mathcal{H}$, with $\operatorname{Law}(X_i \mid \mathcal{H}) = \operatorname{Law}(X_i \mid H_i)$ almost surely for every $1 \leq i \leq s$. Then the pairs $(H_1, X_1), \dots, (H_s, X_s)$ are jointly independent. In particular, $X_1, \dots, X_s$ are jointly independent.*

Proof. We use the following standard properties of conditional independence.

(P1) U and V are conditionally independent given W if and only if

$$\operatorname{Law}(U \mid V, W) = \operatorname{Law}(U \mid W)$$

almost surely (see [37, Theorem 8.9]).

(P2) If U is independent of V, and U and W are conditionally independent given V, then U is independent of (V, W) (see [37, Theorem 8.12]).

The case $s = 1$ is immediate. Fix $2 \leq i \leq s$ and write

$$H_{<i} := (H_1, \dots, H_{i-1}) \quad \text{and} \quad X_{<i} := (X_1, \dots, X_{i-1}).$$

Conditional independence given $\mathcal{H}$ and the hypothesis on the conditional laws imply

$$\operatorname{Law}(X_{<i} \mid \mathcal{H}) = \bigotimes_{j<i} \operatorname{Law}(X_j \mid H_j).$$

The right-hand side is measurable with respect to $\sigma(H_{<i})$, so the tower property gives

$$\operatorname{Law}(X_{<i} \mid H_{<i}, H_i) = \bigotimes_{j<i} \operatorname{Law}(X_j \mid H_j) = \operatorname{Law}(X_{<i} \mid H_{<i}).$$

By (P1), H_i and $X_{<i}$ are conditionally independent given $H_{<i}$. Since H_i is independent of $H_{<i}$, (P2) shows that H_i is independent of $(H_{<i}, X_{<i})$.

Next, conditional independence given $\mathcal{H}$ and (P1) yield

$$\operatorname{Law}(X_i \mid \mathcal{H}, X_{<i}) = \operatorname{Law}(X_i \mid \mathcal{H}) = \operatorname{Law}(X_i \mid H_i).$$

Since H_i and $H_{<i}$ are measurable with respect to $\sigma(\mathcal{H})$, conditioning this identity on $(H_i, H_{<i}, X_{<i})$ gives

$$\mathrm{Law}(X_i \mid H_i, H_{<i}, X_{<i}) = \mathrm{Law}(X_i \mid H_i).$$

Thus (P1) shows that X_i and $(H_{<i}, X_{<i})$ are conditionally independent given H_i. Together with the independence established above, (P2) implies that (H_i, X_i) is independent of $(H_{<i}, X_{<i})$.

Hence each pair (H_i, X_i) is independent of all preceding pairs jointly. This proves joint independence of the pairs, and therefore of their second components. □

The following proposition bounds the dependence range of $(\mathcal{H}_i^{(k)})_{i\in\mathbb{Z}}$ by $4q(4L)^k$ for each $k \geq 0$.

Proposition 4.8. *Let $q = q(L)$ be as in Theorem 4.3. For every $k \geq 0$ and all finite intervals $I, J \subset \mathbb{Z}$ with $\mathrm{dist}(I, J) > 4q(4L)^k$, the random vectors*

$$\left(\mathcal{H}_i^{(k)}\right)_{i\in I} \quad \text{and} \quad \left(\mathcal{H}_j^{(k)}\right)_{j\in J}$$

are independent. In particular, $\mathcal{H}^{(k)}$ is $4q(4L)^k$-dependent.

Proof. We argue by induction. For $k = 0$ the marker configuration is deterministic. Suppose the assertion holds at level $k-1$, and let $\mathrm{dist}(I, J) > 4q(4L)^k$. Set

$$I' := I^{+q(4L)^k} \quad \text{and} \quad J' := J^{+q(4L)^k}.$$

Their distance exceeds $2q(4L)^k > 4q(4L)^{k-1}$, so the old restrictions $\mathcal{H}^{(k-1)}[I']$ and $\mathcal{H}^{(k-1)}[J']$ are independent.

Enumerate $\mathcal{M}_{k-1}$ as $(t_m^{(k-1)})_{m\in\mathbb{Z}}$ and put $K_I := \{m : t_m^{(k-1)} \in I'\}$ and $K_J := \{m : t_m^{(k-1)} \in J'\}$. For $m \in K_I$ and $n \in K_J$, $|t_m^{(k-1)} - t_n^{(k-1)}| > 2q(4L)^k \geq 4(4L)^k$. If $|m-n| \leq 4L$, the interval between $t_m^{(k-1)}$ and $t_n^{(k-1)}$ would consist of at most $4L$ blocks in $\mathcal{P}_{k-1}$. Each has length at most $3(4L)^{k-1}$, including the singleton case $k = 1$, so

$$\left|t_m^{(k-1)} - t_n^{(k-1)}\right| \leq |m-n|\, 3(4L)^{k-1} \leq 3(4L)^k,$$

a contradiction. Thus $\mathrm{dist}(K_I, K_J) > 4L$.

The colouring $V^{(k)}$ is independent of all marker sets constructed at the previous levels. Conditioned on those markers, its law is therefore unchanged, and its finite dependence makes the two colour vectors independent. List each vector in increasing order, without its absolute enumeration indices. Each K_I is an interval of indices. Stationarity of $V^{(k)}$ therefore makes its conditional law depend only on the number of markers in I', which is determined by $\mathcal{H}^{(k-1)}[I']$. The same holds for J'. Lemma 4.7 makes the two pairs consisting of an old local configuration and its colour vector independent. By Lemma 4.6, the new configurations on I and J are functions of the respective pairs. This proves the induction step.

To extend the conclusion to arbitrary index sets, first observe that restrictions to finitely many ordered intervals with successive distances greater than $4q(4L)^k$ are jointly independent. Indeed, each interval is independent of the convex hull of all preceding intervals, so induction gives joint independence. Now let $E, F \subset \mathbb{Z}$ be finite sets with $\mathrm{dist}(E, F) > 4q(4L)^k$. Order the points of $E \cup F$ and group consecutive points according to whether they belong to E or to F. The convex hulls of these groups are ordered intervals with successive distances greater than $4q(4L)^k$. Their restrictions are therefore jointly independent, which implies independence of the restrictions to E and F. For infinite index sets, the conclusion follows because cylinder events generate the corresponding σ-algebras. □

The local dependence of the conditional laws is essential in Lemma 4.7, since conditional independence alone would not justify averaging.

Remark 4.9. The colouring is attached to the indices j in the enumeration $(t_j^{(k-1)})$, not to the positions $t_j^{(k-1)} \in \mathbb{Z}$. The block-length lower bound gives

$$0 < \left|t_i^{(k-1)} - t_j^{(k-1)}\right| \leq (4L)^k \quad \Longrightarrow \quad 0 < |i-j| \leq 4L.$$

Thus the same colouring, with a fixed finite number $q = q(L)$ of colours, can be used at every level. Lemma 4.6 then requires information only within distance $q(4L)^k$, and Proposition 4.8 gives dependence range $4q(4L)^k$. Both distances are fixed multiples of the block scale. This is the

estimate used later to choose a cylinder radius proportional to the separation of the observation times.

Proof of Proposition 4.2. Take the nested marker sets $\mathcal{H} = (\mathcal{M}_k)_{k\geq 0}$ constructed in Section 4.3. Lemma 4.4 gives (H1) and (H2), and Lemma 4.5 gives (H3). Proposition 4.8 gives (H4), with $D := 4q$. □

5. A finite-dimensional biasing lemma and the block laws

Throughout this section and Sections 6–8, we work with the random nested marker sets $\mathcal{H}$ of Proposition 4.2, with law $\mathbb{P}_{\mathcal{H}}$. We use only (H1)–(H4). For each block B of a fixed realisation of $\mathcal{H}$, we construct a law ρ_B on its coordinates. These laws preserve every marginal on at most $L-1$ children (Lemma 5.5) and produce negative L-point correlations across the first L children (Lemma 5.6). The recursive construction follows the principle in [11, Lemma 2.8] of introducing higher-order dependence while preserving lower-order marginals.

The nested partitions are now fixed independently of the coordinate variables. The children of a parent may have different lengths and may be subdivided differently at lower levels. We therefore need a function positively correlated with every coordinate, even when the child laws are not exchangeable.

The following notation identifies a block law with a finite-dimensional law, allowing Lemma 5.2 to be applied to children with different lengths and different laws.

Notation 5.1 (Coordinates on a finite block)**.** For a finite block $B \subset \mathbb{Z}$, write

$$\{-1,1\}^B := \{x = (x_j)_{j\in B} : x_j \in \{-1,1\}\}.$$

For each $j \in B$, define the coordinate map

$$X_j : \{-1,1\}^B \to \{-1,1\} \quad \text{with} \quad X_j(x) := x_j.$$

Thus, whenever ρ is a probability measure on $\{-1,1\}^B$, we regard $X = (X_j)_{j\in B}$ as the corresponding random vector under ρ. For $C \subset B$, write

$$x[C] := (x_j)_{j\in C} \quad \text{and} \quad X[C] := (X_j)_{j\in C}.$$

We identify $\{-1,1\}^B$ with $\{-1,1\}^{|B|}$ by ordering the coordinates of B from left to right.

5.1. Construction of the auxiliary function ψ_ρ.

The following lemma provides the functions used to introduce the dependence between blocks that leads to failure of the CLT.

Lemma 5.2. *Let ρ be a probability measure on $\{-1,1\}^d$, invariant under the global sign reversal $x \mapsto -x$, and let $X = (X_1, \dots, X_d)$ be a random vector with law ρ. Suppose that*

$$\mathbb{E}_\rho(a\cdot X)^2 = \|a\|_2^2 \quad \text{and} \quad \mathbb{E}_\rho(a\cdot X)^4 \leq 3\|a\|_2^4 \quad \text{for every } a \in \mathbb{R}^d. \tag{26}$$

There exists an odd function $\psi_\rho : \{-1,1\}^d \to [-1,1]$ such that

$$\mathbb{E}_\rho[X_j\psi_\rho(X)] = \frac{1}{\sqrt{3d}}, \quad \text{for each } 1 \leq j \leq d. \tag{27}$$

In particular $\mathbb{E}_\rho\psi_\rho = 0$. Both conditions in (26) hold when the coordinates are four-tuplewise independent and satisfy $\rho(X_j = -1) = \rho(X_j = 1) = 1/2$ for every $1 \leq j \leq d$, since then $\mathbb{E}_\rho(a\cdot X)^4 = 3\|a\|_2^4 - 2\sum_i a_i^4$.

Proof. Fix $a \in \mathbb{R}^d \setminus \{0\}$ and set $Y := a\cdot X$. The case $a = 0$ is immediate. Writing $Y^2 = |Y|^{2/3}\cdot|Y|^{4/3}$ and applying Hölder's inequality with the conjugate exponents $3/2$ and 3 gives

$$\mathbb{E}_\rho Y^2 \leq \big(\mathbb{E}_\rho|Y|\big)^{2/3}\big(\mathbb{E}_\rho Y^4\big)^{1/3}, \tag{28}$$

since $\big\||Y|^{2/3}\big\|_{3/2} = (\mathbb{E}_\rho|Y|)^{2/3}$ and $\big\||Y|^{4/3}\big\|_3 = (\mathbb{E}_\rho Y^4)^{1/3}$. Raising both sides of (28) to the power $3/2$ gives $\big(\mathbb{E}_\rho Y^2\big)^{3/2} \leq \mathbb{E}_\rho|Y|\,\big(\mathbb{E}_\rho Y^4\big)^{1/2}$. Thus

$$\mathbb{E}_\rho|a\cdot X| \geq \frac{\big(\mathbb{E}_\rho(a\cdot X)^2\big)^{3/2}}{\big(\mathbb{E}_\rho(a\cdot X)^4\big)^{1/2}} \geq \frac{\|a\|_2^3}{\sqrt{3}\,\|a\|_2^2} = \frac{\|a\|_2}{\sqrt{3}}, \tag{29}$$

where the second inequality uses (26). The bound also holds for $a = 0$.

Consider the set $\mathcal{K} := \big\{\mathbb{E}_\rho[X\psi(X)] : \psi : \{-1,1\}^d \to [-1,1]\big\} \subset \mathbb{R}^d$, which is clearly convex and compact. We claim that $\mathcal{K}$ contains every $v \in \mathbb{R}^d$ with $\|v\|_2 \leq 1/\sqrt{3}$. If some such v were

not in $\mathcal{K}$, then, by the Hahn–Banach separation theorem, there would exist $a \in \mathbb{R}^d$ such that $a \cdot v > \sup_{w \in \mathcal{K}} a \cdot w$. But the choice $\psi = \operatorname{sgn}(a \cdot X)$ produces the point $w := \mathbb{E}_\rho[X \operatorname{sgn}(a \cdot X)]$ of $\mathcal{K}$, and for it

$$a \cdot w = \mathbb{E}_\rho\big[(a \cdot X)\operatorname{sgn}(a \cdot X)\big] = \mathbb{E}_\rho|a \cdot X| \geq \frac{\|a\|_2}{\sqrt{3}} \geq \|a\|_2\|v\|_2 \geq a \cdot v,$$

by (29) and the Cauchy–Schwarz inequality, a contradiction. The vector $v := (1, \dots, 1)/\sqrt{3d}$ satisfies $\|v\|_2 = 1/\sqrt{3}$, so $v \in \mathcal{K}$. Thus there is ψ with values in $[-1, 1]$ such that $\mathbb{E}_\rho[X_j\psi(X)] = 1/\sqrt{3d}$ for every j. Replacing ψ by its odd part $x \mapsto (\psi(x) - \psi(-x))/2$, which again takes values in $[-1, 1]$, leaves these expectations unchanged, because X and $-X$ have the same law. □

Remark 5.3. The convex set $\mathcal{K}$ used in the proof is a scalar multiple of a zonoid [44, Section 2]. A zonoid is a convex set associated with a probability distribution, obtained by collecting weighted averages of the values of a random vector. For a random vector X with law ρ, it is defined by $\mathcal{Z}(\rho) = \big\{\mathbb{E}_\rho[Xg(X)] : g : \mathbb{R}^d \to [0, 1] \text{ measurable}\big\}$. In our proof, $\mathcal{K}$ collects all vectors of correlations with the coordinates that can be obtained from a single function ψ taking values in $[-1, 1]$. Since $\mathbb{E}_\rho X = 0$, the substitution $\psi = 2g - 1$ gives $\mathcal{K} = 2\mathcal{Z}(\rho)$.

The following lemma introduces dependence between factors while preserving every marginal on at most $L - 1$ factors. It uses a product of centred functions to perturb a product measure, a construction used in the Sarmanov family of distributions [65, 46].

Lemma 5.4. *Let $\rho_1, \dots, \rho_r$ be probability measures, with $r \geq L$, and let ψ_i satisfy $|\psi_i| \leq 1$ and $\rho_i(\psi_i) = 0$ for $1 \leq i \leq L$. Then*

$$d\rho = \left(1 - \frac{1}{2}\prod_{i=1}^{L}\psi_i(x_i)\right) d(\rho_1 \otimes \cdots \otimes \rho_r)$$

defines a probability measure with the same null sets as the product measure. For bounded measurable F_i,

$$\rho\left(\prod_{i=1}^{r} F_i(x_i)\right) = \prod_{i=1}^{r}\rho_i(F_i) - \frac{1}{2}\prod_{i=1}^{L}\rho_i(F_i\psi_i)\prod_{i=L+1}^{r}\rho_i(F_i). \tag{30}$$

Any collection of at most $L - 1$ coordinates is independent under ρ, with the original marginal laws. That is,

$$\rho(x_i \in A_i \text{ for all } i \in J) = \prod_{i \in J}\rho_i(A_i)$$

for every nonempty $J \subset \{1, \dots, r\}$ with $|J| \leq L - 1$ and measurable sets A_i.

Proof. Write $\rho_0 := \rho_1 \otimes \cdots \otimes \rho_r$. Since $|\psi_i| \leq 1$, the proposed density lies between $1/2$ and $3/2$. Since $\rho_i(\psi_i) = 0$ for each $1 \leq i \leq L$ we obtain that

$$\int \left(1 - \frac{1}{2}\prod_{i=1}^{L}\psi_i(x_i)\right) d\rho_0 = 1 - \frac{1}{2}\prod_{i=1}^{L}\rho_i(\psi_i) = 1.$$

Thus ρ is a probability measure, and the positive lower bound on the density gives equivalence with ρ_0.

To obtain (30), observe that Fubini's theorem gives

$$\int \prod_{i=1}^{r} F_i(x_i)\, d\rho_0 = \prod_{i=1}^{r}\rho_i(F_i) \text{ and } \int \left(\prod_{i=1}^{r} F_i(x_i)\right)\left(\prod_{i=1}^{L}\psi_i(x_i)\right) d\rho_0 = \prod_{i=1}^{L}\rho_i(F_i\psi_i)\prod_{i=L+1}^{r}\rho_i(F_i).$$

Subtracting half the second term from the first, we obtain (30).

Fix a nonempty $J \subset \{1, \dots, r\}$ with $|J| \leq L - 1$, and choose measurable sets A_i for $i \in J$. In (30), take $F_i = \mathbf{1}_{A_i}$ for $i \in J$ and $F_i = 1$ otherwise. Since J contains at most $L - 1$ indices, there exists $s \in \{1, \dots, L\} \setminus J$. For such an index, $F_s = 1$, hence $\rho_s(F_s\psi_s) = \rho_s(\psi_s) = 0$. Thus the second term in (30) vanishes. The remaining term gives

$$\rho(x_i \in A_i \text{ for all } i \in J) = \prod_{i \in J}\rho_i(A_i),$$

□

5.2. **Recursive definition of the block distributions.** We now define the block laws recursively along the subdivision tree of each block.

Fix a realisation h of the nested marker sets $\mathcal{H}$ given by Proposition 4.2, and write $\mathcal{P}_k(h)$ for its level-k block partition.

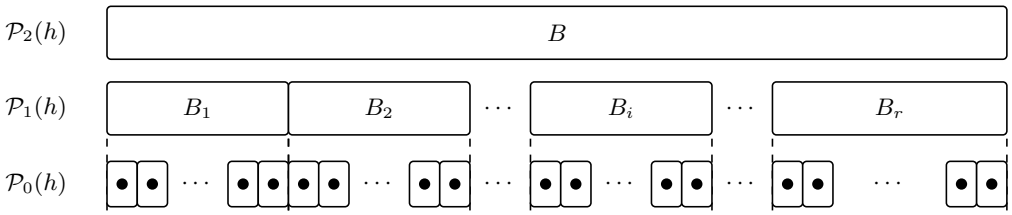
For each $B \in \mathcal{P}_k(h)$, consider the following tree. Its root is B. Each vertex $C \in \mathcal{P}_\ell(h)$ with $\ell \geq 1$ is joined to the blocks of $\mathcal{P}_{\ell-1}(h)$ contained in C. These are its children, ordered from left to right. The subdivision stops at level 0, where each block consists of a single index $j \in B$ (see Figure 5).

We now define, recursively over the levels, a probability measure ρ_B on the coordinates of every block $B \in \mathcal{P}_k(h)$. At level 0, every $B = \{j\} \in \mathcal{P}_0(h)$ receives the uniform law on $\{-1, 1\}$.

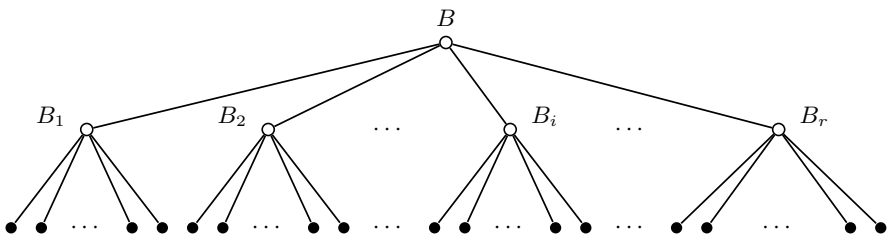
Suppose that the laws have been defined for all blocks in $\mathcal{P}_{k-1}(h)$. Let $B \in \mathcal{P}_k(h)$ have consecutive children $B_1, \dots, B_r \in \mathcal{P}_{k-1}(h)$. For each $1 \leq i \leq r$, let $\psi_i := \psi_{\rho_{B_i}}$ be a function given by Lemma 5.2. Define

$$\rho_B(dx) := \left(1 - \frac{1}{2}\prod_{i=1}^{L} \psi_i(x[B_i])\right) \bigotimes_{i=1}^{r} \rho_{B_i}(dx[B_i]). \tag{31}$$

There are at least L children by (H2), and the density in (31) lies between $1/2$ and $3/2$.



At level 0, the blocks inside B are the singletons $\{j\}$ with $j \in B$, ordered from left to right.



At each leaf, $\rho_{\{j\}} = \frac{1}{2}(\delta_{-1} + \delta_1)$.

Figure 5. A level-2 block B, its children $B_1, \dots, B_r$, and its singleton leaves at level 0, shown as nested intervals above and as a tree below.

The construction of ρ_B depends only on this tree, including the left-to-right order of its children. There are only countably many finite trees of this form. We fix, once and for all, one admissible function ψ_ρ from Lemma 5.2 for each law ρ arising in the construction, and use the same choice whenever that law occurs again. Thus the choice of ψ_{ρ_B} introduces no additional randomness.

The recursion preserves symmetry and every marginal on at most $L-1$ children, and the coordinates remain $(L-1)$-tuplewise independent. These properties ensure that Lemma 5.2 remains applicable at every level and yield exact stabilisation of the finite-dimensional distributions.

Lemma 5.5. *For every $k \geq 0$ and every $B \in \mathcal{P}_k$, the law ρ_B satisfies (i)–(ii) below. For $k \geq 1$, write $B_1, \dots, B_r \in \mathcal{P}_{k-1}$ for its consecutive children. Then (iii) also holds.*

(i) *ρ_B is a probability measure on $\{-1, 1\}^B$ with full support, and*

$$\rho_B(\{x\}) = \rho_B(\{-x\}) \quad \text{for every } x \in \{-1, 1\}^B.$$

(ii) *For every $1 \leq m \leq L-1$, every distinct $j_1, \dots, j_m \in B$, and every $\varepsilon_1, \dots, \varepsilon_m \in \{-1, 1\}$,*

$$\rho_B\big(X_{j_1} = \varepsilon_1, \dots, X_{j_m} = \varepsilon_m\big) = 2^{-m}.$$

(iii) *For every distinct* $i_1,\dots,i_m \in \{1,\dots,r\}$ *with* $1 \le m \le L-1$ *and every* $E_\ell \subset \{-1,1\}^{B_{i_\ell}}$,

$$\rho_B\big(x[B_{i_1}] \in E_1,\dots,x[B_{i_m}] \in E_m\big) = \prod_{\ell=1}^{m} \rho_{B_{i_\ell}}(E_\ell).$$

In particular, for every $1 \le m \le L-1$ *and every distinct* $j_1,\dots,j_m \in B$,

$$\mathbb{E}_{\rho_B}\left[\prod_{\ell=1}^{m} X_{j_\ell}\right] = 0.$$

Proof. We induct on the level of B. At level 0, each singleton block $B=\{p\}$ is assigned the law $\frac{1}{2}(\delta_{-1}+\delta_1)$, so (i)–(ii) hold. Now fix $k \ge 1$ and let $B \in \mathcal{P}_k$ have consecutive children $B_1,\dots,B_r \in \mathcal{P}_{k-1}$. Assume the conclusions for each child law ρ_{B_i}. Since $L-1 \ge 5$, their coordinates are four-tuplewise independent and symmetric. Together with central symmetry, this verifies the hypotheses of Lemma 5.2. Hence the functions ψ_i used in (31) are available and satisfy $\rho_{B_i}(\psi_i)=0$.

Apply Lemma 5.4 with $\rho_i = \rho_{B_i}$. It gives normalisation and the marginal identity (iii). Each child law has full support, so their product assigns positive mass to every configuration. The density is at least $1/2$, proving full support of ρ_B. Under $x \mapsto -x$ each child law is invariant, while

$$\prod_{i=1}^{L} \psi_i(-x[B_i]) = (-1)^L \prod_{i=1}^{L} \psi_i(x[B_i]) = \prod_{i=1}^{L} \psi_i(x[B_i]).$$

Thus the density is also invariant, proving central symmetry and completing (i).

For (ii), fix $E=\{j_1,\dots,j_m\}$ with $m \le L-1$. Let $B_{i_1},\dots,B_{i_s} \in \mathcal{P}_{k-1}$ be the children of $B \in \mathcal{P}_k$ intersecting E. Then $s \le m \le L-1$, so (iii) makes these child vectors independent. Within each child, the selected coordinates are independent and symmetric by induction. Therefore, for any prescribed values on E,

$$\rho_B(X_j = \varepsilon_j \text{ for } j \in E) = \prod_{a=1}^{s} 2^{-|E\cap B_{i_a}|} = 2^{-|E|}.$$

This is (ii). The expectation of a product of distinct selected coordinates is therefore the product of their zero means, proving the final assertion. □

Using Lemmas 5.4 and 5.2, the following lemma computes the L-point correlations across the children $B_i \in \mathcal{P}_{k-1}$ of a parent $B \in \mathcal{P}_k$. These correlations are nonpositive, so their negative contributions accumulate without cancellation. They yield the L-th moment deficit used in Corollary 8.4 to exclude Gaussian weak subsequential limits.

Lemma 5.6. *Let* $k \ge 1$, *let* $B \in \mathcal{P}_k$ *have consecutive children* $B_1,\dots,B_r \in \mathcal{P}_{k-1}$, *and let* $X=(X_j)_{j\in B}$ *have law* ρ_B. *Let* $j_1 < \cdots < j_L$ *be indices in* B. *Assume that some child* $B_i \in \mathcal{P}_{k-1}$ *contains at least one, but not all, of these indices, that is,*

$$1 \le \big|B_i \cap \{j_1,\dots,j_L\}\big| \le L-1 \quad \text{for some } i \in \{1,\dots,r\}.$$

Then

$$\mathbb{E}_{\rho_B}\left[\prod_{\ell=1}^{L} X_{j_\ell}\right] = \begin{cases} -\dfrac{1}{2\,3^{L/2}\sqrt{|B_1|\cdots|B_L|}}, & \text{if } |B_i \cap \{j_1,\dots,j_L\}| = 1 \text{ for every } 1 \le i \le L, \\ 0, & \text{otherwise.} \end{cases} \tag{32}$$

Proof. Recall that $B \in \mathcal{P}_k$ has consecutive children $B_1,\dots,B_r \in \mathcal{P}_{k-1}$, ordered from left to right. Put $E := \{j_1,\dots,j_L\}$ and $F_i := \prod_{j\in E\cap B_i} X_j$, interpreting an empty product as one. Lemma 5.4 gives

$$\mathbb{E}_{\rho_B}\prod_{\ell=1}^{L} X_{j_\ell} = \prod_{i=1}^{r} \rho_{B_i}(F_i) - \frac{1}{2}\prod_{i=1}^{L} \rho_{B_i}(F_i\psi_i) \prod_{i=L+1}^{r} \rho_{B_i}(F_i). \tag{33}$$

By hypothesis some child $B_s \in \mathcal{P}_{k-1}$ contains between 1 and $L-1$ selected indices. Lemma 5.5 (ii) makes their coordinates independent with zero means, so

$$\rho_{B_s}(F_s) = \prod_{j\in E\cap B_s} \rho_{B_s}(X_j) = 0.$$

Hence the independent term in (33) vanishes. Hence,

$$\mathbb{E}_{\rho_B}\prod_{\ell=1}^{L} X_{j_\ell} = -\frac{1}{2}\prod_{i=1}^{L}\rho_{B_i}(F_i\psi_i)\prod_{i=L+1}^{r}\rho_{B_i}(F_i).$$

If one of the first L children $B_i \in \mathcal{P}_{k-1}$ contains no selected coordinate, its factor in the correction is $\rho_{B_i}(\psi_i) = 0$, so the correlation is zero. In the remaining case, each of the blocks $B_1, \dots, B_L$ contains at least one of the indices $j_1, \dots, j_L$. There are only L selected coordinates, so each of those children contains exactly one and none lies in a later child. In particular,

$$E \subset B_1 \cup \cdots \cup B_L \quad \text{and} \quad |E \cap B_i| = 1 \quad \text{for every } 1 \le i \le L.$$

Since both the children and the selected indices are ordered, $j_i \in B_i$ for $1 \le i \le L$. Equation (27) then gives

$$\rho_{B_i}(F_i\psi_i) = \rho_{B_i}(X_{j_i}\psi_i) = \frac{1}{\sqrt{3|B_i|}}.$$

For every later child $B_i \in \mathcal{P}_{k-1}$ with $i > L$, the factor $\rho_{B_i}(F_i)$ equals one. Substituting into (33) therefore yields

$$\mathbb{E}_{\rho_B}\prod_{\ell=1}^{L} X_{j_\ell} = -\frac{1}{2}\prod_{i=1}^{L}\frac{1}{\sqrt{3|B_i|}} = -\frac{1}{2\,3^{L/2}\sqrt{|B_1|\cdots|B_L|}}.$$

This proves both cases of (32). □

Remark 5.7. The non-zero value in (32) depends only on the lengths of the first L children $B_1, \dots, B_L \in \mathcal{P}_{k-1}$, and not on the particular coordinate chosen in each child. Moreover, every L-point correlation covered by Lemma 5.6 is nonpositive. Thus, when these correlations are summed over a block, there is no cancellation between non-zero terms.

6. Construction of the stationary process

For fixed markers, we combine the level-k block laws independently. Exact stabilisation of their finite-dimensional marginals defines a limiting law. Averaging over the markers then gives the stationary process.

Related ideas appear in finitary coding. The markers and skeleta of del Junco [14, Sections 2–5] and the cells of Spinka's cell process [69, Sections 2 and 4] play a role analogous to that of our nested marker sets and interval partitions. Here we use nested interval partitions to construct a probability measure on $\{-1,1\}^{\mathbb{Z}}$ recursively. The preservation of marginals on at most $L-1$ blocks ensures consistency throughout the construction.

6.1. Finite-level processes and stabilisation of finite-dimensional distributions.

Fix a realisation h of $\mathcal{H} = (\mathcal{M}_k)_{k\ge 0}$, and write $\mathcal{P}_k(h)$ for its level-k block partition. For each $B \in \mathcal{P}_k(h)$ with $k \ge 1$, the law ρ_B was constructed from the laws of its consecutive children $B_1, \dots, B_r \in \mathcal{P}_{k-1}(h)$. At level 0, the law on each singleton block $\{p\} \in \mathcal{P}_0(h)$ is $\rho_{\{p\}} = \frac{1}{2}(\delta_{-1} + \delta_1)$. On $\Sigma = \{-1,1\}^{\mathbb{Z}}$, define

$$\nu_h^{(k)} := \bigotimes_{B\in\mathcal{P}_k(h)} \rho_B \quad \text{and} \quad \nu^{(k)} := \int \nu_h^{(k)}\,\mathbb{P}_{\mathcal{H}}(dh). \tag{34}$$

Write $Z^{(k)}$ for the coordinate process under $\nu^{(k)}$. For $m \in \mathbb{Z}$, denote by $h+m$ the configuration obtained by translating every marker set by m. Translation preserves the subdivision of each block into children at every level, including their left-to-right order. Thus the laws of the translated blocks are translates of the original laws. Hence,

$$\nu_{h+m}^{(k)} = (\sigma^{-m})_*\nu_h^{(k)}.$$

By Lemma 4.5, the nested marker sets are stationary, so

$$(\mathcal{M}_0 + m, \dots, \mathcal{M}_k + m) \overset{d}{=} (\mathcal{M}_0, \dots, \mathcal{M}_k) \quad \text{for } m \in \mathbb{Z}. \tag{35}$$

Averaging gives

$$(\sigma^{-m})_*\nu^{(k)} = \int \nu_{h+m}^{(k)}\,\mathbb{P}_{\mathcal{H}}(dh) = \nu^{(k)},$$

which proves stationarity.

The following lemma shows that each finite-level process is finitely dependent after averaging over the marker sets, with dependence range bounded by a fixed multiple of the block scale.

Lemma 6.1. *For every $k \geq 0$ and finite interval I, the marginal $\nu_h^{(k)}|_I$ depends only on the marker configuration through level k in $I^{+3(4L)^k}$. Moreover, $Z^{(k)}$ is $(D+6)(4L)^k$-dependent.*

Proof. For a realisation $h = (M_j)_{j\geq 0}$ of $\mathcal{H}$, write

$$h_i^{(k)} := \big(\mathbf{1}_{\{i\in M_0\}}, \dots, \mathbf{1}_{\{i\in M_k\}}\big) \quad \text{and} \quad h^{(k)} := (h_i^{(k)})_{i\in\mathbb{Z}}.$$

Every block $B \in \mathcal{P}_k(h)$ intersecting I has length at most $3(4L)^k$, including the singleton case $k = 0$. Both endpoints therefore lie in $I^{+3(4L)^k}$. The marker indicators through level k in this enlarged interval determine those endpoints and all lower-level markers between them. Thus they determine all subdivisions of each block intersecting I, and hence its law. Since

$$\nu_h^{(k)}|_I = \bigotimes_{\substack{B\in\mathcal{P}_k(h)\\ B\cap I\neq\varnothing}} \rho_B|_{B\cap I},$$

they also determine the marginal on I.

Now suppose $\operatorname{dist}(I,J) > (D+6)(4L)^k$. No block $B \in \mathcal{P}_k(h)$ can intersect both intervals, because its length would exceed $3(4L)^k$. Under the joint law $\mathbb{P}_{\mathcal{H}}(dh)\nu_h^{(k)}(d\omega)$, the coordinate vectors on I and J are therefore independent conditionally on $\mathcal{H}$. Their respective conditional laws depend only on $\mathcal{H}^{(k)}[I^{+3(4L)^k}]$ and $\mathcal{H}^{(k)}[J^{+3(4L)^k}]$, by the locality just proved. These local configurations are independent by (H4), since

$$\operatorname{dist}\big(I^{+3(4L)^k}, J^{+3(4L)^k}\big) \geq \operatorname{dist}(I,J) - 6(4L)^k > D(4L)^k.$$

These are precisely the hypotheses of Lemma 4.7, with the two local marker configurations as the H_i and the coordinate vectors as the X_i. It gives independence under the averaged law $\nu^{(k)}$. For bounded functions F, G of the coordinates on I, J, respectively, the factorisation reads

$$\nu^{(k)}(FG) = \int \nu_h^{(k)}(F)\nu_h^{(k)}(G)\,\mathbb{P}_{\mathcal{H}}(dh) = \left(\int \nu_h^{(k)}(F)\,\mathbb{P}_{\mathcal{H}}(dh)\right)\left(\int \nu_h^{(k)}(G)\,\mathbb{P}_{\mathcal{H}}(dh)\right)$$
$$= \nu^{(k)}(F)\nu^{(k)}(G).$$

The extension to arbitrary index sets follows by the same argument as in the proof of Proposition 4.8. □

In particular, restrictions to any finite family of ordered intervals with successive distances greater than $(D+6)(4L)^k$ are jointly independent.

For fixed markers, the joint law on at most $L-1$ level-k blocks is unchanged at all higher levels, regardless of their separation. This preservation allows us to define the limiting law and identify its finite-dimensional distributions with those of a finite-level process.

Lemma 6.2. *Fix a realisation h of the nested marker sets $\mathcal{H}$ and integers $\ell \geq k \geq 0$. Suppose that $E \subset \mathbb{Z}$ intersects at most $L-1$ level-k blocks, that is,*

$$\#\{B \in \mathcal{P}_k(h) : E \cap B \neq \varnothing\} \leq L-1.$$

Then

$$\nu_h^{(\ell)}\big|_E = \nu_h^{(k)}\big|_E = \bigotimes_{\substack{B\in\mathcal{P}_k(h)\\ E\cap B\neq\varnothing}} \rho_B\big|_{E\cap B}. \tag{36}$$

Proof. We prove by induction on $\ell - k$ that the marginal on any $t \leq L-1$ distinct level-k blocks $B_1, \dots, B_t$ is $\bigotimes_{i=1}^t \rho_{B_i}$. For $\ell = k$, this is the definition of $\nu_h^{(k)}$.

Suppose the assertion holds for $\ell-1$, where $\ell > k$. First consider the selected blocks contained in one level-ℓ block B. Let $C_1, \dots, C_m \in \mathcal{P}_{\ell-1}(h)$ be the distinct children of B containing them. Since $m \leq t \leq L-1$, Lemma 5.5(iii) gives

$$\rho_B\big|_{C_1\cup\cdots\cup C_m} = \bigotimes_{a=1}^m \rho_{C_a}.$$

Within each C_a, the induction hypothesis applies to the selected level-k blocks it contains. Therefore

$$\rho_B\big|_{B_1\cup\cdots\cup B_t} = \bigotimes_{a=1}^{m}\left(\bigotimes_{\{i:B_i\subset C_a\}}\rho_{B_i}\right) = \bigotimes_{i=1}^{t}\rho_{B_i}.$$

If the selected blocks belong to different level-ℓ blocks, apply the same calculation in each of those blocks and use their independence under $\nu_h^{(\ell)}$. This completes the induction.

Now take the level-k blocks intersecting E. There are at most $L-1$ of them by hypothesis, so the identity just proved applies to their union. Taking the marginal on E gives (36). □

Two cases are used below. Since every level-k block has length at least $(4L)^k$, by (H1) for $k \geq 1$ and by the singleton construction for $k = 0$, a set contained in an interval of length at most $(4L)^k$ intersects at most two level-k blocks, and a union of $s \leq R$ such intervals intersects at most $2s \leq 2R = L-2$ of them. Lemma 6.2 applies to each fixed marker configuration. Since the block lengths and subdivisions are random, conditional independence does not imply independence after averaging over the markers. Lemma 6.1 establishes independence after averaging for coordinate vectors on sufficiently separated intervals.

6.2. **Existence and stationarity of the limiting process.** The following proposition constructs the stationary process by averaging the limiting laws over the marker configurations. It establishes full support and $(L-1)$-tuplewise independence. For each fixed marker configuration h, the coordinate vector on every block $B \in \mathcal{P}_k(h)$ has law ρ_B under ν_h.

Proposition 6.3. *There exists a family $(\nu_h)_h$ of Borel probability measures on $\Sigma = \{-1,1\}^{\mathbb{Z}}$ such that $h \mapsto \nu_h(A)$ is measurable for every Borel set $A \subset \Sigma$. Define the averaged probability measure on Σ by*

$$\nu(A) := \int \nu_h(A)\,\mathbb{P}_{\mathcal{H}}(dh), \quad \text{for } A \in \mathcal{B}(\Sigma).$$

Then the following hold.

(i) *For every finite set $E \subset \mathbb{Z}$, there exists an integer $k_E \geq 0$, depending only on E and not on h, such that, for $\mathbb{P}_{\mathcal{H}}$-almost every h,*

$$\nu_h^{(\ell)}\big|_E = \nu_h\big|_E \quad \text{for every } \ell \geq k_E.$$

Thus the joint distribution of the coordinates indexed by E is unchanged after level k_E and equals their joint distribution under ν_h.

(ii) *The measure ν is σ-invariant and has full support. More precisely, $\nu(\sigma^{-1}A) = \nu(A)$ for every Borel set $A \subset \Sigma$, and $\nu(\omega_i = \varepsilon_i \text{ for every } i \in E) > 0$ for every finite $E \subset \mathbb{Z}$ and every choice $(\varepsilon_i)_{i\in E} \in \{-1,1\}^E$.*

(iii) *Let $Z_i(\omega) := \omega_i$. For every $1 \leq m \leq L-1$, distinct indices $j_1,\dots,j_m \in \mathbb{Z}$, and values $\varepsilon_1,\dots,\varepsilon_m \in \{-1,1\}$, one has, for $\mathbb{P}_{\mathcal{H}}$-almost every h,*

$$\nu_h(Z_{j_1} = \varepsilon_1,\dots,Z_{j_m} = \varepsilon_m) = 2^{-m}.$$

Therefore, $\nu(Z_{j_1} = \varepsilon_1,\dots,Z_{j_m} = \varepsilon_m) = 2^{-m}$. Hence the coordinates are $(L-1)$-tuplewise independent under both the conditional laws ν_h and the averaged law ν, and take the values -1 and 1 with probability $1/2$ each.

(iv) *For $\mathbb{P}_{\mathcal{H}}$-almost every h, every $k \geq 0$ and every block $B \in \mathcal{P}_k(h)$, $\nu_h\big|_B = \rho_B$, where ρ_B is the block law constructed in Section 5, using the uniform law on $\{-1,1\}$ at level 0 and the recursion (31) at higher levels.*

Proof. We divide the proof into four steps. We work with realisations h of $\mathcal{H} = (\mathcal{M}_k)_{k\geq 0}$ for which $\mathcal{M}_0 = \mathbb{Z}$, the marker sets are nested and bi-infinite, and the bounds (H1)–(H2) hold at every level. By Proposition 4.2, these properties hold $\mathbb{P}_{\mathcal{H}}$-almost surely and define a measurable set of marker configurations.

Step 1. *We construct ν_h, establish its measurable dependence on h, and prove (i).*

Proof of Step 1. Fix a nonempty finite set $E \subset \mathbb{Z}$, let $I = [\min E, \max E] \cap \mathbb{Z}$, and choose an integer $k_E \geq 1$ such that $|I| \leq (4L)^{k_E}$. This choice depends only on E, not on h. Every level-k_E block has length greater than $(4L)^{k_E}$ by (H1). Hence, I intersects at most two such blocks.

Otherwise it would contain a complete block between the first and last blocks it intersects, contradicting $|I| \le (4L)^{k_E}$. Hence

$$\#\{B \in \mathcal{P}_{k_E}(h) : B \cap E \neq \varnothing\} \le 2 \le L - 1.$$

Lemma 6.2 gives

$$\nu_h^{(\ell)}\big|_E = \nu_h^{(k_E)}\big|_E \quad \text{for every } \ell \ge k_E.$$

We take $\nu_h^{(k_E)}|_E$ as the proposed limiting marginal on $\{-1,1\}^E$. The identity above shows that choosing a larger level gives the same marginal.

These proposed marginals are consistent. Indeed, if $E \subset F$ are finite and nonempty, choose $\ell \ge \max\{k_E, k_F\}$. Then

$$\left(\nu_h^{(k_F)}\big|_F\right)\big|_E = \left(\nu_h^{(\ell)}\big|_F\right)\big|_E = \nu_h^{(\ell)}\big|_E = \nu_h^{(k_E)}\big|_E.$$

The Kolmogorov extension theorem [37, Theorem 8.23] therefore gives a unique Borel probability measure ν_h on Σ whose marginal on every finite nonempty E is $\nu_h^{(k_E)}|_E$. Thus

$$\nu_h^{(\ell)}\big|_E = \nu_h\big|_E \quad \text{for every } \ell \ge k_E,$$

proving (i).

We next verify measurability. Fix a cylinder event

$$A := \{\omega \in \Sigma : \omega_i = \varepsilon_i \text{ for every } i \in E\},$$

where E is finite and nonempty. By the construction of ν_h and the product definition of $\nu_h^{(k_E)}$,

$$\nu_h(A) = \nu_h^{(k_E)}(A) = \prod_{\substack{B \in \mathcal{P}_{k_E}(h) \\ B \cap E \neq \varnothing}} \rho_B(X_j = \varepsilon_j \text{ for every } j \in B \cap E).$$

The locality assertion in Lemma 6.1 shows that this quantity is determined by the finite configuration $h^{(k_E)}[I^{+3(4L)^{k_E}}]$. That configuration consists of finitely many measurable marker indicators and takes only finitely many values. Therefore $h \mapsto \nu_h(A)$ is measurable on the full-measure set under consideration.

Outside the measurable set of marker configurations specified at the start of the proof, define ν_h as the product of the uniform laws on $\{-1,1\}$. This choice on a null set does not change the averaged measure.

Since cylinder sets form a generating class for the Borel σ-algebra of Σ, the kernel criterion in [37, Lemma 3.1] implies that $h \mapsto \nu_h$ is a probability kernel. Hence

$$\nu(A) := \int \nu_h(A)\, \mathbb{P}_{\mathcal{H}}(dh), \quad \text{for } A \in \mathcal{B}(\Sigma),$$

defines a Borel probability measure on Σ. ■

Step 2. *We prove (ii). The coordinate process $(Z_i)_{i\in\mathbb{Z}}$ is stationary under ν, and ν has full support on Σ.*

Proof of Step 2. Let A be a cylinder event depending on coordinates in a finite nonempty set E, and fix $m \in \mathbb{Z}$. The event $\sigma^{-m}A$ depends on coordinates in $E+m$. Choose $\ell \ge \max\{k_E, k_{E+m}\}$. By Step 1, the conditional probabilities of both events agree with their level-ℓ probabilities for almost every h. Integrating and using (34) gives

$$\nu(A) = \nu^{(\ell)}(A) \quad \text{and} \quad \nu(\sigma^{-m}A) = \nu^{(\ell)}(\sigma^{-m}A).$$

The coordinate process $Z^{(\ell)}$ is stationary under $\nu^{(\ell)}$, as proved after (35). Hence

$$\nu(\sigma^{-m}A) = \nu^{(\ell)}(\sigma^{-m}A) = \nu^{(\ell)}(A) = \nu(A).$$

The probability measures $(\sigma^m)_*\nu$ and ν agree on cylinder events. Since these form a π-system generating $\mathcal{B}(\Sigma)$, the two measures agree on every Borel set [37, Lemma 1.18]. This proves stationarity.

For full support, fix a nonempty cylinder event $A = \{\omega : \omega_i = \varepsilon_i \text{ for every } i \in E\}$. For $\mathbb{P}_{\mathcal{H}}$-almost every realisation h of $\mathcal{H}$, the probability $\nu_h(A)$ is the finite product displayed in Step 1. Each factor is positive because Lemma 5.5(i) gives full support of ρ_B, and any prescription of

coordinate values on $B \cap E$ extends to a coordinate configuration on B. Thus $\nu_h(A) > 0$ for $\mathbb{P}_{\mathcal{H}}$-almost every h, and

$$\nu(A) = \int \nu_h(A)\, \mathbb{P}_{\mathcal{H}}(dh) > 0.$$

Every nonempty open subset of Σ contains a nonempty cylinder event, so supp $\nu = \Sigma$. ■

Step 3. *We prove (iii). For any distinct indices $j_1, \dots, j_m$, with $1 \le m \le L-1$, the variables $Z_{j_1}, \dots, Z_{j_m}$ are independent and have mean zero. This holds with respect to ν and, for $\mathbb{P}_{\mathcal{H}}$-almost every realisation h of $\mathcal{H}$, with respect to ν_h.*

Proof of Step 3. Fix $1 \le m \le L-1$, distinct indices $j_1, \dots, j_m \in \mathbb{Z}$, and values $\varepsilon_1, \dots, \varepsilon_m \in \{-1, 1\}$. Put $E = \{j_1, \dots, j_m\}$. Since the level-0 blocks are singletons, E intersects exactly $m \le L-1$ of them. Lemma 6.2, applied with $k = 0$, therefore gives

$$\nu_h^{(\ell)}\big|_E = \nu_h^{(0)}\big|_E = \bigotimes_{i=1}^{m} \rho_{\{j_i\}} \quad \text{for every } \ell \ge 0.$$

For $\ell \ge k_E$, Step 1 identifies the left-hand side with $\nu_h|_E$. Each singleton law is the uniform law on $\{-1, 1\}$, so

$$\nu_h(Z_{j_1} = \varepsilon_1, \dots, Z_{j_m} = \varepsilon_m) = \prod_{i=1}^{m} \rho_{\{j_i\}}(X_{j_i} = \varepsilon_i) = 2^{-m}.$$

Integrating this identity over h yields

$$\nu(Z_{j_1} = \varepsilon_1, \dots, Z_{j_m} = \varepsilon_m) = \int 2^{-m}\, \mathbb{P}_{\mathcal{H}}(dh) = 2^{-m}.$$

These are precisely the joint distributions asserted in (iii). ■

Step 4. *We prove (iv). For every $k \ge 0$, every block $B \in \mathcal{P}_k(h)$ and every $\varepsilon = (\varepsilon_i)_{i \in B} \in \{-1, 1\}^B$,*

$$\nu_h(Z_i = \varepsilon_i \text{ for every } i \in B) = \rho_B(\{\varepsilon\}).$$

Proof of Step 4. Fix h, an integer $k \ge 0$, and a block $B \in \mathcal{P}_k(h)$. The set B intersects only one level-k block, namely itself. Lemma 6.2, applied with $E = B$, gives

$$\nu_h^{(\ell)}\big|_B = \nu_h^{(k)}\big|_B = \rho_B \quad \text{for every } \ell \ge k,$$

where the last equality follows from the product definition of $\nu_h^{(k)}$. Since B is finite, from Step 1 we obtain that there exists k_B such that $\nu_h^{(\ell)}|_B = \nu_h|_B$ for every $\ell \ge k_B$. Taking $\ell \ge \max\{k, k_B\}$ proves

$$\nu_h\big|_B = \nu_h^{(\ell)}\big|_B = \rho_B.$$

■

This completes the proof. □

We now consider $\mathcal{H}$ and Z on the same probability space, with joint probability measure

$$\mathbb{P}(dh, d\omega) := \mathbb{P}_{\mathcal{H}}(dh)\, \nu_h(d\omega).$$

The process $Z = (Z_i)_{i \in \mathbb{Z}}$ has law ν. Knowing $\mathcal{H}$ determines the blocks, but the coordinates Z_i remain random. For a realisation h of $\mathcal{H}$, their law is ν_h. Thus the conditional expectation (see [37, Chapter 8] for a review) is

$$\mathbb{E}[F(Z) \mid \mathcal{H}] = \int_{\Sigma} F(\omega)\, \nu_{\mathcal{H}}(d\omega) \quad \text{almost surely,}$$

for every bounded Borel function $F : \Sigma \to \mathbb{R}$. Taking $F = \mathbf{1}_A$ gives the probability that $Z \in A$ when the marker sets are known, namely

$$\mathbb{P}(Z \in A \mid \mathcal{H}) = \mathbb{E}[\mathbf{1}_A(Z) \mid \mathcal{H}] = \nu_{\mathcal{H}}(A) \quad \text{almost surely,}$$

for every Borel set $A \subset \Sigma$.

The following lemma is the counterpart of Lemma 5.6 for the limiting process $(Z_i)_{i \in \mathbb{Z}}$, transferring the correlation formula for the block coordinates X_i to this process. The L-point correlations are nonpositive, so their negative contributions add without cancellation and are used to prove failure of the CLT for $(Z_i)_{i \in \mathbb{Z}}$.

Lemma 6.4. *For $\mathbb{P}_{\mathcal{H}}$-almost every realisation h of $\mathcal{H}$, the following statements hold for every $j_1 < \cdots < j_L$ in $\mathbb{Z}$. Write $E := \{j_1, \dots, j_L\}$.*

Suppose that $E \subset B$ for some block $B \in \mathcal{P}_k(h)$. Choose k minimal, and write $B_1, \dots, B_r \in \mathcal{P}_{k-1}(h)$ for the children of the unique block $B \in \mathcal{P}_k(h)$ containing E, ordered from left to right. Then

$$\int_\Sigma \prod_{i=1}^{L} \omega_{j_i}\, \nu_h(d\omega) = \begin{cases} -\dfrac{1}{2\,3^{L/2}\sqrt{|B_1|\cdots|B_L|}}, & \text{if } |E \cap B_i| = 1 \text{ for every } 1 \le i \le L, \\ 0, & \text{otherwise.} \end{cases} \tag{37}$$

If no block at any level contains E, the integral is zero.

In particular, for every $j_1 < \cdots < j_L$,

$$\mathbb{E}\left[\prod_{i=1}^{L} Z_{j_i} \,\middle|\, \mathcal{H}\right] \le 0 \quad \text{almost surely, and } \mathbb{E}\left[\prod_{i=1}^{L} Z_{j_i}\right] \le 0.$$

Proof. Fix a realisation h of $\mathcal{H}$ for which Proposition 6.3 holds and put $E = \{j_1, \dots, j_L\}$. Suppose first that a block contains E, and choose the smallest such level k. The block $B \in \mathcal{P}_k(h)$ is unique because $\mathcal{P}_k(h)$ is a partition. Moreover, $k \ge 1$, since a singleton cannot contain L distinct indices. Write $B_1, \dots, B_r \in \mathcal{P}_{k-1}(h)$ for the children of $B \in \mathcal{P}_k(h)$, ordered from left to right. The set E intersects at least two children of B. Otherwise one child $B_i \in \mathcal{P}_{k-1}(h)$ would contain it, contradicting minimality.

Proposition 6.3(iv) identifies the marginal on B with ρ_B. Thus

$$\int_\Sigma \prod_{i=1}^{L} \omega_{j_i}\, \nu_h(d\omega) = \mathbb{E}_{\rho_B} \prod_{i=1}^{L} X_{j_i}.$$

The selected coordinates intersect at least two children, so Lemma 5.6 applies and gives (37).

Now suppose that no block contains E. At each level ℓ, the blocks $C \in \mathcal{P}_\ell(h)$ intersecting E satisfy $1 \le |C \cap E| \le L - 1$. By Lemma 5.5(ii), the selected coordinates within each such block are independent and centred, so

$$\mathbb{E}_{\rho_C} \prod_{j \in C \cap E} X_j = \prod_{j \in C \cap E} \mathbb{E}_{\rho_C} X_j = 0.$$

Since the blocks are independent under $\nu_h^{(\ell)}$,

$$\int_\Sigma \prod_{j \in E} \omega_j\, \nu_h^{(\ell)}(d\omega) = \prod_{\substack{C \in \mathcal{P}_\ell(h) \\ C \cap E \neq \varnothing}} \mathbb{E}_{\rho_C} \prod_{j \in C \cap E} X_j = 0.$$

For $\ell \ge k_E$, the marginal on E agrees with that of ν_h, so the same integral under ν_h is zero.

In both cases the integral under ν_h is nonpositive. The conditional-expectation identity above therefore gives $\mathbb{E}[\prod_i Z_{j_i} \mid \mathcal{H}] \le 0$ almost surely, and

$$\mathbb{E}\prod_{i=1}^{L} Z_{j_i} = \int \left(\int_\Sigma \prod_{i=1}^{L} \omega_{j_i}\, \nu_h(d\omega) \right) \mathbb{P}_{\mathcal{H}}(dh) \le 0.$$

There are countably many increasing integer tuples. Intersecting their probability-one sets makes all these assertions simultaneous. □

The simultaneous assertion allows (37) to be applied to a block selected as a function of the markers.

7. Exponential mixing of order R

We study the process $Z = (Z_i)_{i \in \mathbb{Z}}$ of Proposition 6.3, with law ν, using the notation introduced before Theorem C. The finite-level dependence of Lemma 6.1 gives joint independence of the coordinate vectors on at most R intervals separated by a fixed multiple of their lengths (Proposition 7.1). This implies that σ is mixing with respect to ν (Corollary 3.2) and yields exponential mixing of order R for regular observables (Proposition 7.2).

Proposition 7.1. *Let $(Z_i)_{i\in\mathbb{Z}}$ be the process given in Proposition 6.3 and define $K := 4L(D+7)$, where D is the constant in Proposition 4.2. Let $2 \le s \le R$ and let $I_1, \dots, I_s$ be ordered disjoint nonempty intervals. If*

$$\min_{i<s} \operatorname{dist}(I_i, I_{i+1}) > K \max_i |I_i|, \tag{38}$$

then $Z[I_1], \dots, Z[I_s]$ are jointly independent.

Proof. Put $m := \max_i |I_i|$ and choose the least $k \ge 0$ with $(4L)^k \ge m$. Minimality gives

$$m \le (4L)^k \le 4Lm.$$

The same inequalities hold when $k = 0$, in which case $m = 1$. For $k \ge 1$, each interval intersects at most two blocks of $\mathcal{P}_k(h)$ by their length lower bound. For $k = 0$, each window is a singleton. Hence $E := I_1 \cup \cdots \cup I_s$ intersects at most $2s \le 2R = L-2$ blocks. Lemma 6.2 gives $\nu_h^{(\ell)}|_E = \nu_h^{(k)}|_E$ for every $\ell \ge k$. Taking ℓ large enough to apply Proposition 6.3(i), and then averaging, we obtain

$$\nu_h|_E = \nu_h^{(k)}|_E \quad \text{and} \quad \nu|_E = \nu^{(k)}|_E.$$

On the other hand, (38) implies

$$\min_{i<s} \operatorname{dist}(I_i, I_{i+1}) > 4L(D+7)m \ge (D+7)(4L)^k > (D+6)(4L)^k.$$

Lemma 6.1 makes the finite-level restrictions to these windows jointly independent. For events $A_i \subset \{-1, 1\}^{I_i}$, equality of the joint laws on E and of their marginals gives

$$\begin{aligned}\nu(Z[I_i] \in A_i \text{ for all } i) &= \nu^{(k)}(Z^{(k)}[I_i] \in A_i \text{ for all } i) \\ &= \prod_{i=1}^{s} \nu^{(k)}(Z^{(k)}[I_i] \in A_i) = \prod_{i=1}^{s} \nu(Z[I_i] \in A_i).\end{aligned}$$

This is the required independence. □

To pass from window independence to exponential mixing, we approximate observables in $\mathcal{F}_\theta(\Sigma)$ by cylinder functions with window radius proportional to the smallest time gap. The cylinder functions are independent and the approximation error decays exponentially, giving the bounds needed for the horseshoe realisations.

Proposition 7.2. *For every $0 < \theta < 1$, the measure ν satisfies*

$$\left| \nu\left(\prod_{i=1}^{R} F_i \circ \sigma^{n_i} \right) - \prod_{i=1}^{R} \nu(F_i) \right| \le C_{R,\theta}\, e^{-c_\theta \Delta(\mathbf{n})} \prod_{i=1}^{R} \|F_i\|_\theta \quad \text{where} \quad c_\theta := \frac{|\log \theta|}{4(K+1)}.$$

Proof. Set $s := R$ and fix $F_1, \dots, F_s \in \mathcal{F}_\theta(\Sigma)$ and the observation times. For $\ell \ge 0$, let $\omega^{(\ell)}$ agree with ω on $[-\ell, \ell]$ and equal 1 elsewhere, and put $F_{i,\ell}(\omega) := F_i(\omega^{(\ell)})$. This function depends only on the coordinates in that window. Since $\omega^{(\ell)} \in \Sigma$ and $d_\theta(\omega, \omega^{(\ell)}) \le \theta^{\ell+1}$,

$$\|F_{i,\ell}\|_\infty \le \|F_i\|_\infty \quad \text{and} \quad \|F_i - F_{i,\ell}\|_\infty \le \theta^{\ell+1} \|F_i\|_\theta. \tag{39}$$

Write $\Delta := \Delta(\mathbf{n})$. For $\Delta > 2(K+1)$, choose $\ell = \lfloor \Delta/(4(K+1)) \rfloor$. The function $F_{i,\ell} \circ \sigma^{n_i}$ depends on $I_i = [n_i - \ell, n_i + \ell] \cap \mathbb{Z}$. These windows have length $2\ell + 1$ and successive distances at least $\Delta - 2\ell$. The choice of ℓ gives

$$\Delta - 2\ell - K(2\ell + 1) = \Delta - 2(K+1)\ell - K \ge \Delta/2 - K > 0.$$

Thus Proposition 7.1 applies. Independence and stationarity yield

$$\nu\left(\prod_{i=1}^{s} F_{i,\ell} \circ \sigma^{n_i} \right) = \prod_{i=1}^{s} \nu(F_{i,\ell}).$$

We now estimate the two approximation errors. For any $a_1, \dots, a_s, b_1, \dots, b_s$, the telescoping identity is

$$\prod_{i=1}^{s} a_i - \prod_{i=1}^{s} b_i = \sum_{i=1}^{s} \left(\prod_{j<i} a_j \right) (a_i - b_i) \left(\prod_{j>i} b_j \right).$$

Apply it pointwise with $a_i = F_i \circ \sigma^{n_i}$ and $b_i = F_{i,\ell} \circ \sigma^{n_i}$. Using (39), we obtain

$$\left|\nu\left(\prod_i F_i \circ \sigma^{n_i}\right) - \nu\left(\prod_i F_{i,\ell} \circ \sigma^{n_i}\right)\right| \le \sum_{i=1}^{s} \|F_i - F_{i,\ell}\|_\infty \prod_{j\neq i} \|F_j\|_\infty \le s\theta^{\ell+1} \prod_{i=1}^{s} \|F_i\|_\theta.$$

For the product of expectations, use the same identity with $a_i = \nu(F_i)$ and $b_i = \nu(F_{i,\ell})$. Here $|a_i|, |b_i| \le \|F_i\|_\infty$ and $|a_i - b_i| \le \|F_i - F_{i,\ell}\|_\infty$, so

$$\left|\prod_i \nu(F_i) - \prod_i \nu(F_{i,\ell})\right| \le s\theta^{\ell+1} \prod_i \|F_i\|_\theta.$$

The cylinder factorisation and triangle inequality now give

$$\left|\nu\left(\prod_i F_i \circ \sigma^{n_i}\right) - \prod_i \nu(F_i)\right| \le 2s\theta^{\ell+1} \prod_i \|F_i\|_\theta \le 2se^{-c_\theta \Delta} \prod_i \|F_i\|_\theta,$$

where $\ell + 1 > \Delta/(4(K+1))$ gives $c_\theta = |\log\theta|/(4(K+1))$. For $\Delta \le 2(K+1)$, the left side is at most $2\prod_i \|F_i\|_\theta$, so all cases are covered by $C_{s,\theta} := \max\{2s, 2e^{2c_\theta(K+1)}\}$. □

8. Failure of the CLT and of higher-order exponential mixing

Recall that $(Z_i)_{i\in\mathbb{Z}}$ is the stationary process of Proposition 6.3, with $Z_i(\omega) = \omega_i$ under ν. We use $S_n = \sum_{i=0}^{n-1} Z_i$ and $L = 2R+2$. Proposition 8.1 finds a contribution of order $n^{L/2}$ to the sum of negative L-point correlations, using tuples whose successive gaps are proportional to n. This gives both the moment deficit excluding Gaussian weak subsequential limits and a polynomial lower bound for a separated correlation, excluding exponential mixing of order L.

Proposition 8.1. *Let $n \ge 6(4L)^2$ and let $k \ge 2$ be the integer with*

$$6(4L)^k \le n < 6(4L)^{k+1}. \tag{40}$$

Then, writing $g_n := (4L)^{k-1}/2$,

$$-\sum_{\substack{0\le j_1<\cdots<j_L<n \\ \Delta(\mathbf{j})\ge g_n}} \mathbb{E}\left[\prod_{i=1}^{L} Z_{j_i}\right] \ge \frac{(4L)^{(k-1)L/2}}{2^{L+1}3^{L/2}} \ge \frac{1}{2^{L+1}3^{L/2}} \left(\frac{n}{6(4L)^2}\right)^{L/2}, \tag{41}$$

and $g_n \ge n/(12(4L)^2)$.

Proof. The proof is divided into two steps.

Step 1. *Choose a complete block and separated subintervals of its children.*

Proof of Step 1. Let $A = [u, v) \cap \mathbb{Z} \in \mathcal{P}_k$ be the block containing 0. Since $u \le 0 < v$, (H1) gives $v \le v - u \le 3(4L)^k$. The next block $A' \in \mathcal{P}_k$ has length at most $3(4L)^k$, so its right endpoint is at most $6(4L)^k \le n$. Thus at least one complete level-k block lies in $[0, n)$. Choose the leftmost such block $B = B(\mathcal{H}) \in \mathcal{P}_k$. Its first L children are $B_1, \dots, B_L \in \mathcal{P}_{k-1}$, ordered from left to right. Write $B_i = [u_i, v_i) \cap \mathbb{Z}$. There are at least L children by (H2), and consecutive children satisfy $v_i = u_{i+1}$.

Set $a := (4L)^{k-1}/4$. This is an integer because $k \ge 2$ and $4L$ is divisible by 4. Define

$$C_i := [u_i + a, v_i - a) \cap \mathbb{Z}.$$

Since $|B_i| > (4L)^{k-1} = 4a$,

$$|C_i| = |B_i| - 2a \ge |B_i|/2 > 0.$$

If $j_i \in C_i$, then $u_i + a \le j_i \le v_i - a - 1$. Hence

$$j_{i+1} - j_i \ge (u_{i+1} + a) - (v_i - a - 1) = 2a + 1 = g_n + 1.$$

Every selected coordinate lies in $B \subset [0, n)$, so these tuples are increasing and have $\Delta(\mathbf{j}) \ge g_n$. The upper bound in (40) also gives

$$g_n = \frac{(4L)^{k-1}}{2} > \frac{n}{12(4L)^2}.$$

■

Step 2. *Sum the conditional correlations and average over the markers.*

Proof of Step 2. Fix a realisation h of $\mathcal{H}$ for which Step 1 and Lemma 6.4 hold. Let $B = B(h) \in \mathcal{P}_k(h)$ be the block chosen in Step 1, with first L children $B_1, \dots, B_L \in \mathcal{P}_{k-1}(h)$ and trimmed intervals $C_i \subset B_i$. For this fixed h, these blocks and intervals are fixed. If $j_i \in C_i$, then the selected coordinates intersect distinct children $B_i \in \mathcal{P}_{k-1}(h)$, so no block of that level or below contains all of them. Their smallest containing block is therefore $B \in \mathcal{P}_k(h)$. The correlation classification gives

$$\mathbb{E}_{\nu_h} \prod_{i=1}^{L} Z_{j_i} = -\frac{1}{2\,3^{L/2} \prod_{i=1}^{L} |B_i|^{1/2}}.$$

This value is the same for all $\prod_i |C_i|$ choices. Using $|C_i| \geq |B_i|/2$ and $|B_i| > (4L)^{k-1}$, we obtain

$$-\sum_{j_1 \in C_1, \dots, j_L \in C_L} \mathbb{E}_{\nu_h} \prod_{i=1}^{L} Z_{j_i} = \frac{\prod_i |C_i|}{2\,3^{L/2} \prod_i |B_i|^{1/2}} \geq \frac{\prod_i |B_i|^{1/2}}{2^{L+1} 3^{L/2}} \geq \frac{(4L)^{(k-1)L/2}}{2^{L+1} 3^{L/2}}.$$

By Step 1, all these tuples belong to the deterministic set

$$\mathcal{T}_n := \{\mathbf{j} : 0 \leq j_1 < \cdots < j_L < n,\ \Delta(\mathbf{j}) \geq g_n\}.$$

Every other conditional L-point correlation is also nonpositive. Adding its negative therefore preserves the lower bound and gives

$$-\sum_{\mathbf{j} \in \mathcal{T}_n} \mathbb{E}_{\nu_h} \prod_{i=1}^{L} Z_{j_i} \geq \frac{(4L)^{(k-1)L/2}}{2^{L+1} 3^{L/2}}.$$

This holds for almost every h. Although the block and the C_i depend on h, the enlarged index set $\mathcal{T}_n$ does not. We may therefore interchange its finite sum with the integral over the marker law, obtaining

$$-\sum_{\mathbf{j} \in \mathcal{T}_n} \mathbb{E} \prod_{i=1}^{L} Z_{j_i} = \int \left(-\sum_{\mathbf{j} \in \mathcal{T}_n} \mathbb{E}_{\nu_h} \prod_{i=1}^{L} Z_{j_i} \right) \mathbb{P}_{\mathcal{H}}(dh) \geq \frac{(4L)^{(k-1)L/2}}{2^{L+1} 3^{L/2}}.$$

Finally, $n < 6(4L)^{k+1}$ implies $(4L)^{k-1} > n/(6(4L)^2)$, proving the second inequality in (41). ■

□

8.1. **No Gaussian subsequential limits.** Tuplewise independence identifies the low-order moments, while Proposition 8.1 gives a uniform deficit in the normalised L-th moment. Together these exclude every Gaussian weak subsequential limit.

The Khintchine inequality bounds the moments of independent symmetric $\{-1, 1\}$-valued random variables by the corresponding Gaussian moments. We combine this bound with the negative L-point correlations of our process to obtain a strict L-th moment deficit.

Lemma 8.2 ([28]). *Let $\xi_1, \dots, \xi_n$ be independent random variables such that*

$$\mathbb{P}(\xi_i = 1) = \mathbb{P}(\xi_i = -1) = \frac{1}{2} \quad \text{for each } 1 \leq i \leq n.$$

For every $p \geq 2$ and every $a_1, \dots, a_n \in \mathbb{R}$,

$$\left(\mathbb{E} \left| \sum_{i=1}^{n} a_i \xi_i \right|^p \right)^{1/p} \leq (\mathbb{E}|G|^p)^{1/p} \left(\sum_{i=1}^{n} a_i^2 \right)^{1/2},$$

where $G \sim \mathcal{N}(0,1)$. In particular, if $L \geq 2$ is even, then $\mathbb{E}(\xi_1 + \cdots + \xi_n)^L \leq (L-1)!!\, n^{L/2}$.

Corollary 8.3. *Let $n \geq 1$ and let $\xi_1, \dots, \xi_n$ be independent random variables such that*

$$\mathbb{P}(\xi_i = 1) = \mathbb{P}(\xi_i = -1) = \frac{1}{2} \quad \text{for each } 1 \leq i \leq n.$$

Then

$$\mathbb{E}S_n^p = \mathbb{E}(\xi_1 + \cdots + \xi_n)^p \quad \text{for every integer } 1 \leq p \leq L-1.$$

In particular,

$$\mathbb{E}S_n = 0, \quad \mathbb{E}S_n^2 = n \quad \text{and} \quad \mathbb{E}S_n^4 = 3n^2 - 2n. \tag{42}$$

Moreover, for every $n \geq 6(4L)^2$,

$$\mathbb{E}\left(\frac{S_n}{\sqrt{n}}\right)^L \leq (L-1)!! - \delta_L \quad \textit{and} \quad \delta_L := \frac{L!}{2^{L+1}3^{L/2}\big(6(4L)^2\big)^{L/2}} > 0. \tag{43}$$

Proof. Fix $1 \leq p \leq L-1$. Each monomial $Z_{j_1} \cdots Z_{j_p}$ in S_n^p involves at most $L-1$ distinct coordinates. By Proposition 6.3(iii), the corresponding expectations agree even when some indices are repeated:

$$\mathbb{E}[Z_{j_1} \cdots Z_{j_p}] = \mathbb{E}[\xi_{j_1+1} \cdots \xi_{j_p+1}].$$

Summing over $j_1, \ldots, j_p \in \{0, \ldots, n-1\}$ gives the asserted moment equality. Taking $p = 1, 2, 4$ gives (42).

For the L-th moment, subtracting the two expansions gives

$$\mathbb{E}S_n^L - \mathbb{E}(\xi_1 + \cdots + \xi_n)^L = \sum_{j_1,\ldots,j_L=0}^{n-1} \left(\mathbb{E}\prod_{i=1}^{L} Z_{j_i} - \mathbb{E}\prod_{i=1}^{L} \xi_{j_i+1}\right).$$

Terms with repeated indices involve at most $L-1$ distinct coordinates and cancel by the same argument. For L distinct indices, independence and centring give $\mathbb{E}\prod_i \xi_{j_i+1} = 0$. Each increasing tuple has $L!$ permutations, all with the same coordinate product. Thus

$$\mathbb{E}S_n^L = \mathbb{E}(\xi_1 + \cdots + \xi_n)^L + L! \sum_{0 \leq j_1 < \cdots < j_L < n} \mathbb{E}\prod_{i=1}^{L} Z_{j_i}. \tag{44}$$

By Lemma 6.4, all summands in the last sum are nonpositive. Restricting to the separated tuples in Proposition 8.1 therefore gives the upper bound

$$\sum_{0 \leq j_1 < \cdots < j_L < n} \mathbb{E}\prod_{i} Z_{j_i} \leq -\frac{1}{2^{L+1}3^{L/2}} \left(\frac{n}{6(4L)^2}\right)^{L/2}.$$

Lemma 8.2 bounds the independent moment by $(L-1)!!\, n^{L/2}$. Substituting both bounds into (44) yields

$$\mathbb{E}S_n^L \leq (L-1)!!\, n^{L/2} - \frac{L!}{2^{L+1}3^{L/2}} \left(\frac{n}{6(4L)^2}\right)^{L/2}.$$

Dividing by $n^{L/2}$ proves (43). □

The exclusion of Gaussian weak subsequential limits from the moment estimates follows closely the argument in [11, Step 3.11]. The fourth-moment bound identifies the mean and variance of any hypothetical Gaussian limit, while the strict deficit in the L-th moment rules out that limit. Our estimate on correlations at widely separated coordinates also yields the failure of exponential mixing at order L.

Corollary 8.4. *The sequence* $S_n/\sqrt{n}$ *has no Gaussian weak subsequential limit.*

Proof. Suppose $Y_j := S_{n_j}/\sqrt{n_j}$ converges in distribution to a Gaussian random variable G. By (42),

$$\mathbb{E}Y_j = 0, \quad \mathbb{E}Y_j^2 = 1 \quad \text{and} \quad \mathbb{E}Y_j^4 = 3 - \frac{2}{n_j} \leq 3.$$

In particular, for $A > 0$,

$$\mathbb{E}\big[|Y_j|\mathbf{1}_{\{|Y_j|>A\}}\big] \leq \frac{3}{A^3} \quad \text{and} \quad \mathbb{E}\big[Y_j^2\mathbf{1}_{\{|Y_j|>A\}}\big] \leq \frac{3}{A^2}.$$

These uniform tail bounds give uniform integrability of both (Y_j) and (Y_j^2) by [2, condition (3.18)]. The continuous mapping theorem [2, Theorem 2.7] gives $Y_j^2 \xrightarrow{j\to\infty} G^2$. Uniform integrability then passes the first two moments to the limit by [2, Theorem 3.5], giving

$$\mathbb{E}G = \lim_j \mathbb{E}Y_j = 0 \quad \text{and} \quad \mathbb{E}G^2 = \lim_j \mathbb{E}Y_j^2 = 1.$$

Thus the only possible Gaussian limit is $\mathcal{N}(0,1)$. In particular, a degenerate Gaussian is already excluded.

For the higher moment, fix $M > 0$. The function $x \mapsto \min\{|x|^L, M\}$ is bounded and continuous, so the continuous mapping theorem [2, Theorem 2.7], boundedness and (43) imply

$$\mathbb{E}\min\{|G|^L, M\} = \lim_j \mathbb{E}\min\{|Y_j|^L, M\} \le (L-1)!! - \delta_L.$$

Let $M \to \infty$ and apply monotone convergence. This is the usual lower-semicontinuity argument for a nonnegative moment [2, Theorem 3.4], and it gives

$$\mathbb{E}|G|^L \le (L-1)!! - \delta_L.$$

But L is even and $G \sim \mathcal{N}(0,1)$, so $\mathbb{E}|G|^L = (L-1)!!$, which yields a contradiction. □

Remark 8.5. For $R = 2$, Proposition 7.2 gives exponential two-point mixing, while Corollary 8.4 excludes the CLT for the coordinate observable $\varphi(\omega) = \omega_0$.

8.2. **Failure of exponential mixing at order L.** Dividing the lower bound in Proposition 8.1 by the number of available tuples gives a lower bound for a single correlation. We again use the coordinate observable $\varphi(\omega) = \omega_0$.

Corollary 8.6. *For every $N \ge 6(4L)^2$ there are integers $0 \le n_1 < \cdots < n_L < N$ with*

$$\Delta(\mathbf{n}) \;\ge\; \frac{N}{12(4L)^2} \quad \text{and} \quad -\mathbb{E}\left[\prod_{i=1}^{L} Z_{n_i}\right] \;\ge\; \frac{N^{-L/2}}{2^{L+1}3^{L/2}\big(6(4L)^2\big)^{L/2}}. \tag{45}$$

Therefore (Σ, ν, σ) is not exponentially mixing of order L in $\mathcal{F}_\theta(\Sigma)$ for any $0 < \theta < 1$.

Proof. We divide the proof into two steps.

Step 1. *We find a tuple $\mathbf{n} = (n_1, \dots, n_L)$ with $0 \le n_1 < \cdots < n_L < N$ satisfying (45).*

Proof of Step 1. Let $\mathcal{T}_N := \big\{(j_1, \dots, j_L) : 0 \le j_1 < \cdots < j_L < N,\ \Delta(\mathbf{j}) \ge g_N\big\}$. Recall the definition of g_N from Proposition 8.1. From Proposition 8.1, we obtain

$$\sum_{\mathbf{j}\in\mathcal{T}_N} \left(-\mathbb{E}\left[\prod_{i=1}^{L} Z_{j_i}\right]\right) \ge \frac{N^{L/2}}{2^{L+1}3^{L/2}\big(6(4L)^2\big)^{L/2}}. \tag{46}$$

Each summand is nonnegative by Lemma 6.4. Since $\mathcal{T}_N$ contains at most N^L tuples, the sum is at most N^L times its largest summand,

$$\sum_{\mathbf{j}\in\mathcal{T}_N} \left(-\mathbb{E}\left[\prod_{i=1}^{L} Z_{j_i}\right]\right) \le N^L \max_{\mathbf{j}\in\mathcal{T}_N} \left(-\mathbb{E}\left[\prod_{i=1}^{L} Z_{j_i}\right]\right). \tag{47}$$

Choose $\mathbf{n} = (n_1, \dots, n_L) \in \mathcal{T}_N$ at which this maximum is attained. Combining (46) and (47) and dividing by N^L we obtain

$$-\mathbb{E}\left[\prod_{i=1}^{L} Z_{n_i}\right] \ge \frac{N^{-L/2}}{2^{L+1}3^{L/2}\big(6(4L)^2\big)^{L/2}}.$$

From the definition of $\mathcal{T}_N$ and Proposition 8.1 we obtain $\Delta(\mathbf{n}) \ge g_N \ge N/(12(4L)^2)$. These are precisely the two inequalities in (45). ■

Step 2. *The system (Σ, ν, σ) is not exponentially mixing of order L.*

Proof of Step 2. Fix $0 < \theta < 1$. The coordinate observable $\varphi(\omega) := \omega_0$ belongs to $\mathcal{F}_\theta(\Sigma)$ and has mean zero by Proposition 6.3(iii). Since $\varphi \circ \sigma^{n_i} = Z_{n_i}$, we have

$$\nu\left(\prod_{i=1}^{L} \varphi \circ \sigma^{n_i}\right) - \nu(\varphi)^L = \mathbb{E}\left[\prod_{i=1}^{L} Z_{n_i}\right].$$

If exponential mixing of order L held, applying its defining estimate to the tuples from Step 1 would give

$$\frac{N^{-L/2}}{2^{L+1}3^{L/2}\big(6(4L)^2\big)^{L/2}} \le \left|\mathbb{E}\left[\prod_{i=1}^{L} Z_{n_i}\right]\right| \le Ce^{-c\Delta(\mathbf{n})}\|\varphi\|_\theta^L \le Ce^{-cN/(12(4L)^2)}\|\varphi\|_\theta^L,$$

with $C, c > 0$ independent of N. Multiplying by $N^{L/2}$ gives a fixed positive number bounded above by a quantity tending to zero. This contradiction proves the assertion. ■

□

Remark 8.7. Corollary 8.6 excludes exponential mixing at order L, not qualitative mixing at that order. The subintervals C_i in Proposition 8.1 force the successive gaps to be proportional to the observation scale, a property that does not follow from the moment deficit alone.

Proof of Theorem C. Fix $R \geq 2$, set $L := 2R + 2$, and take the measure ν constructed in Proposition 6.3.

Proposition 6.3(ii)–(iii) gives stationarity, full support and the coordinate distributions and tuplewise independence in item (i). Propositions 7.1 and 7.2 give items (ii) and (iii), with $K_R = K$.

Corollary 8.3 gives the moment bound in item (iv), with $\delta = \delta_L$ and $n_0 = 6(4L)^2$, and Corollary 8.4 excludes Gaussian weak subsequential limits. Finally, Corollary 8.6 gives item (v), including failure of exponential mixing at order L. All properties hold for the same measure ν, completing the proof. □

9. Lean formalisation

Theorems A, B and C have been formalised in Lean 4 using Mathlib. The complete source code is available at `https://github.com/Manzatto/finite-order-mixing-does-not-imply-clt`.

Each main theorem file contains its statement and proof. All declarations below belong to the namespace `FiniteOrderMixing`, and the paths are relative to `FiniteOrderMixing/`.

- Theorem A: statement `TheoremA` and proof `theoremA`, in `TheoremA.lean`.
- Theorem B: statement `TheoremB` and proof `theoremB`, in `TheoremB.lean`.
- Theorem C: statement `TheoremC` and proof `theoremC`, in `TheoremC.lean`.

The formalisation establishes all six assertions of Theorems A and B, and all five assertions of Theorem C.

The external results cited in the paper and used in the formal proofs are also formalised in Lean, in the forms required by our argument rather than in their full published generality. Their hypotheses are verified at each application. For example, the Kolmogorov extension theorem is proved for a finite alphabet, which suffices to construct the symbolic process.

The project contains no `sorry` placeholders or additional axioms. The three main proofs depend only on Lean's standard axioms `propext`, `Classical.choice` and `Quot.sound`. With the versions pinned by `lean-toolchain` and `lake-manifest.json`, compilation and axiom dependencies can be checked from the repository root by running

```
lake build
lake env lean CheckAxioms.lean
```

Declaration of AI use. OpenAI's ChatGPT 6.0 (Astra) was used as a research assistant during the development of this work. AI assistance was also used to develop and revise the Lean formalisation of the main results. The resulting proofs were checked by Lean's kernel, with the axiom dependencies described above. The authors take full responsibility for the correctness of all mathematical statements and proofs, the accuracy of the references, and the final content of the manuscript.

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Appendix A. Smooth physical realisation of symbolic measures

In this appendix, we prove Proposition 3.11, which realises every prescribed ergodic invariant measure of the full two-symbol shift as a physical measure of a $\mathcal{C}^\infty$ diffeomorphism on any compact smooth surface without boundary.

Colli and Vargas construct a $\mathcal{C}^\infty$ surface diffeomorphism with a wandering domain whose orbits are all generic for a particular invariant measure supported on a horseshoe [13, Theorem 1]. Kiriki, Li, Nakano, Soma and Vargas [43, Theorem A] develop methods for realising any prescribed ergodic invariant measure of the full two-symbol shift as a physical measure of a $\mathcal{C}^r$ surface diffeomorphism, for $2 \leq r < \infty$ (see also [41, 42]). Our proof adapts these methods to a symmetric affine horseshoe.

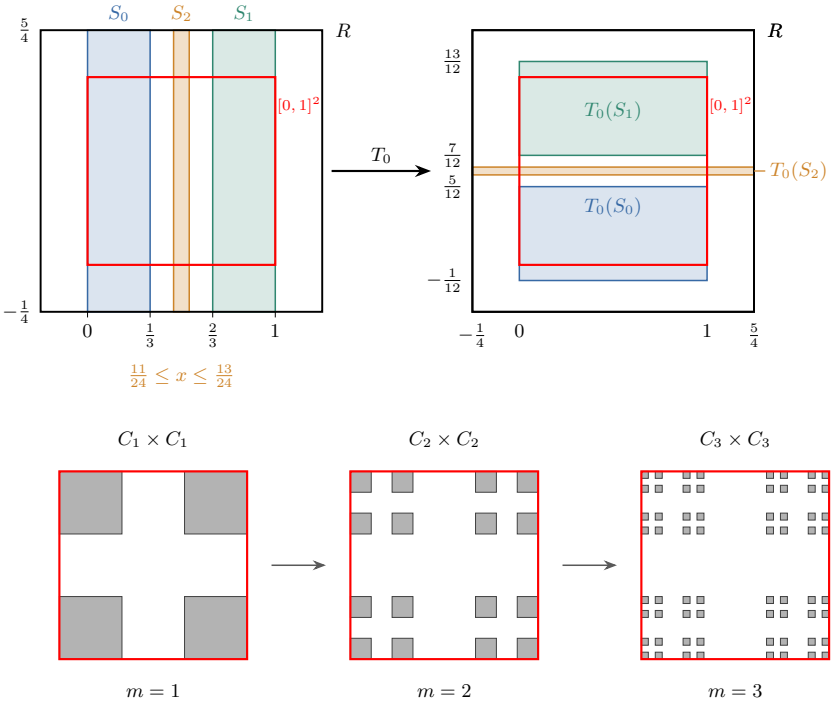


Figure 6. The strips S_0, S_1, S_2 and their images under T_0 (top), together with the sets $\bigcap_{i=-m}^{m-1} T_0^{-i}(S_0 \cup S_1 \cap [0,1]) = C_m \times C_m$ for $m = 1, 2, 3$ (bottom). The black frame is $R = [-\frac{1}{4}, \frac{5}{4}]^2$ and the red frame is $[0,1]^2$.

We first explain the idea of the construction. Fix a nonempty compact smooth surface M without boundary. Set $\gamma = 1/225$ and consider the disjoint rectangles

$$S_0 = \left[0, \frac{1}{3}\right] \times \left[-\frac{1}{4}, \frac{5}{4}\right], \quad S_1 = \left[\frac{2}{3}, 1\right] \times \left[-\frac{1}{4}, \frac{5}{4}\right], \text{ and } S_2 = \left[\frac{11}{24}, \frac{13}{24}\right] \times \left[-\frac{1}{4}, \frac{5}{4}\right].$$

Choose a smooth diffeomorphism $T_0 : \mathbb{R}^2 \to \mathbb{R}^2$ equal to the identity outside a compact set (R in Figure 6), with the following expressions near S_0, S_1 and S_2

$$T_0(x,y) = \begin{cases} (3x, y/3)\,, & (x,y) \in S_0, \\ (3x-2, (y+2)/3)\,, & (x,y) \in S_1, \\ \left(1-y, \frac{1}{2} + \gamma\left(x - \frac{1}{2}\right)\right), & (x,y) \in S_2. \end{cases} \tag{48}$$

Choose a chart $\chi : V \subset M \to \Omega \subset \mathbb{R}^2$, where Ω is a disc in which $[-2,2]^2$ and the closure of the support of T_0 are compactly contained. Define $F_0 = \chi^{-1} \circ T_0 \circ \chi$ on V and extend it by the identity outside V. Then $F_0 : M \to M$ is a $\mathcal{C}^\infty$ diffeomorphism satisfying $F_0(V) = V$.

Let $C \subset [0,1]$ be the (classical) middle-thirds Cantor set and let $\Sigma = \{-1,1\}^{\mathbb{Z}}$, with $(\sigma\omega)_i = \omega_{i+1}$. The two outer branches give the affine horseshoe $\Lambda = \chi^{-1}(C \times C)$, with coding

$$\pi(\omega) = \chi^{-1}\left(\sum_{i\geq 0} \frac{\omega_i + 1}{3^{i+1}}, \sum_{i \geq 0} \frac{\omega_{-(i+1)} + 1}{3^{i+1}}\right). \tag{49}$$

Thus $F_0(\Lambda) = \Lambda$ and $F_0 \circ \pi = \pi \circ \sigma$. Moreover, Λ is a uniformly hyperbolic set for F_0. The map π is a homeomorphism, and the derivative on the horseshoe is $\mathrm{diag}(3, 1/3)$ in these coordinates. For $0 < \theta < 1$, use $d_\theta(\omega, \omega') = \theta^{\min\{|i| : \omega_i \neq \omega'_i\}}$ and $\|u\|_\theta = \|u\|_\infty + \mathrm{Lip}_{d_\theta}(u)$.

Let ν be the prescribed ergodic invariant measure. The conjugacy π transfers the statistical properties of (Σ, σ, ν) to (Λ, F_0, μ), where $\mu = \pi_*\nu$. We choose a ν-generic point ω and set $x = \pi(\omega)$, so that the time averages along the F_0-orbit of x converge to the integrals against μ.

We then perturb F_0 away from Λ to obtain a smooth diffeomorphism T and a nonempty open set W. Since $T = F_0$ near Λ, we have $T^i(x) = F_0^i(x)$ for every $i \geq 0$. The orbits of points in W under T follow the orbit $(T^i(x))_{i\in\mathbb{N}}$ most of the time. Hence, they have the same statistics as the orbit of x in the sense that, for every continuous observable, their time averages converge uniformly on W to its integral against μ. Thus μ is physical.

Geometrically, this resembles the figure-eight example described by Young [73, Section 2.2, Figure 1]. Here the prescribed statistical behaviour comes from a generic orbit in a horseshoe.

Proof of Proposition 3.11. Since ν is ergodic, Birkhoff's ergodic theorem allows us to choose $\omega = (\omega_j)_{j\in\mathbb{Z}} \in \Sigma$ in the basin of ν under σ. We use a fixed Riemannian distance d on M. Until the transfer to M, all constructions take place in the coordinates of (48). The sequence $\omega \in \Sigma$ is fixed throughout the proof.

Step 1. *For every integer $N_0 \geq 3$, there exist a strictly increasing sequence of positive integers $(n_k)_{k\geq 0}$, sequences of points $((x_k, y_k))_{k\geq 0}$ in $[0,1]^2$ and $((v_k, z_k))_{k\geq 0}$ in S_2, and a sequence of real numbers $(\varepsilon_k)_{k\geq 0}$, such that, for every $k \geq 0$,*

- $T_0^j(x_k, y_k) \in S_0 \cup S_1$ *for* $0 \leq j < n_k$.
- $T_0^{n_k}(x_k, y_k) = (v_k, z_k)$.
- $T_0(v_k, z_k + \varepsilon_k) = (x_{k+1}, y_{k+1})$.
- $|\varepsilon_k| \leq 3^{-N_0 - \lfloor\sqrt{k+2}\rfloor}$.

Proof of Step 1. Fix an integer $N_0 \geq 3$ and set

$$h_0 = 0, \quad h_{k+1} = N_0 + \lfloor\sqrt{k+2}\rfloor, \quad n_k = k + 1 + h_{k+1}, \quad t_0 = 0, \quad t_{k+1} = t_k + n_k + 1. \tag{50}$$

It follows from (50) that $h_{k+1} \leq n_k$, $1 \leq n_{k+1} - n_k \leq 2$ and $h_k = O(\sqrt{k})$.

We call a finite word $\omega^{(k)} = (\omega_0^{(k)}, \dots, \omega_{n_k-1}^{(k)}) \in \{-1,1\}^{n_k}$ a block of length n_k. We will choose (x_k, y_k) so that, for $0 \leq j < n_k$,

$$T_0^j(x_k, y_k) \in \begin{cases} S_0, & \omega_j^{(k)} = -1 \\ S_1, & \omega_j^{(k)} = 1 \end{cases}, \quad \text{and} \quad T_0^{n_k}(x_k, y_k) \in S_2.$$

Define the words by $\omega^{(0)} \in \{-1,1\}^{n_0}$ with $\omega_j^{(0)} := \omega_j$ for $0 \leq j < n_0$. Moreover, we recursively define

$$\omega_j^{(k+1)} := \begin{cases} -\omega_{n_k-1-j}^{(k)}, & 0 \leq j < h_{k+1}, \\ \omega_{t_{k+1}+j}, & h_{k+1} \leq j < n_{k+1}. \end{cases} \tag{51}$$

Observe that $\omega^{(k+1)}$ is obtained by reversing the order and changing the signs of the last h_{k+1} symbols of $\omega^{(k)}$, then completing the block with symbols from ω.

For $s \in \{-1, 1\}$, write $\phi_s(u) := (u + 1 + s)/3$. The maps ϕ_{-1} and ϕ_1 contract $[0, 1]$ onto its left and right thirds, respectively. By (48),

$$T_0(x, y) = \begin{cases} \big(\phi_{-1}^{-1}(x), \phi_{-1}(y)\big), & (x, y) \in S_0, \\ \big(\phi_1^{-1}(x), \phi_1(y)\big), & (x, y) \in S_1. \end{cases} \tag{52}$$

Set

$$v_k = \frac{1}{2} + \frac{1}{48(k+1)}, \quad y_0 = \frac{1}{2} \quad \text{and} \quad y_{k+1} = \frac{1}{2} + \gamma\left(v_k - \frac{1}{2}\right), \tag{53}$$

and define

$$x_k = \phi_{\omega_0^{(k)}} \circ \cdots \circ \phi_{\omega_{n_k-1}^{(k)}}(v_k), \; z_k = \phi_{\omega_{n_k-1}^{(k)}} \circ \cdots \circ \phi_{\omega_0^{(k)}}(y_k) \text{ and } \varepsilon_k = 1 - z_k - x_{k+1}. \tag{54}$$

Since $\phi_s([0, 1]) \subset [0, 1]$, (53) and (54) give

$$(x_k, y_k) \in [0, 1]^2 \quad \text{and} \quad (v_k, z_k) \in \left[\frac{1}{2}, \frac{25}{48}\right] \times [0, 1] \subset S_2.$$

The maps ϕ_{-1} and ϕ_1 send $[0, 1]$ into $[0, 1/3]$ and $[2/3, 1]$, respectively.

We claim that, for $0 \le j \le n_k$,

$$T_0^j(x_k, y_k) = \left(\phi_{\omega_j^{(k)}} \circ \cdots \circ \phi_{\omega_{n_k-1}^{(k)}}(v_k), \phi_{\omega_{j-1}^{(k)}} \circ \cdots \circ \phi_{\omega_0^{(k)}}(y_k)\right), \tag{55}$$

where an empty composition should be understood as the identity. For $j = 0$, (55) follows from the definition of x_k in (54).

Suppose (55) holds for some $j < n_k$. Since $v_k, y_k \in [0, 1]$, the first coordinate of $T_0^j(x_k, y_k)$ lies in

$$\phi_{\omega_j^{(k)}}([0, 1]) = \begin{cases} \left[0, \frac{1}{3}\right], & \omega_j^{(k)} = -1, \\ \left[\frac{2}{3}, 1\right], & \omega_j^{(k)} = 1, \end{cases}$$

and its second coordinate lies in $[0, 1]$. Hence, $T_0^j(x_k, y_k) \in S_0$ if $\omega_j^{(k)} = -1$, and $T_0^j(x_k, y_k) \in S_1$ if $\omega_j^{(k)} = 1$. Using (52) and $\phi_{\omega_j^{(k)}}^{-1} \circ \phi_{\omega_j^{(k)}} = \mathrm{id}$, we obtain

$$T_0^{j+1}(x_k, y_k) = \left(\phi_{\omega_{j+1}^{(k)}} \circ \cdots \circ \phi_{\omega_{n_k-1}^{(k)}}(v_k), \phi_{\omega_j^{(k)}} \circ \cdots \circ \phi_{\omega_0^{(k)}}(y_k)\right).$$

This proves (55) by induction, together with $T_0^j(x_k, y_k) \in S_0 \cup S_1$ for $0 \le j < n_k$. Taking $j = n_k$ in (55) and using the definition of z_k in (54) gives

$$T_0^{n_k}(x_k, y_k) = \left(v_k, \phi_{\omega_{n_k-1}^{(k)}} \circ \cdots \circ \phi_{\omega_0^{(k)}}(y_k)\right) = (v_k, z_k).$$

From (54), $z_k + \varepsilon_k = 1 - x_{k+1} \in [0, 1]$. Since $v_k \in [1/2, 25/48]$ from (53), the point $(v_k, z_k + \varepsilon_k)$ lies in S_2. Using (48), (53) and (54), we obtain

$$T_0(v_k, z_k + \varepsilon_k) = \left(1 - z_k - \varepsilon_k, \frac{1}{2} + \gamma\left(v_k - \frac{1}{2}\right)\right) = (x_{k+1}, y_{k+1}),$$

which proves the second identity.

Finally, the identity $1 - \phi_s(u) = \phi_{-s}(1 - u)$, together with (51) and (54), implies

$$1 - z_k, \; x_{k+1} \in \left(\phi_{\omega_0^{(k+1)}} \circ \cdots \circ \phi_{\omega_{h_{k+1}-1}^{(k+1)}}\right)([0, 1]).$$

This interval has length $3^{-h_{k+1}}$. Since $\varepsilon_k = 1 - z_k - x_{k+1}$ by (54), we obtain, using (50),

$$|\varepsilon_k| \le 3^{-h_{k+1}} = 3^{-N_0 - \lfloor\sqrt{k+2}\rfloor}. \tag{56}$$

■

Step 2. *For N_0 sufficiently large, there exists a smooth diffeomorphism P of $\mathbb{R}^2$, equal to the identity outside* $\operatorname{int} S_2$*, such that, for every $k \ge 0$, $P(x, y) = (x, y + \varepsilon_k)$ on a neighbourhood of (v_k, z_k).*

Proof of Step 2. Define the constants

$$\rho_k = \frac{1}{100(k+2)^2}. \tag{57}$$

Define the closed squares

$$Q_k = (v_k, z_k) + [-\rho_k, \rho_k]^2 \quad \text{and} \quad Q_k' = (v_k, z_k) + [-\rho_k/2, \rho_k/2]^2.$$

They are centred at (v_k, z_k) and have sides $2\rho_k$ and ρ_k, respectively. Choose a smooth function $\beta : \mathbb{R}^2 \to [0,1]$, equal to one on $[-1/2, 1/2]^2$ and supported in $(-1,1)^2$. Define

$$f(z) = \sum_{k\geq 0} \varepsilon_k \beta\left(\frac{z-(v_k,z_k)}{\rho_k}\right) \quad \text{and} \quad P(x,y) = (x, y + f(x,y)).$$

For each k, the term $\varepsilon_k\beta((z-(v_k,z_k))/\rho_k)$ in the sum above is zero outside Q_k. Since $\beta(x) = 1$, for each $x \in [-1/2,1/2]^2$, it follows that

$$\varepsilon_k \beta\left(\frac{z-(v_k,z_k)}{\rho_k}\right) = \varepsilon_k$$

for every $z \in Q_k'$. From (53) and (57), the squares Q_k are pairwise disjoint. Indeed, if $j < k$, then $v_k \leq v_{j+1}$, so

$$v_j - v_k \geq v_j - v_{j+1} = \frac{1}{48(j+1)(j+2)}.$$

Also, $\rho_k \leq \rho_j$, giving

$$\rho_j + \rho_k \leq 2\rho_j = \frac{1}{50(j+2)^2} < \frac{1}{48(j+1)(j+2)}.$$

Comparing these estimates gives $v_k + \rho_k < v_j - \rho_j$, and therefore Q_k and Q_j are disjoint. The definition of v_k gives $1/2 < v_k \leq 25/48$. Also, $z_k \in [0,1]$, because $y_k \in [0,1]$ and each of the maps ϕ_{-1} and ϕ_1 maps $[0,1]$ into itself. Since $\rho_k \leq 1/400$, it follows that

$$Q_k \subset \left[\frac{1}{2} - \frac{1}{400}, \frac{25}{48} + \frac{1}{400}\right] \times \left[-\frac{1}{400}, 1 + \frac{1}{400}\right] \text{ for every } k \geq 0.$$

Moreover, $\overline{Q_k} \subset \operatorname{int} S_2$. Thus $\bigcup_{k\geq 0} Q_k$ is compactly contained in $\operatorname{int} S_2$.

For each integer $m \geq 0$, we differentiate the function $z \mapsto \varepsilon_k\beta((z-(v_k,z_k))/\rho_k)$, m times with respect to $z = (x,y)$, keeping k fixed. The chain rule gives

$$D^m\left[\varepsilon_k\beta\left(\frac{z-(v_k,z_k)}{\rho_k}\right)\right] = \frac{\varepsilon_k}{\rho_k^m} D^m\beta\left(\frac{z-(v_k,z_k)}{\rho_k}\right).$$

Thus,

$$\left\| D^m\left[\varepsilon_k\beta\left(\frac{\cdot-(v_k,z_k)}{\rho_k}\right)\right]\right\|_\infty \leq C_m|\varepsilon_k|\rho_k^{-m}, \text{ where } C_m = \|D^m\beta\|_\infty$$

is finite and depends only on β and m. Using (56) and (57), we obtain

$$C_m \sum_{k\geq 0} |\varepsilon_k|\rho_k^{-m} \leq C_m 100^m 3^{-N_0} \sum_{k\geq 0} (k+2)^{2m} 3^{-\lfloor\sqrt{k+2}\rfloor} < \infty. \tag{58}$$

Therefore the series defining f, and the series obtained by differentiating its summands any number of times, converge uniformly. Hence f is smooth. Taking $m = 1$ in (58) allows us to fix N_0 sufficiently large that $\|Df\|_\infty < 1/2$. For each fixed x, the map $y \mapsto y + f(x,y)$ is then strictly increasing and equals the identity for large $|y|$. Thus P is a smooth diffeomorphism.

For $z \in Q_k'$, we have $\varepsilon_k\beta\left((z-(v_k,z_k))/\rho_k\right) = \varepsilon_k$. For $j \neq k$, the squares Q_j and Q_k are disjoint. Since $Q_k' \subset Q_k$, every $z \in Q_k'$ lies outside Q_j, implying $\varepsilon_j\beta\left((z-(v_j,z_j))/\rho_j\right) = 0$. Thus, for every $z \in Q_k'$,

$$f(z) = \sum_{j\geq 0} \varepsilon_j\beta\left(\frac{z-(v_j,z_j)}{\rho_j}\right) = \varepsilon_k\beta\left(\frac{z-(v_k,z_k)}{\rho_k}\right) + \sum_{\substack{j\geq 0\\ j\neq k}} \varepsilon_j\beta\left(\frac{z-(v_j,z_j)}{\rho_j}\right) = \varepsilon_k.$$

This implies

$$P(z) = z + (0, \varepsilon_k) \quad \text{for } z \in Q_k'. \tag{59}$$

This proves the stated local translation property. ∎

Step 3. *Choose N_0 and P as in Step 2 and let $G = T_0 \circ P$. There exists a sequence of closed rectangles $(R_k)_{k\geq 0}$, centred at (x_k, y_k) and with nonempty interiors, such that*

$$G^{n_k+1}(R_k) = R_{k+1} \quad \text{for every } k \geq 0. \tag{60}$$

Moreover, for $0 \leq j < n_k$, one has $G^j(R_k) \subset S_0$ if $\omega_j^{(k)} = -1$, and $G^j(R_k) \subset S_1$ if $\omega_j^{(k)} = 1$.

Proof of Step 3. Set $\delta = 10^{-3}$ and define positive numbers

$$p_0 = \delta 3^{-n_0}, \quad q_0 = \frac{\delta}{45}, \quad p_{k+1} = 3^{-n_k} q_k \quad \text{and} \quad q_{k+1} = \gamma 3^{n_k} p_k. \tag{61}$$

Let $R_k := (x_k, y_k) + [-p_k, p_k] \times [-q_k, q_k]$.

Since $n_{k+1} - n_k \leq 2$ by (50) and $\gamma = 1/225$, (61) gives

$$p_{k+1} = 3^{-n_k} q_k \leq 9 \cdot 3^{-n_{k+1}} q_k \quad \text{and} \quad q_{k+1} = \frac{3^{n_k} p_k}{225}.$$

Starting from $p_0 = \delta 3^{-n_0}$ and $q_0 = \delta/45$, induction gives

$$p_k \leq \delta 3^{-n_k} 5^{-k} \quad \text{and} \quad q_k \leq \frac{\delta}{45} 5^{-k}. \tag{62}$$

We prove the claimed inclusions by induction on j. Assume that they hold for $0 \leq i < j$. Then $G^j(R_k) = T_0^j(R_k)$ is a rectangle centred at $T_0^j(x_k, y_k)$, with horizontal side $2 \cdot 3^j p_k$. By (55),

$$\left(T_0^j(x_k, y_k)\right)_1 = \phi_{\omega_j^{(k)}} \circ \cdots \circ \phi_{\omega_{n_k-1}^{(k)}}(v_k),$$

where $(x, y)_1 := x$. Since each ϕ_s is increasing with derivative $1/3$, and $\min\{v_k, 1 - v_k\} \geq 23/48$, if $\omega_j^{(k)} = -1$, then

$$\frac{23}{48} 3^{-(n_k-j)} \leq \left(T_0^j(x_k, y_k)\right)_1 \leq \frac{1}{3} - \frac{23}{48} 3^{-(n_k-j)}.$$

If $\omega_j^{(k)} = 1$, then

$$\frac{2}{3} + \frac{23}{48} 3^{-(n_k-j)} \leq \left(T_0^j(x_k, y_k)\right)_1 \leq 1 - \frac{23}{48} 3^{-(n_k-j)}.$$

From (62),

$$2 \cdot 3^j p_k \leq 2\delta 5^{-k} 3^{-(n_k-j)} < \frac{23}{48} 3^{-(n_k-j)}.$$

Hence the horizontal projection of $G^j(R_k)$ lies in $[0, 1/3]$ if $\omega_j^{(k)} = -1$, and in $[2/3, 1]$ if $\omega_j^{(k)} = 1$.

For the vertical coordinates, (53) and (62) give

$$[y_k - q_k, y_k + q_k] \subset \left[\frac{1}{2} - \frac{\delta}{45}, \frac{1}{2} + \frac{\gamma}{48} + \frac{\delta}{45}\right] \subset [0, 1].$$

Moreover,

$$\phi_{-1}([0, 1]) = \left[0, \frac{1}{3}\right] \quad \text{and} \quad \phi_1([0, 1]) = \left[\frac{2}{3}, 1\right].$$

Thus the vertical projection of $G^j(R_k)$ remains in $[0, 1]$. Therefore, $G^j(R_k) \subset S_0$ if $\omega_j^{(k)} = -1$, and $G^j(R_k) \subset S_1$ if $\omega_j^{(k)} = 1$, completing the induction.

The inclusions proved above show that all points of R_k follow the same sequence of branches in S_0 and S_1 for the first n_k iterates. Hence $G^{n_k}(R_k) = T_0^{n_k}(R_k)$. Taking $j = n_k$ in (55) gives

$$G^{n_k}(x_k, y_k) = T_0^{n_k}(x_k, y_k) = (v_k, z_k).$$

On both S_0 and S_1, the affine formula for G has linear part $\operatorname{diag}(3, 1/3)$. Thus, for $|u| \leq p_k$ and $|w| \leq q_k$,

$$G^{n_k}(x_k + u, y_k + w) = G^{n_k}(x_k, y_k) + (3^{n_k} u, 3^{-n_k} w) = (v_k + 3^{n_k} u, z_k + 3^{-n_k} w).$$

Since $R_k = (x_k, y_k) + [-p_k, p_k] \times [-q_k, q_k]$, letting u and w vary over these intervals gives

$$G^{n_k}(R_k) = (v_k, z_k) + [-3^{n_k} p_k, 3^{n_k} p_k] \times [-3^{-n_k} q_k, 3^{-n_k} q_k].$$

By (62), its sides satisfy

$$2 \cdot 3^{n_k} p_k \leq 2\delta 5^{-k} \quad \text{and} \quad 2 \cdot 3^{-n_k} q_k \leq \frac{2\delta}{45} 3^{-n_k} 5^{-k} \leq 2\delta 5^{-k}.$$

Since $\delta = 10^{-3}$ and $(k+2)^2 \le 5^{k+1}$,

$$2 \cdot \delta \cdot 5^{-k} = \frac{1}{500 \cdot 5^k} \le \frac{1}{100(k+2)^2} = \rho_k.$$

Thus $G^{n_k}(R_k) \subset (v_k, z_k) + [-\rho_k/2, \rho_k/2]^2 = Q'_k$. From (59), P translates this rectangle by $(0, \varepsilon_k)$. Its centre is therefore $(v_k, z_k + \varepsilon_k) = (v_k, 1 - x_{k+1})$, where we used $\varepsilon_k = 1 - z_k - x_{k+1}$. Thus

$$P\big(G^{n_k}(R_k)\big) = (v_k, 1 - x_{k+1}) + [-3^{n_k} p_k, 3^{n_k} p_k] \times [-3^{-n_k} q_k, 3^{-n_k} q_k].$$

Since $(v_k, 1 - x_{k+1}) \in [1/2, 25/48] \times [0, 1]$, and both sides of this rectangle are at most 2δ, we obtain

$$P\big(G^{n_k}(R_k)\big) \subset \left[\frac{1}{2} - \delta, \frac{25}{48} + \delta\right] \times [-\delta, 1 + \delta] \subset S_2.$$

We can therefore apply the formula for T_0 on S_2, which sends $(v_k, 1 - x_{k+1})$, the centre of $P(G^{n_k}(R_k))$, to

$$T_0(v_k, 1 - x_{k+1}) = \left(x_{k+1}, \frac{1}{2} + \gamma\left(v_k - \frac{1}{2}\right)\right) = (x_{k+1}, y_{k+1}).$$

The first coordinate of $T_0(x, y)$ is $1 - y$, and the second coordinate is $1/2 + \gamma(x - 1/2)$. Thus

$$\begin{aligned} G^{n_k+1}(R_k) &= T_0\big(P(G^{n_k}(R_k))\big) = (x_{k+1}, y_{k+1}) + [-3^{-n_k} q_k, 3^{-n_k} q_k] \times [-\gamma 3^{n_k} p_k, \gamma 3^{n_k} p_k] \\ &= (x_{k+1}, y_{k+1}) + [-p_{k+1}, p_{k+1}] \times [-q_{k+1}, q_{k+1}] = R_{k+1}, \end{aligned}$$

where we used (61). Hence (60) holds. ∎

Step 4. *There exists a $\mathcal{C}^\infty$ diffeomorphism $T : M \to M$, equal to F_0 near Λ, and a nonempty open set $W \subset M$ such that*

$$\lim_{N\to\infty} \frac{1}{N} \sum_{j=0}^{N-1} \sup_{y \in W} d(T^j y, \pi(\sigma^j \omega)) = 0. \tag{63}$$

Proof of Step 4. Define T on M by

$$T(x) = \begin{cases} \chi^{-1} \circ G \circ \chi(x), & x \in V, \\ x, & x \in M \setminus V. \end{cases}$$

Since G equals the identity outside a compact subset of Ω, this defines a $\mathcal{C}^\infty$ diffeomorphism $T : M \to M$. The equality $G = T_0$ near $C \times C$ gives $T = F_0$ near Λ. Define

$$W := \chi^{-1}(\operatorname{int} R_0),$$

which is nonempty and open.

Fix an integer $q \ge 1$ and let $j = t_k + \ell$, where $h_k + q \le \ell < n_k - q$. For every integer s with $-q \le s \le q$, we have $h_k \le \ell + s < n_k$. Thus (51) gives

$$\omega^{(k)}_{\ell+s} = \omega_{t_k+\ell+s} = \omega_{j+s}.$$

Now let $x \in W$. Since $\chi(x) \in R_0$, (60) implies $\chi(T^{t_k} x) = G^{t_k}(\chi(x)) \in R_k$ and therefore $\chi(T^{j+s}x) = G^{\ell+s}\big(\chi(T^{t_k}x)\big) \in G^{\ell+s}(R_k)$. From Step 3,

$$\chi(T^{j+s}x) \in S_0 \text{ if } \omega^{(k)}_{\ell+s} = \omega_{j+s} = -1 \quad \text{and} \quad \chi(T^{j+s}x) \in S_1 \text{ if } \omega^{(k)}_{\ell+s} = \omega_{j+s} = 1.$$

For $-q \le s \le q$, write

$$\chi(T^{j+s}x) = (u_s, v_s) \quad \text{and} \quad \chi(\pi(\sigma^{j+s}\omega)) = (\bar{u}_s, \bar{v}_s).$$

Since $(u_s, v_s) \in G^{\ell+s}(R_k) \subset [0, 1]^2$ by Step 3, we have $u_s, v_s \in [0, 1]$. Also, $\bar{u}_s, \bar{v}_s \in [0, 1]$, because $\chi(\Lambda) = C \times C \subset [0, 1]^2$. By (49), $(\bar{u}_s, \bar{v}_s)$ belongs to S_0 if $\omega_{j+s} = -1$, and to S_1 if $\omega_{j+s} = 1$. Since $G = T_0$ on $S_0 \cup S_1$, subtracting the coordinate formulas in (48) for $G(u_s, v_s)$ and $G(\bar{u}_s, \bar{v}_s)$ gives, for $-q \le s < q$,

$$u_{s+1} - \bar{u}_{s+1} = 3(u_s - \bar{u}_s) \quad \text{and} \quad v_{s+1} - \bar{v}_{s+1} = \frac{v_s - \bar{v}_s}{3}.$$

Iterating these identities gives

$$|u_0 - \bar{u}_0| = 3^{-q}|u_q - \bar{u}_q| \le 3^{-q} \quad \text{and} \quad |v_0 - \bar{v}_0| = 3^{-q}|v_{-q} - \bar{v}_{-q}| \le 3^{-q}.$$

Hence $\|\chi(T^j x) - \chi(\pi(\sigma^j\omega))\|_{\mathbb{R}^2} \le \sqrt{2}\,3^{-q}$. Since χ^{-1} is Lipschitz on $[0,1]^2$ and $x \in W$ was arbitrary, we obtain

$$\sup_{y\in W} d(T^j y, \pi(\sigma^j\omega)) \le C3^{-q}, \tag{64}$$

where C is independent of k, q and j. Fix $q \ge 1$ and, for each $N \ge 1$, let E_N be the set of times $j = t_k + \ell \in \{0,\dots,N-1\}$, with $0 \le \ell \le n_k$, for which $\ell < h_k + q$ or $\ell \ge n_k - q$. Thus (64) holds for every $j \in \{0,\dots,N-1\} \setminus E_N$. We now show that $\#E_N/N \to 0$ and deduce (63).

Each interval of times $t_k,\dots,t_k+n_k$ contains at most h_k+q excluded times at its beginning, q at its end, and the return time t_k+n_k. Hence it contains at most h_k+2q+1 excluded times. Choose K such that $t_K \le N < t_{K+1}$. Counting the completed intervals $0,\dots,K-1$ and bounding the contribution from the unfinished interval by n_K+1 gives

$$\#E_N \le \sum_{k=0}^{K-1}(h_k+2q+1) + (n_K+1) \le K(N_0+\sqrt{K}+2q+1) + (n_K+1).$$

By (50),

$$t_K = \sum_{k=0}^{K-1}(n_k+1) \ge \frac{K(K+1)}{2} \quad \text{and} \quad n_K + 1 = K + 2 + N_0 + \lfloor\sqrt{K+2}\rfloor.$$

Since $N \ge t_K$, for $K \ge 1$ we obtain

$$\frac{\#E_N}{N} \le \frac{2(N_0+\sqrt{K}+2q+1)}{K+1} + \frac{2(n_K+1)}{K(K+1)} \longrightarrow 0 \quad \text{as } N\to\infty,$$

with q fixed. For $j \notin E_N$, (64) bounds the distance by $C3^{-q}$. For $j \in E_N$, the distance is at most $\operatorname{diam}(M)$. Therefore

$$\begin{aligned}\frac{1}{N}\sum_{j=0}^{N-1}\sup_{y\in W} d(T^j y,\pi(\sigma^j\omega)) &= \frac{1}{N}\sum_{\substack{0\le j<N\\ j\notin E_N}}\sup_{y\in W} d(T^j y,\pi(\sigma^j\omega)) + \frac{1}{N}\sum_{j\in E_N}\sup_{y\in W} d(T^j y,\pi(\sigma^j\omega))\\ &\le C3^{-q}\frac{N-\#E_N}{N} + \operatorname{diam}(M)\frac{\#E_N}{N}.\end{aligned}$$

Taking the limit superior as $N\to\infty$ gives

$$\limsup_{N\to\infty}\frac{1}{N}\sum_{j=0}^{N-1}\sup_{y\in W} d(T^j y,\pi(\sigma^j\omega)) \le C3^{-q}.$$

Letting $q\to\infty$ proves (63). The final averaging argument follows [42, proof of Theorem A]. ■

Step 5. *The measure $\mu = \pi_*\nu$ is T-invariant and, for every $v \in \mathcal{C}^0(M)$,*

$$\lim_{N\to\infty}\sup_{y\in W}\left|\frac{1}{N}\sum_{j=0}^{N-1} v(T^j y) - \int v\,d\mu\right| = 0.$$

In particular, μ is physical and W is contained in its basin. Moreover, $T(\Lambda)=\Lambda$, $T\circ\pi = \pi\circ\sigma$, and there exists $C_1 < \infty$ such that $\|v\circ\pi\|_{1/3} \le C_1\|v\|_{\mathcal{C}^1(M)}$ for every $v \in \mathcal{C}^1(M)$.

Proof of Step 5. By Step 4, $T = F_0$ near Λ. Hence $T(\Lambda)=\Lambda$ and $T\circ\pi = \pi\circ\sigma$. Since ν is σ-invariant, $\mu = \pi_*\nu$ is T-invariant.

Since ω lies in the basin of ν, for every $v \in \mathcal{C}^0(M)$,

$$\frac{1}{N}\sum_{j=0}^{N-1} v(\pi(\sigma^j\omega)) \xrightarrow{N\to\infty} \int v\,d\mu. \tag{65}$$

For Lipschitz v, (63) transfers the convergence in (65) uniformly to W. Uniform approximation by Lipschitz functions gives the same conclusion for every continuous v, proving (15). The set W is nonempty and open, hence has positive smooth volume. Thus μ is physical.

Finally, the coding of Λ and the affine formulas in (48) give $d\big(\pi(\omega),\pi(\omega')\big) \le C d_{1/3}(\omega,\omega')$ for all $\omega,\omega' \in \Sigma$. Composing with $v \in \mathcal{C}^1(M)$ gives the stated norm bound for a constant C_1 independent of v. ■

□